

**MASTER STATISTICAL ANALYSIS PLAN (M-SAP)**

|                                          |                                                                                                                                                                                                                          |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Title</b>                             | Platform Trial for the Treatment of Amyotrophic Lateral Sclerosis (ALS): A perpetual multi-center, multi-regimen, clinical trial evaluating the safety and efficacy of investigational products for the treatment of ALS |
| <b>Regulatory Sponsor</b>                | Merit E. Cudkowicz, MD                                                                                                                                                                                                   |
| <b>Current Protocol Version</b>          | 5.0                                                                                                                                                                                                                      |
| <b>Current Protocol Date</b>             | 15 Dec 2022                                                                                                                                                                                                              |
| <b>Statistical Analysis Plan Version</b> | 3.0                                                                                                                                                                                                                      |
| <b>Statistical Analysis Plan Date</b>    | 06 Feb 2023                                                                                                                                                                                                              |

## SAP APPROVAL SIGNATURES

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## SAP REVISION HISTORY

| Version | Date        | Description of Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | 24 Jun 2020 | Initial version                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2.0     | 01 Aug 2022 | Revision of Section 5.2 SVC to specify that a minimum of 2 completed SVC maneuvers are required to calculate an estimate of maximum SVC for a given participant visit. A single maximum vital capacity reported by a pulmonary function lab and obtained by either a slow or forced maneuver is also acceptable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3.0     | 06 Feb 2023 | <p>Revision of Section 1.0 Governing Documents to reference Master Protocol version 5.0.</p> <p>Revision of Section 2.3 Study Population to update eligibility criteria.</p> <p>Revision of Section 2.5 Regimen Allocation to specify additional stratification by use vs. non-use of sodium phenylbutyrate/taurursodiol at the time of screening for the Master Protocol.</p> <p>Revision of Section 2.6 Treatment Allocation to specify additional stratification by use vs. non-use of sodium phenylbutyrate/taurursodiol at the time of screening for the Master Protocol.</p> <p>Revision of Section 2.9 Master Protocol Schedule of Assessments to align with Master Protocol version 5.0.</p> <p>Revision of Section 3.6 Missing Data to specify that mean imputation of missing baseline covariates occurs after transformation, where applicable.</p> <p>Revision of Section 4 Study Endpoints to align with Master Protocol version 5.0.</p> <p>Revision of Section 5.1 ALSFRS-R to specify details of the calculation of pre-baseline slope.</p> <p>Revision of Section 5.2 SVC to specify details of the calculation of normal values.</p> <p>Revision of Section 5.3 HHD and Grip Strength to define HHD0[2] and to specify that HHD0 and HHD0[2] are composite endpoints with death or death-equivalent.</p> <p>Removal of former Section 5.4 Voice Characteristics to align with Master Protocol version 5.0.</p> <p>Revision of Section 5.4 Biofluid Biomarkers to specify the assay techniques used to quantify serum creatinine and serum and CSF NfL</p> |

| Version | Date | Description of Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |      | <p>and to specify that levels of serum and CSF NfL will be log-transformed in all analyses.</p> <p>Revision of Section 5.5 Patient-reported Outcomes to align with Master Protocol version 5.0.</p> <p>Revision of Section 5.6 Survival to add an additional survival endpoint of death alone, to specify that the date of PAV initiation, where applicable, will be imputed as the fifteenth day of a month if not specified more precisely, and to specify that time at risk begins at each participant's Baseline Visit and ends at the Week 24 Visit if completed or at the earlier of the end of the Week 24 Visit window or date last known alive if a Week 24 visit is not completed.</p> <p>Addition of Section 5.7 King's ALS Clinical Staging System to specify determination of King's stage.</p> <p>Addition of Section 5.8 Clinical Safety Laboratory Tests to specify clinical safety laboratory tests and derivation of measures of drug-induced liver injury.</p> <p>Revision of Section 6.1 Analysis Sets to define the Full Analysis Set, to add the Efficacy Common Mode of Administration (ECM) analysis set, and to specify the time point at which data is excluded from the Efficacy Per-protocol analysis set.</p> <p>Revision of Section 6.2 Baseline Characteristics to include additional baseline characteristics.</p> <p>Revision of Section 6.3 Primary Efficacy Analysis and Supportive Analyses to specify the estimand.</p> <p>Revision of Section 6.4 Interim Analysis to specify that the default design for a regimen will include stopping early only for futility.</p> <p>Revision of Section 6.5 Hierarchical Testing to fully specify the secondary endpoints.</p> <p>Revision of Section 6.5.2 Repeated-measures Model to add a fixed term for treatment group (removing the shared-baseline assumption at the recommendation of the FDA), to add fixed terms for baseline use of sodium phenylbutyrate/taurursodiol, baseline serum NfL level, and their interactions with visit, and to specify the estimand.</p> <p>Revision of Section 6.5.3 Random-slopes Model to add a fixed term for treatment group (removing the shared-baseline assumption at the recommendation of the FDA), to add fixed terms for baseline serum NfL level, baseline use of sodium phenylbutyrate/taurursodiol, and their interactions with visit, and to specify the estimand.</p> |

| Version | Date | Description of Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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|         |      | <p>Revision of Section 6.5.4 Survival to specify the analysis sets, to include baseline age, baseline serum NfL level, and baseline use of sodium phenylbutyrate/taursodiol as additional covariates in adjusted models.</p> <p>Revision of Section 6.5.5 CAFS to provide additional detail on the calculation of the rank-sums and to specify analyses of additional survival endpoints, functional endpoints, analysis methods.</p> <p>Revision of Section 6.5.6 HHD0 and HHD0[2] to specify that analysis requires methods for interval-censored data and to add baseline serum NfL level and baseline use of sodium phenylbutyrate/taursodiol as additional covariates,</p> <p>Revision of Section 6.5.7 Placebo Multiple Imputation to specify analysis primary and secondary functional endpoints, to add baseline serum NfL level and baseline use of sodium phenylbutyrate/taursodiol as additional covariates, and to revise specification of template SAS code,</p> <p>Revision of Section 6.5.9 Subgroup Analyses to add subgroups, specify how rare subgroup levels should be handled, and specify the estimates to be obtained.</p> <p>Revision of Section 6.5.10 Comparison of Controls across Regimens to add baseline serum NfL level and baseline use of sodium phenylbutyrate/taursodiol as additional covariates.</p> <p>Revision of Section 6.6.1 Treatment-emergent Adverse Events to define treatment-emergence, to include summaries of fatal TEAEs, to include summary of TEAEs observed in at least 5% of participants, to specify summaries of TEAEs by subgroups, to indicate that treatment-dependent differences in TEAE incidence rates rather than rate ratios will be estimated, and to remove testing of treatment-dependent differences in the distribution of TEAE characteristics.</p> <p>Revision of Section 6.6.2 Safety Labs to specify that clinical safety laboratory abnormalities will be summarized by CTCAE toxicity grade and to include an aggregate summary of abnormalities observed over all post-baseline visits in shift tables.</p> <p>Revision of Section 6.6.4 Vital Signs and Weight to clarify the analysis sets and analysis methods used to analyze change in weight.</p> <p>Revision of Section 6.7.1 Participant Disposition to specify more completely the summaries of participant disposition.</p> <p>Revision of Section 6.7.2 Study Drug Compliance Exposure and Tolerance to restrict summaries to the ERO and SRO analysis sets and</p> |

| Version | Date | Description of Changes                                                                                                                                                             |
|---------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |      | to change measures of exposure from counts of doses to durations of exposure.                                                                                                      |
|         |      | Addition of Section 6.7.4 Medical History to specify summaries of medical history data.                                                                                            |
|         |      | Addition of Section 6.7.5 Blindedness to specify summaries and analyses of blindedness assessments.                                                                                |
|         |      | Addition of Section 6.7.6 Protocol Deviations to specify summaries of protocol deviations.                                                                                         |
|         |      | Revision of Section 8 References to add new references and remove references that are no longer cited.                                                                             |
|         |      | Addition of Appendix 1 ALS Master Protocol Recommended Statistical Analysis, Design and Simulation Report; and a Summary of Changes from the previous version, dated May 14, 2021. |

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## 1. Governing Documents

This Master Statistical Analysis Plan (M-SAP) defines the default outcome measures, analysis samples, and planned analyses for the placebo-controlled portion of regimens tested in the HEALEY ALS Platform Trial for Regimen-specific Appendices (RSAs) initiated after implementation of version 5.0 of the HEALEY ALS Platform Trial Master Protocol in February 2023. RSAs initiated on earlier versions of the HEALEY ALS Platform Trial Master Protocol will refer to version 2.0 of the M-SAP dated 06 Mar 2020. Each RSA will have a regimen-specific SAP (R-SAP) that specifies any deviations or further details that supersede the specifications given here. The M-SAP and each R-SAP supplement the Master Protocol, the "ALS Master Protocol Recommended Statistical Analysis, Design and Simulation Report" (MPRDR, Appendix 1 to this M-SAP), and the applicable RSA. Please refer to the Master Protocol and applicable RSA for details on the rationale for the study design, eligibility criteria, conduct of the trial, clinical assessments and schedule of assessments, definitions and reporting of adverse events, data management conventions, and regulatory oversight and compliance procedures. The MPRDR and any regimen-specific deviations described in the applicable RSA and R-SAP are authoritative in defining the primary and interim analyses. In case of discrepancies between either SAP and the Master Protocol and applicable RSA concerning other matters of analysis, the M-SAP and applicable R-SAP are authoritative. In case of discrepancies between the M-SAP and R-SAP, the R-SAP is authoritative. On all other matters, the Master Protocol and applicable RSA are authoritative.

## 2. Study Design

### 2.1 Overview

The HEALEY ALS Platform Trial is a perpetual multi-center, multi-regimen clinical trial evaluating the safety and efficacy of investigational products for the treatment of ALS. Investigational products will be evaluated according to details specified in distinct RSAs. Each RSA will describe the nature of the intervention and its mechanism of action, including the mode and frequency of administration, number of doses, the specific target population (to be selected within the pre-defined subsets of the Master Protocol), additional enrollment criteria (if any), sample size, and other specific intervention-related information and assessments (safety or other assessments that may be in addition to those outlined in the Master Protocol).

### 2.2 Study Objectives

Primary Efficacy Objective:

- To evaluate the efficacy of multiple investigational products as compared to placebo on ALS disease progression.

Secondary Efficacy Objectives:

- To evaluate the effect of multiple investigational products on selected secondary measures of disease progression.

Safety Objectives:

- To evaluate the safety of multiple investigational products for people with ALS.



Exploratory Efficacy Objectives:

- To evaluate the effect of multiple investigational products on selected biomarkers and endpoints.

Each regimen will test the efficacy of a distinct treatment.

## 2.3 Study Population

Individuals eligible for trial participation are men or women at least 18 years old who meet the El Escorial criteria of possible, laboratory-supported probable, probable, or definite for a diagnosis of sporadic or familial ALS, report onset of weakness within 36 months of screening, possess a slow vital capacity (SVC) at least 50% of that expected based on age, sex, and height, are treatment naïve or on a stable dose of riluzole, edaravone, or both, are naïve to or have started treatment with sodium phenylbutyrate/taurursodiol at least 30 days prior to Master Protocol Screening, lack unstable clinical or psychiatric conditions, and were not recently exposed to other investigational treatments or prior gene therapy. Detailed inclusion and exclusion criteria are specified in the Master Protocol and applicable RSA.

Participants will be recruited from approximately 60 centers located throughout the US that are part of the Northeast ALS (NEALS) Consortium.

## 2.4 Participant Flow

Participants that consent to the Master Protocol will be screened for Master Protocol-level inclusion and exclusion criteria. Those determined eligible will be randomly assigned to an applicable regimen. Participants that consent to their assigned regimen will be screened for any regimen-specific inclusion and exclusion criteria. Those determined eligible will be randomized to either active or placebo treatment for their assigned regimen.

This perpetual platform trial will continue enrolling research participants for as long as there are regimens that are enrolling. Enrollment into a specific regimen will continue until the target number of randomized participants in a regimen has been reached unless stopped earlier due to a determination of regimen success or futility.

Each regimen will compare double-blind active and placebo treatment groups for a maximum of up to 28 weeks or as otherwise specified in the RSA. Participants will complete a baseline visit prior to initiating study drug. After the baseline visit, participants will be seen in clinic at 4, 8, 16, and 24 weeks and will complete telephone visits at 2, 12, and 20 weeks and 28 days after last dose of study drug or as otherwise specified in the RSA.

An active treatment extension (ATE) will be available to participants who complete the double-blind phase as described in their respective RSAs. Participants may also opt to re-screen for the Master Platform and enroll in a second regimen, if eligible.

Detailed descriptions of study procedures and timing are specified in the Master Protocol and applicable RSA.

## 2.5 Regimen Allocation

Participants determined eligible for Master Protocol-level inclusion and exclusion criteria will be randomized with equal probability to all applicable regimens. Applicable regimens are those for

which the participant's site is active and for which the participant is not known to be ineligible, e.g., due to prior screen failure. Randomizations are stratified for all combinations of use vs. non-use of riluzole, edaravone, and sodium phenylbutyrate/taurursodiol at the time of screening for the Master Protocol. Details of regimen assignment are described in the Platform Trial Regimen Assignment Plan.

## 2.6 Treatment Allocation

Prior to the baseline visit, eligible participants for a given regimen will be randomly allocated in a 3:1 ratio to active or placebo treatment. Active participants may be further divided at random to more than one dose group, if applicable. Randomizations follow a permuted-block schedule, stratified for all combinations of use vs. non-use of riluzole, edaravone, and sodium phenylbutyrate/taurursodiol at the time of screening for the Master Protocol. The randomization schedule will be prepared by computer program by an unblinded study statistician who will not contribute to decisions concerning the course, conduct, or analysis of the HEALEY ALS Platform Trial.

## 2.7 Treatment Administration

Details of treatment administration are described in the applicable RSA.

## 2.8 Allocation Concealment

Allocation to regimen is not concealed. Allocation to treatment groups within a regimen is concealed to participants and investigators. The regimen randomization schedule is known only to the unblinded study statistician who generated the schedule, any unblinded staff responsible for managing the interactive response technology (IRT) used for randomization, all staff at the central pharmacy responsible for distribution of study drug, and the unblinded Independent Statistical Analysis Committee (ISAC) responsible for interim analyses. An independent Data and Safety Monitoring Board (DSMB) will receive unblinded data as described in the DSMB Charter. Concealment of the true regimen treatment assignment of specific participants is achieved by using anonymous study drug kit identifiers linked at time of dispensation to participants. Member of the HEALEY ALS Platform Trial Executive Committee, members of the Regimen Steering Committee, site investigators and other site staff including research pharmacists, clinical coordination and data management staff, the medical monitor, and all participants are blinded to regimen treatment assignments of participants. Additional details concerning blinding are described in Data Sharing Plan and the Clintrex/CMSU Firewall Plan.

## 2.9 Master Protocol Schedule of Assessments

| Activity                                | Master Protocol Screening <sup>1</sup> | Regimen Specific Screening <sup>1</sup> | Baseline | Week 2    | Week 4 <sup>13</sup> | Week 8 <sup>13</sup> | Week 12   | Week 16 <sup>13</sup> | Week 20    | Week 24    | Follow-Up Safety Call |
|-----------------------------------------|----------------------------------------|-----------------------------------------|----------|-----------|----------------------|----------------------|-----------|-----------------------|------------|------------|-----------------------|
|                                         | Clinic                                 | Clinic                                  | Clinic   | Phone     | Clinic               | Clinic               | Phone     | Clinic                | Phone      | Clinic     | Phone                 |
|                                         | -42 to -1 Days                         | Refer to RSA                            | Day 0    | Day 14 ±3 | Day 28 ±7            | Day 56 ±7            | Day 84 ±3 | Day 112 ±7            | Day 140 ±3 | Day 168 ±7 | Refer to RSA          |
| Written Informed Consent <sup>2</sup>   | X                                      | X                                       |          |           |                      |                      |           |                       |            |            |                       |
| Inclusion/Exclusion Review              | X                                      | X <sup>3</sup>                          |          |           |                      |                      |           |                       |            |            |                       |
| Regimen Specific Screening procedure(s) |                                        | X                                       |          |           |                      |                      |           |                       |            |            |                       |

| Activity                                | Master Protocol Screening <sup>1</sup> | Regimen Specific Screening <sup>1</sup> | Baseline       | Week 2          | Week 4 <sup>13</sup> | Week 8 <sup>13</sup> | Week 12         | Week 16 <sup>13</sup> | Week 20         | Week 24        | Follow-Up Safety Call |
|-----------------------------------------|----------------------------------------|-----------------------------------------|----------------|-----------------|----------------------|----------------------|-----------------|-----------------------|-----------------|----------------|-----------------------|
|                                         | Clinic                                 | Clinic                                  | Clinic         | Phone           | Clinic               | Clinic               | Phone           | Clinic                | Phone           | Clinic         | Phone                 |
|                                         | -42 to -1 Days                         | Refer to RSA                            | Day 0          | Day 14 ±3       | Day 28 ±7            | Day 56 ±7            | Day 84 ±3       | Day 112 ±7            | Day 140 ±3      | Day 168 ±7     | Refer to RSA          |
| ALS & Medical History                   | X                                      |                                         |                |                 |                      |                      |                 |                       |                 |                |                       |
| Demographics                            | X                                      |                                         |                |                 |                      |                      |                 |                       |                 |                |                       |
| Physical Examination                    | X                                      |                                         |                |                 |                      |                      |                 |                       |                 |                |                       |
| Neurological Exam                       | X                                      |                                         |                |                 |                      |                      |                 |                       |                 |                |                       |
| Vital Signs <sup>4</sup>                | X                                      |                                         | X              |                 | X                    | X                    |                 | X                     |                 | X              |                       |
| Slow Vital Capacity                     | X <sup>12</sup>                        |                                         | X              |                 |                      | X                    |                 | X                     |                 | X              |                       |
| Muscle Strength Assessment              |                                        |                                         | X              |                 |                      | X                    |                 | X                     |                 | X              |                       |
| ALSFRS-R                                | X                                      |                                         | X              |                 | X                    | X                    | X               | X                     | X               | X              |                       |
| Patient Reported Outcomes <sup>10</sup> |                                        |                                         | X              |                 |                      | X                    |                 | X                     |                 | X              |                       |
| 12-Lead ECG                             | X                                      |                                         |                |                 |                      |                      |                 |                       |                 | X              |                       |
| Clinical Safety Labs <sup>5</sup>       | X                                      |                                         | X              |                 | X                    | X                    |                 | X                     |                 | X              |                       |
| Concomitant Medication Review           | X                                      | X                                       | X              |                 | X                    | X                    | X               | X                     | X               | X              | X                     |
| Adverse Event Review <sup>6</sup>       | X                                      | X                                       | X              | X               | X                    | X                    | X               | X                     | X               | X              | X                     |
| Columbia-Suicide Severity Rating Scale  |                                        |                                         | X              |                 | X                    | X                    |                 | X                     |                 | X              |                       |
| Biomarker Blood Collection              |                                        |                                         | X              |                 |                      | X                    |                 | X                     |                 | X              |                       |
| Biomarker Urine Collection              |                                        |                                         | X              |                 |                      | X                    |                 | X                     |                 | X              |                       |
| DNA Collection <sup>7</sup> (optional)  |                                        |                                         | X              |                 |                      |                      |                 |                       |                 |                |                       |
| CSF Collection <sup>11</sup>            |                                        |                                         | X              |                 |                      |                      |                 | X                     |                 |                |                       |
| Assignment to a regimen                 | X                                      |                                         |                |                 |                      |                      |                 |                       |                 |                |                       |
| Randomization within a regimen          |                                        | X                                       |                |                 |                      |                      |                 |                       |                 |                |                       |
| Administer/Dispense Study Drug          |                                        |                                         | X <sup>8</sup> |                 |                      | X                    |                 | X                     |                 |                |                       |
| Study Drug Accountability/Compliance    |                                        |                                         |                | X <sup>14</sup> | X                    | X                    | X <sup>14</sup> | X                     | X <sup>14</sup> | X              |                       |
| Exit Questionnaire                      |                                        |                                         |                |                 |                      |                      |                 |                       |                 | X              |                       |
| Vital Status Determination              |                                        |                                         |                |                 |                      |                      |                 |                       |                 | X <sup>9</sup> |                       |

<sup>1</sup> Master Protocol Screening procedures must be completed within 42 days to 1 day prior to the Baseline Visit. Refer to RSA for details about RSA Specific Screening procedures, if any, should be performed.

<sup>2</sup> During the Master Protocol Screening Visit, participants will be consented via the Master Protocol informed consent form (ICF). After a participant is assigned to a regimen, participants will be consented a second time via the regimen-specific ICF.

<sup>3</sup> At the Regimen Specific Screening Visit, participants will have regimen-specific inclusion and exclusion criteria assessed, if applicable.

<sup>4</sup> Vital signs include weight, systolic and diastolic pressure, respiratory rate, heart rate and temperature. Height in cm measured at Master Protocol Screening Visit only.

<sup>5</sup> Clinical safety labs include hematology (CBC with differential), complete chemistry panel, liver function tests, thyroid function and urinalysis. Coagulation panel will occur at the Master Protocol Screening and Week 16 Visits or as in the RSA. Serum pregnancy testing will occur in women of child-bearing potential at the Master Protocol Screening Visit and as necessary during the study. Pregnancy testing is only repeated as applicable if there is a concern for pregnancy.

<sup>6</sup> Adverse events that occur after signing Master Protocol consent form will be recorded.

<sup>7</sup> The DNA sample can be collected after the Baseline Visit if a baseline sample is not obtained or the sample is not usable.

<sup>8</sup> Administer first dose of study drug only after Baseline Visit procedures are completed

<sup>9</sup> Vital status, defined as a determination of date of death or death equivalent or date last known alive, will be determined for each randomized participant at the end of the placebo-controlled period of their follow-up (generally the Week 24 Visit, as indicated). If at that time the participant is alive, his or her vital status should be determined again at the time of the last participant's last visit (LPLV) of the placebo-controlled period of a given regimen. We may also ascertain vital status at later time points by using publicly available data sources as described in section 8.15.b

<sup>10</sup> Each RSA will detail which PRO is collected at each Visit. Specific PRO collection may differ from this Schedule of Activities.

<sup>11</sup> CSF collection may differ from the Schedule of Activities and will be performed as described in each RSA.

<sup>12</sup> If required due to pandemic-related or other restrictions, Forced Vital Capacity (FVC) performed by a Pulmonary Function Laboratory evaluator may be used for eligibility (Master Protocol Screening ONLY).

<sup>13</sup> Visit may be conducted via phone or telemedicine with remote services instead of in-person if needed to protect the safety of the participant due to a pandemic, disability or other medical reason.

<sup>14</sup> Drug accountability will not be done at phone visits. A drug compliance check in will be conducted during phone visits to ensure participant is taking drug per dose regimen and to note any report of missed doses.

### **3. General Considerations for Data Analysis**

#### **3.1 Statistical Software**

Statistical analyses will be performed using SAS (SAS Institute, NC, USA), R (R Foundation for Statistical Computing, Vienna, Austria), or JAGS (Plummer, SourceForge).

#### **3.2 Summary Statistics**

Data will be summarized with respect to disposition, demographics, pre-treatment characteristics, efficacy endpoints, safety endpoints, and tolerability. Summary statistics for continuous variables will include the number of observations, the mean, median, standard deviation, inter-quartile range, and range. Summaries of categorical data will include counts, denominators, and percentages.

#### **3.3 Precision**

Results will generally be reported to 3 significant figures. Percentages will generally be reported to 0.1 percentage points. P-values will generally be reported to two digits when greater than or equal to 0.095, to three digits when greater than or equal to 0.00095 and less than 0.095, and as <0.001 for all smaller values.

#### **3.4 Transformations**

Data that are strictly positive, continuous, and strongly right skewed will typically be log-transformed prior to any inferential testing. Skewness greater than 3 will be used as a guide in determining which variables to transform. Which variables warrant transformation and how they will be transformed will be determined prior to regimen data lock. Original, untransformed values will be used for all summaries. Any data transformations used in the primary and interim analyses are described in the MPRDR (Appendix 1 to this M-SAP).

#### **3.5 Multiplicity Adjustments**

A single primary analysis is planned. The default criterion for significance at the final analysis is adjusted to ensure an overall one-sided type 1 error rate less than 0.025 in the absence of early stopping for futility. Secondary efficacy endpoints will be tested in a pre-specified order using a closed testing procedure to maintain an overall two-tailed family-wise error rate less than 0.05 under the condition that the primary and all secondary analyses are null. Nominal comparison-wise p-values will also be reported for all analyses unless otherwise specified.

#### **3.6 Missing Data**

If baseline values of a given measure are missing, the last observed pre-treatment value will be used. Missing baseline covariates will be imputed using the mean of the respective covariate. For any baseline covariate that is transformed for analysis, means will be imputed after transformation. Off-schedule results, e.g., those collected as part of an Early Termination Visit, will be used in place of the closest missing scheduled visit (while preserving visit sequence) for analyses that depend on visit-specific data. Other results will not be carried forward.

The planned mixed model analyses, where applied, yield estimates that are unbiased conditional on the observed values under a missing at random assumption. Secondary analyses of efficacy endpoints that accommodate missing values are described below (see Sections 6.5.5 and 6.5.6).

## **4. Study Endpoints**

### **4.1 Efficacy Endpoints**

The primary efficacy endpoint is change from baseline through Week 24 in disease severity as measured by the ALS Functional Rating Scale-Revised (ALSFRS-R) total score and survival.

Secondary efficacy endpoints include change from baseline through Week 24 in muscle strength as measured isometrically using hand-held dynamometry (HHD) and grip strength, respiratory function as assessed by slow vital capacity (SVC), and survival.

### **4.2 Exploratory Efficacy Endpoints**

The following categories of exploratory endpoints will be evaluated:

- Biofluid biomarkers
- Patient-reported outcomes

### **4.3 Safety Endpoints**

The following safety endpoints will be evaluated:

- Treatment-emergent adverse and serious adverse events.
- Changes in laboratory values and treatment-emergent and clinically significant laboratory abnormalities.
- Changes in ECG parameters and treatment-emergent and clinically significant ECG abnormalities.
- Treatment-emergent suicidal ideation and suicidal behavior.

Reported proportions will use as their denominator all participants in the applicable Safety sample (see Section 6.1 below).

## **5. Measurement Definitions**

### **5.1 ALSFRS-R**

The ALSFRS-R (Cedarbaum et al. 1999) is a 12-item instrument completed by clinician interview for assessing participants' function in four domains: bulbar, fine motor, gross motor, and respiratory. Each item is scored from 0 to 4 with higher scores indicating greater function. The ALSFRS-R total score is the sum of all items (range 0 to 48). Domain scores are the sum of their respective items (range 0 to 12): bulbar function = Q1-Q3, fine motor function = Q4-Q6, gross motor function = Q7-Q9, and respiratory function = QR1-QR3 (Q10-Q12).

The average rate of disease progression prior to the Baseline Visit as measured by ALSFRS-R (pre-baseline slope of ALSFRS-R or delta-FRS) will be calculated as 48 minus the ALSFRS-R

total score measured at the Baseline Visit, that difference divided by the number of months from onset of symptom weakness due to ALS to the date of the Baseline Visit. The number of months will be calculated as the difference in days from onset of symptomatic weakness to the Baseline Visit multiplied by 12 / 365.25. The date of onset of symptomatic weakness will be imputed as the fifteenth day of a month if not specified more precisely.

## 5.2 SVC

Slow vital capacity (SVC) is the maximum volume of air that can be slowly exhaled after slow, maximal inhalation. Trained technicians coach participants through 3 to 5 maneuvers using a spirometer. A minimum of 2 completed SVC maneuvers are required to calculate an estimate of maximum SVC for a given participant visit. A single maximum vital capacity reported by a pulmonary function lab and obtained by either a slow or forced maneuver is also acceptable. The maximum volume expired is converted to percent of predicted normal using normal values for forced vital capacity (FVC). FVC normal values are calculated based on sex, age, height, and ethnicity using equations published by the Global Lung Function Initiative (GLI, Quanjer et al. 2012). Higher values indicate greater respiratory function.

Age will be calculated as number of days from date of birth to the date of a given SVC assessment divided by 365.25. The following correspondence between self-identified race and race defined by GLI classification will be used:

| Self-identified Race                      | GLI-defined Race |
|-------------------------------------------|------------------|
| American Indian or Alaska Native          | Mixed/Other      |
| Asian                                     | South East Asian |
| Black or African American                 | African American |
| Native Hawaiian or Other Pacific Islander | Mixed/Other      |
| White                                     | Caucasian        |
| Unknown                                   | Caucasian        |
| Not reported                              | Caucasian        |
| More than one race indicated              | Mixed/Other      |

## 5.3 HHD and Grip Strength

Manual muscle testing by hand-held dynamometry (HHD) assesses isometric muscle strength by recording force generated against a hand-held dynamometer placed against a given limb by a trained evaluator. Each muscle is tested with a standard beginning position, usually midway between maximum flexion or extension of the muscle being studied, and with the dynamometer placed at a defined position. Trained evaluators coach participants through 3 to 5 maximal exertions for each muscle and record the maximum force observed in pounds. Evaluators note if participants are not able to overcome the force exerted for a given muscle. Conversely, a value of 0 is recorded for a given muscle if the participant cannot assume the testing position due to weakness or can attain the position but cannot exert measurable force.

The following 11 muscles are tested bilaterally by HHD: shoulder flexion, elbow flexion, elbow extension, wrist extension, abductor pollicis brevis contraction, abductor digiti minimi contraction, first dorsal interosseous contraction, hip flexion, knee flexion, knee extension, and ankle dorsiflexion. Grip strength is tested bilaterally by Jamar hand dynamometer. Between HHD and hand grip, 24 muscles or muscle groups are tested. The force recorded for each muscle



is converted to a percentage change from baseline among muscles with non-zero strength at baseline. Individual percentages are averaged together to yield upper extremity, lower extremity, and global average percentages. Higher values indicate greater strength retention.

HHD data will also be summarized as time to loss of strength. HHD0 and HHD0[2] will be defined as the times from the Baseline Visit to the first and second post-baseline occurrence of a muscle with a strength recording of 0, respectively, among those muscles that were non-zero at baseline. All 11 muscles assessed by HHD as well as grip strength will be considered. Both HHD0 and HHD0[2] are composite endpoints with death or death-equivalent, censored at the Week 24 Visit if completed or at the earlier of the end of the Week 24 Visit window or date last known alive if a Week 24 visit is not completed. Larger HHD0 and HHD0[2] times indicate longer preservation of strength.

#### **5.4 Biofluid Biomarkers**

Blood and urine biomarkers will be assayed. Biomarkers will include serum creatinine and serum and cerebrospinal fluid (CSF) neurofilament light chain (NfL). Serum creatinine will be assayed by the kinetic Jaffe method (test 001370, Labcorp, Burlington, NC). NfL will be assayed by single-molecule array (Simoa; Quanterix, Billerica, MA). Levels of serum and CSF NfL that are reported to be below the limit of quantitation will be imputed at the limit of quantitation. Levels of serum and CSF NfL will be log-transformed in all analyses. Other biomarkers may be analyzed.

#### **5.5 Patient-reported Outcomes**

Patient-reported outcomes may be included as regimen-specific assessments. The description and scoring of each instrument will be specified in the R-SAP.

#### **5.6 Survival**

The primary survival endpoint will be the composite of death or death equivalent. A secondary survival endpoint of death alone independent of any death equivalent (including a death equivalent that occurs prior to baseline) is also defined. Death equivalent will be defined as use of permanent assisted ventilation (PAV), where PAV is defined as more than 22 hours per day of noninvasive or invasive mechanical ventilation for more than seven consecutive days. The date of PAV initiation is the first day of the consecutive days. The date of PAV initiation, where applicable, will be imputed as the fifteenth day of a month if not specified more precisely. Any participant on PAV at baseline will be censored at baseline for the composite survival endpoint.

Time at risk for both the composite survival endpoint of death or death equivalent and the survival endpoint of death alone will be measured from each participant's Baseline Visit and will be censored at the Week 24 Visit if completed or at the earlier of the end of the Week 24 Visit window or date last known alive if a Week 24 visit is not completed.

Additional secondary analyses of survival may consider different durations of follow-up.

#### **5.7 King's ALS Clinical Staging System**

The King's ALS Clinical Staging System (Roche et al. 2012) is a 4-level ordinal scale with the first three levels indicating the number (1, 2, or 3) of distinct central nervous system regions

(bulbar, upper limb, and lower limb) with neuromuscular dysfunction and levels 4a and 4b indicating nutritional or respiratory failure secondary to ALS, respectively.

Participants will be classified to King's stage 1, 2, 3, 4a, or 4b based on scores from ALSFRS-R assessments according to a published derivation (Balendra et al. 2014). Bulbar involvement is defined as a score less than 4 on any of the ALSFRS-R questions in the bulbar domain (questions 1, 2, and 3). Upper limb involvement is defined as a score less than 4 on either of the ALSFRS-R questions related to hand function (questions 4 and 5A). Lower limb involvement is defined as a score less than 4 on the ALSFRS-R question about walking (question 8). Nutritional failure is defined as responding that the participant uses gastrostomy for greater than 50% of their nutrition. Respiratory failure is defined as a score of 0 on the ALSFRS-R question addressing dyspnea (question 10 or R-1) or a score less than 4 on the ALSFRS-R question about use of mechanical ventilation (question 12 or R-3). Participants without evidence by ALSFRS-R scores of involvement of any of the three central nervous system regions will be scored as King's stage 1 due to their confirmed diagnosis with ALS. Participant may meet criteria for both King's stage 4a and 4b.

## 5.8 Clinical Safety Laboratory Tests

Clinical safety labs include hematology, blood chemistry panel, liver function tests, thyroid function, urinalysis, and pregnancy testing in women of childbearing potential as specified in Section 9.1.2 Clinical Safety Laboratory Tests of the Master Protocol:

- Hematology with differential panel: complete blood count with differential (hematocrit, hemoglobin, platelet count, RBC indices, Total RBC, Total WBC, and WBC & differential)
- Blood chemistry panel/Liver function tests (LFTs): alanine aminotransferase (ALT (SGPT)), aspartate aminotransferase (AST (SGOT)), albumin, alkaline phosphatase, bicarbonate, blood urea nitrogen, calcium, chloride, creatinine (estimated Glomerular Filtration Rate [eGFR] will be calculated using the MDRD equation and creatinine clearance will be calculated using the Cockcroft-Gault equation), glucose, magnesium, phosphate, potassium, sodium, thyroid-stimulating hormone (TSH), total bilirubin and total protein
- Urinalysis: albumin, bilirubin, blood, clarity, color, glucose, ketones, nitrate, pH, protein, specific gravity, urobilinogen and WBC screen
- Serum human chorionic gonadotrophin (hCG) for WOCBP (collected only at Master Protocol Screening Visit, and as necessary throughout course of study)
- Coagulation parameters: Prothrombin time, International normalized ratio, Activated partial thromboplastin time

Clinical safety labs will also include derived measures of potential drug-induced liver injury (DILI), including those that potentially meet the Hy's law criteria, as distinct safety lab outcomes.

Three potential DILI criteria will be defined:

- ALT or AST >3x ULN with TBL >1.5x ULN



- AST or ALT  $>3\times$  ULN with TBL  $>2\times$  ULN
  - AST or ALT  $>3\times$  ULN with TBL  $>2\times$  ULN and ALP  $<2\times$  ULN (potential Hy's Law cases)
- where ULN is upper level of normal and all levels are measured on the same day.

## 6. Statistical Methodology

### 6.1 Analysis Sets

The following analysis sets will be used for testing efficacy, safety, and tolerability endpoints of a specific focal regimen:

- Efficacy Full Analysis Set (FAS): Participants who were randomized within the focal regimen plus placebo participants from specified regimens who had an opportunity to complete placebo-controlled follow-up, classified according to their randomized treatment assignment. Observations made after premature permanent discontinuation of study drug are included in this sample, should such participants remain on study. Observations completed after regimen data lock will be excluded. Participants determined to not meet ALS diagnostic criteria are excluded.
- Efficacy Concurrent-controls (ECC) Set: The subset of participants in the FAS analysis set who were randomized within 180 days prior to the first randomization for the applicable regimen and within 180 days after the last randomization for the applicable regimen, exclusive.
- Efficacy Regimen-only (ERO) Set: The subset of participants in the FAS analysis set who were randomized within a given regimen.
- Efficacy Common Mode of Administration (ECM) Set: The subset of participants in the FAS analysis set who are in regimens in which study drug is administered by the same route as the focal regimen (including oral, subcutaneous injection, intravenous infusion).
- Efficacy Per-protocol (EPP) Set: The subset of participants in the FAS analysis set who initiated study treatment and who meet other requirements for protocol compliance as specified in the R-SAP, classified according to the treatment they actually received. This sample may include placebo participants from no other, some other, or all other regimens and may be restricted to those who would have been eligible for the applicable regimen as specified in the R-SAP. If a participant's data are truncated for inclusion in the EPP analysis set due to a time-dependent event (e.g., non-adherence to protocol-specified dosing, initiation of a prohibited medication, initiation of a medication approved for treatment of ALS), clinical events observed after the censoring event may be included in the EPP analysis set as specified in the R-SAP. Data from placebo participants shared from other regimens will not be truncated due to non-adherence to protocol-specified dosing.
- Safety Full (STF) Set: Participants who initiated treatment within the focal regimen plus placebo participants from specified regimens who are not known to be ineligible for the focal regimen and who initiated treatment in their respective regimen, classified according to the treatment they actually received. Observations made after premature permanent discontinuation of study drug are included in this sample, should such participants remain on study. Observations completed after regimen data lock are excluded.

- Safety Narrow (STN) Set: The subset of participants in the STF analysis set who were exposed to interventions with the same mode of administration.
- Safety Regimen-only (SRO) Set: The subset of participants in the STF analysis set who were randomized within a given regimen.

The meaning of regimen data lock and the criteria for inclusion of data for placebo participants assigned to other regimens at the time of regimen data lock will be defined in the HEALEY ALS Platform Trial Data Management Plan and the HEALEY ALS Platform Trial Data Sharing Plan.

## 6.2 Baseline Characterization

Each analysis sample will be summarized overall and by treatment group for the following characteristics: randomization site; age, sex, race, ethnicity, El Escorial diagnosis, months since ALS symptom onset, delay between symptom onset and ALS diagnosis, site of ALS symptom onset, use at baseline of riluzole, edaravone, or sodium phenylbutyrate/taurursodiol and combinations of use and non-use of these medications, ALSFRS-R total score, delta-FRS, SVC, King's stage, weight, body mass index (BMI), serum creatinine concentration, and serum NfL concentration.

## 6.3 Primary Efficacy Analysis and Supportive Analyses

Unless otherwise specified in the RSA, the primary analysis for each regimen is a Bayesian shared-parameter model of function and survival. Details of the model, including documentation of operating characteristics under a range of scenarios, are provided in the MPRDR (Appendix 1 to this M-SAP). Any regimen-specific deviations are described in the applicable RSA and R-SAP.

The estimand of the primary analysis is the relative rate of disease progression (the "disease rate ratio" or DRR) of active treatment relative to placebo in the FAS population under an assumption that active treatment slows mean time to death or death equivalent by the same proportion as treatment slows the mean rate of functional progression as measured by change in ALSFRS-R total score over time. The estimand is defined by the following attributes:

- Treatment: active treatment vs. placebo.
- Population: FAS population as defined in Section 6.1.
- Variables: time to death or death equivalent and rate of change in ALSFRS-R total score from baseline to the Week 24 Visit.
- Intercurrent event 1: treatment discontinuation due to death or death equivalent: no ALSFRS-R data from participants who reach the death or death equivalent endpoint are included in the analysis, handled via mortality component in model, composite variable strategy approach.
- Intercurrent event 2: treatment discontinuation not due to death or death equivalent: handled via treatment policy approach, all data will be used including data collected during the placebo-controlled period after treatment discontinuation regardless of concomitant medication, for those participants who have not been censored due to mortality. Missing data post-treatment will not be imputed, handled via missing at random assumption.
- Population-level summary: mean ratio of hazard or progression rate of active treatment relative to placebo.

## 6.4 Interim Analysis

Unless otherwise specified in the RSA, regimens will be considered for early stopping for success or futility at regular intervals. The default design for a regimen will include stopping early only for futility. Details of the interim analysis schedule, the early stopping rules, and documentation of operating characteristics under a range of scenarios, are provided in the MPRDR (Appendix 1 to this M-SAP). Any regimen-specific deviations are described in the applicable RSA and R-SAP.

## 6.5 Secondary Efficacy Analyses

### 6.5.1 Hierarchical Testing

Primary inference for secondary efficacy endpoints will be based on analysis of the FAS analysis set using a repeated-measures linear mixed model for functional endpoints (see Section 6.5.2 below) and by Kaplan-Meier product-limit estimates and log-rank test for survival (see Section 6.5.5 below). The default sequence for testing is the following:

1. Change from baseline to Week 24 in upper limb muscle strength as assessed isometrically using HHD and grip strength and measured as average percent change from baseline among maneuvers with non-zero strength at baseline in the FAS analysis set,
2. Change from baseline to Week 24 in respiratory function as assessed by slow vital capacity (SVC) and measured as percent predicted using GLI normal values in the FAS analysis set,
3. Change from baseline to Week 24 in lower limb muscle strength as assessed isometrically using HHD and measured as average percent change from baseline among maneuvers with non-zero strength at baseline in the FAS analysis set, and
4. Survival defined as freedom from death and permanent assisted ventilation (PAV) from baseline to Week 24 in the FAS analysis set.

RSAs may specify a different sequence of secondary endpoints to evaluate using a closed testing procedure or another method for controlling family-wise type 1 error rate.

If the primary analysis indicates a significant slowing in disease progression from the Bayesian shared-parameter model of function and survival, then each secondary efficacy endpoint would be declared significant in the specified sequence using a criterion of two-tailed  $p < 0.05$ . After the first failure to declare significance, no endpoints lower in the hierarchy can be significant. Nominal comparison-wise p-values for secondary efficacy endpoints will also be reported.

### 6.5.2 Repeated-measures Model

Treatment groups of a given regimen will also be compared for progression of continuous efficacy and selected safety endpoints (ALSFRS-R total score, ALSFRS-R domain scores, HHD upper extremity, lower extremity, and global average percentages, SVC percent-predicted, serum creatinine, serum NfL, and weight) using a repeated-measures linear mixed model in the FAS, ECC, ERO, ECM, and EPP samples. The model will include fixed terms for discrete visit, treatment group, treatment group  $\times$  visit interaction, time since symptom onset and time since symptom onset  $\times$  visit interaction, delta-FRS and delta-FRS  $\times$  visit interaction, baseline log-transformed serum NfL level and baseline log-transformed serum NfL level  $\times$  visit interaction, baseline riluzole use and baseline riluzole  $\times$  visit interaction, baseline edaravone use and baseline

edaravone  $\times$  visit interaction, and baseline sodium phenylbutyrate/taurursodiol use and baseline sodium phenylbutyrate/taurursodiol  $\times$  visit interaction. Fixed terms other than treatment group and visit will be centered by subtracting the sample mean of a given covariate for a given analysis set. The model will assume unstructured covariance among participant-specific repeated measures. In analysis sets in which placebo participants are drawn from more than three regimens, the model will include random regimen-specific intercepts. In analysis samples with three or fewer regimens, the model will include fixed effects for regimen.

The following equations describe the model with regimen random effects:

$$Y_{ij} = a_{k(i)} + \gamma_1 t_i + \gamma_{2,j} v_j + \gamma'_3 \mathbf{z}_i + \gamma_{4,j} t_i v_j + \gamma'_{5,j} \mathbf{z}_i v_j + \epsilon_{ij} \quad (\text{eqn. 1})$$

$$a_k \sim N(0, \sigma_r^2), \epsilon_i \sim N(\mathbf{0}, \mathbf{R}), \text{Cov}(b_{k(i)}, \epsilon_{ij}) = 0$$

where  $Y_{ij}$  is a given efficacy endpoint measured for participant  $i$  at visit  $j$ ,  $a_{k(i)}$  is a random intercept for regimen  $k$  to which participant  $i$  was assigned,  $v_j$  is an indicator variable for visit  $j$ ,  $\mathbf{z}_i$  is the vector of covariates (centered time since onset, centered delta-FRS, centered baseline log-transformed serum NfL level, centered baseline riluzole use, centered baseline edaravone use, and centered baseline sodium phenylbutyrate/taurursodiol use) for participant  $i$ ,  $t_i$  is an indicator variable for treatment  $t$  to which participant  $i$  was assigned,  $\gamma_1$ ,  $\gamma_{2,j}$ ,  $\gamma_3$ ,  $\gamma_{4,j}$ , and  $\gamma_{5,j}$  are estimated parameters and vectors of parameters for the fixed effects, and  $\epsilon_{ij}$  is the residual for participant  $i$  at visit  $j$ . The regimen-specific random effects are normally distributed with mean 0 and variance  $\sigma_r^2$ . The vector of residuals for a given participant are normally distributed with mean  $\mathbf{0}$  and an unstructured covariance matrix  $\mathbf{R}$ . The regimen-specific random effect for a given participant and residuals for that participant are uncorrelated.

The following SAS code specifies the model:

```
proc mixed data=xxx method=reml;
  class regimen id trtrnd visit;
  model Value = trtrnd|visit
              sx2b1|visit dFRS|visit nfl|visit rlz|visit edv|visit
              amx|visit / solution cl;
  random intercept / subject=regimen type=vc;
  repeated visit / subject=id type=un;
```

where id is a participant study identifier, trtrnd is the randomly assigned treatment group, visit is the visit identifier, Value is value of the efficacy endpoint being tested for a given participant at a given visit, sx2b1 is months since ALS symptom onset, dFRS is pre-baseline slope, nfl is baseline log-transformed serum NfL level, rlz is an indicator of riluzole use at baseline, edv is an indicator of edaravone use at baseline, and amx is an indicator of sodium phenylbutyrate/taurursodiol use at baseline.

The primary estimate will be the treatment-dependent difference in change from baseline to the Week 24 Visit. The estimate and its 95% Wald confidence bounds will be obtained by a linear contrast of adjusted means. The following SAS code specifies the linear contrast for a regimen with one active treatment assuming an endpoint measured every 8 weeks and that the sort order for treatment group has the active group last and visits are sorted chronologically:

```
estimate "3|Act vs Plb|dWk 24" trtrnd*visit 1 0 0 -1 -1 0 0 1 / cl;
```

The estimand estimated by the primary linear contrast of the repeated-measures linear mixed model is the mean difference in 24-week change from baseline of a given continuous efficacy endpoint in the active treatment group relative to the placebo group in the FAS population. The estimand is defined by the following attributes:

- Treatment: active treatment vs. placebo.
- Population: FAS population as defined in Section 6.1.
- Variables: absolute change in endpoint from baseline to the Week 24 Visit.
- Intercurrent event: treatment discontinuation: handled via treatment policy approach, all data will be used including data collected during the placebo-controlled period after treatment discontinuation. Missing data post-treatment, including data missing due to death, will not be imputed, handled via missing at random assumption.
- Population-level summary: difference in conditional means of active treatment relative to placebo.

A significant positive difference in 24-week change from baseline in the direction of greater improvement or less worsening among participants randomized to active treatment would support inference of benefit from active treatment for the efficacy endpoint being tested. Inference from this analysis is supportive of inference from the Bayesian shared-parameter, repeated-measures model for the primary endpoint.

### 6.5.3 Random-slopes Model

Treatment groups of a given regimen will be compared for mean rate of progression of continuous efficacy and selected safety endpoints (ALSFRS-R total score, ALSFRS-R domain scores, HHD upper extremity, lower extremity, and global average percentages, SVC percent-predicted, serum creatinine, serum NfL, and weight) using a random-slopes linear mixed model in the FAS, ECC, ERO, ECM, and EPP samples. The model will include fixed terms for month since the Baseline Visit, treatment group, treatment group  $\times$  month interaction, time since ALS symptom onset and time since ALS symptom onset  $\times$  month interaction, delta-FRS and delta-FRS  $\times$  month interaction, baseline riluzole use and baseline riluzole use  $\times$  month interaction, baseline edaravone use and baseline edaravone use  $\times$  month interaction, baseline sodium phenylbutyrate/taurursodiol use and baseline sodium phenylbutyrate/taurursodiol  $\times$  month interaction, and baseline log-transformed serum NfL level and baseline log-transformed serum NfL level  $\times$  month interaction. Study months will be calculated as the difference in days from the Baseline Visit to the date of assessment of a given endpoint multiplied by 12 / 365.25. Fixed terms other than treatment group and month will be centered by subtracting the sample mean of a given covariate for a given analysis set. The model will include random participant-specific intercepts and slopes with unstructured covariance. In analysis samples in which placebo participants are drawn from more than three regimens, the model will include random regimen-specific intercepts and slopes with unstructured covariance. In analysis samples with three or fewer regimens, the model will include fixed effects for regimen and regimen  $\times$  month interaction.

The following equations describe the model with regimen random effects:

$$\begin{aligned}
Y_{ij} &= \gamma_1 + a_{k(i)}^0 + b_i^0 + \gamma_2 t_i + \gamma_3' z_i \\
&+ (\gamma_4 + a_{k(i)}^1 + b_i^1 + \gamma_5 t_i + \gamma_6' z_i) m_{ij} + \epsilon_{ij} \\
\{a_k^0, a_k^1\} &\sim N(\mathbf{0}, \Sigma_r), \{b_k^0, b_k^1\} \sim N(\mathbf{0}, \Sigma_p), \epsilon_{ij} \sim N(0, \sigma_\epsilon^2) \\
\text{Cov}(\mathbf{a}_k, \mathbf{b}_k) &= \mathbf{0}, \text{Cov}(\mathbf{a}_k, \epsilon_{i.}) = \mathbf{0}, \text{and } \text{Cov}(\mathbf{b}_k, \epsilon_{i.}) = \mathbf{0}
\end{aligned} \tag{eqn. 2}$$

where  $Y_{ij}$  is a given efficacy endpoint measured for participant  $i$  at visit  $j$ ,  $a_{k(i)}^0$  and  $a_{k(i)}^1$  are random intercept and slope for regimen  $k$  to which participant  $i$  was assigned,  $b_i^0$  and  $b_i^1$  are random intercept and slope for participant  $i$ ,  $z_i$  is the vector of covariates (centered time since onset, centered delta-FRS, centered baseline riluzole use, centered baseline edaravone use, centered for sodium phenylbutyrate/taurursodiol use, and baseline log-transformed serum NfL level) for participant  $i$ ,  $m_{ij}$  is the time from baseline to observation  $j$  for participant  $i$  in months calculated as days x 12 / 365.25,  $t_i$  is an indicator variable for treatment  $t$  to which participant  $i$  was assigned,  $\gamma_1, \gamma_2, \gamma_3, \gamma_4, \gamma_5$ , and  $\gamma_6$  are estimated parameters and vectors of parameters for the fixed effects, and  $\epsilon_{ij}$  is the residual for observation  $j$  for participant  $i$ . The regimen-specific random effects are normally distributed with mean  $\mathbf{0}$  and unstructured covariance matrix  $\Sigma_r$ . The participant-specific random effects are normally distributed with mean  $\mathbf{0}$  and unstructured covariance matrix  $\Sigma_p$ . The residuals for a given participant are normally distributed with mean 0 and variance  $\sigma_\epsilon^2$ . The regimen-specific random effects, participant-specific random effects, and residuals are uncorrelated.

The following SAS code specifies the model:

```
proc mixed data=xxx method=reml;
  class regimen id trtrnd;
  model Value = month trtrnd*month
              sx2b1|month dFRS|month nfl|month rlz|month edv|month
              amx|month / solution;
  random intercept month / subject=regimen type=un;
  random intercept month / subject=id type=un;
```

where month is time in months from the Baseline Visit (assuming 12 months in an average of 365.25 days per year) and other fields are the same as identified above in Section 6.5.2.

The primary estimate will be the treatment-dependent difference in slopes. The estimate and its 95% Wald confidence bounds will be obtained by a linear contrast of adjusted means. The following SAS code specifies the linear contrast for a regimen with one active treatment assuming that the sort order for treatment group has the active group last:

```
estimate "3|Act vs Plb|Slope (/mn)" month 0 trtrnd*month -1 1 / cl;
```

The estimand estimated by the primary linear contrast of the repeated-measures linear mixed model is the mean difference in 24-week change from baseline of a given continuous efficacy endpoint in the active treatment group relative to the placebo group in the FAS population. The estimand is defined by the following attributes:

- Treatment: active treatment vs. placebo.
- Population: FAS population as defined in Section 6.1.
- Variables: absolute change in endpoint from baseline to the Week 24 Visit.



- Intercurrent event: treatment discontinuation: handled via treatment policy approach, all data will be used including data collected during the placebo-controlled period after treatment discontinuation. Missing data post-treatment, including data missing due to death, will not be imputed, handled via missing at random assumption.
- Population-level summary: difference in conditional means of active treatment relative to placebo.

A significant positive difference in slopes in the direction of greater improvement or less worsening among participants randomized to active treatment would support inference of benefit from active treatment for the efficacy endpoint being tested. Inference from this analysis is supportive of inference from the Bayesian shared-parameter, repeated-measures model for the primary endpoint.

#### 6.5.4 Survival

The survival endpoints will be analyzed in the FAS, ECC, ERO, ECM, EPP, STF, STN, and SRO analysis sets. The proportion of participants who reach each survival endpoint and the time from baseline to that endpoint will be summarized by treatment group in all analysis samples. If more than 10% of a given sample reaches a survival endpoint, then the estimated time to a 10% event rate will be estimated by Kaplan-Meier product-limit estimates with complimentary log-log confidence bounds by treatment group. Time to each survival endpoint will be compared between treatment groups by log-rank test. If at least one occurrence of a given survival endpoint is observed in each treatment group, then the treatment-dependent hazard ratio with profile likelihood confidence bounds will be estimated by unadjusted Cox proportional hazards regression and by Cox proportional hazards regression adjusting for age at baseline, time since ALS symptom onset, delta-FRS, baseline log-transformed serum NfL level, use of riluzole at baseline, use of edaravone at baseline, and use of sodium phenylbutyrate/taurursodiol at baseline with a random gamma-distributed regimen-specific frailty term. If convergence is not obtained or the analysis sample includes 3 or fewer regimens, then the frailty term will be dropped.

#### 6.5.5 CAFS

To account for potentially informative loss to follow-up due to death, treatment groups of a given regimen will be compared by the Combined Assessment of Function and Survival (CAFS, Berry et al. 2013) in the FAS, ECC, ERO, ECM, and EPP samples. CAFS is a joint-rank test (Finkelstein and Schoenfeld 1999) of a composite endpoint combining function and survival. The test consists of calculating a rank-sum score for each individual relative to pair-wise comparisons with all other participants. Participants are ranked according to time to the survival endpoint when that is observed for both members of a pair or when one is censored after the observed event time for the other. Pairs that cannot be ranked by time to the survival endpoint are ranked by absolute change from baseline in the functional endpoint at the maximum follow-up time at which both participants have an observation. Pair-wise rankings are scored -1 for the member of the pair with shorter time to the survival endpoint or greater decline in function and 1 for the other member or 0 for both members when time to the survival endpoint is equal or not evaluable and change in function is equal, e.g., when one participant is lost to follow-up after baseline. Pair-wise ranking scores are summed by participant to calculate a rank-sum. Inference of treatment-dependent difference in the hierarchical composite of death and disease progression is based on a Wilcoxon rank-sum test of the participant-specific rank-sum scores or on a regression of the rank-sum scores adjusting for covariates.



All combinations of the following elements will be tested:

1. Survival endpoint: (a) death or death-equivalent, and (b) death alone.
2. Function endpoint: (a) ALSFRS-R total score, (b) HHD upper extremity percentage, (c) SVC percent-predicted, and (d) HHD lower extremity percentage.
3. Analysis: (a) Wilcoxon rank-sum test by t-distribution approximation, and (b) adjusted mixed model analysis controlling for time since ALS symptom onset, delta-FRS, baseline log-transformed serum NfL level, use of riluzole at baseline, use of edaravone at baseline, and use of sodium phenylbutyrate/taurursodiol at baseline and including a random effect of regimen in all analyses that include at least three regimens or a fixed effect of regimen otherwise.

### 6.5.6 HHD0 and HHD0[2]

The time to first and second loss of measurable strength among muscles assessed by HHD and grip strength will be compared between treatments by interval-censored nonparametric maximum likelihood methods and interval-censored Cox proportional hazards regression in the FAS, ECC, ERO, ECM, and EPP samples. The model will adjust for time since ALS symptom onset, delta-FRS, baseline log-transformed serum NfL level, use of riluzole at baseline, use of edaravone at baseline, and use of sodium phenylbutyrate/taurursodiol at baseline.

The first HHD0 and HHD0[2] analyses will consider all 11 muscles and grip strength with non-zero strength at baseline (up to 24 bilateral strength assessments in total). A second pair of HHD0 and HHD0[2] analyses will restrict to the four bilateral muscle groups or grip strength that reach zero most frequently in a specific analysis sample (up to 8 bilateral strength assessments in total).

### 6.5.7 Placebo Multiple Imputation

To account for possible treatment-dependent loss to follow-up whether due to death, intolerance, or other causes, the effect of treatment on ALSFRS-R total score, HHD upper extremity percentage, SVC percent-predicted, and HHD lower extremity percentage will be estimated in the repeated-measures and random-slopes models using placebo multiple imputation for missing data (Ayele et al. 2014) in the FAS and ERO samples. Fifty imputed datasets will be generated for a given endpoint. Missing values for each visit will be imputed from those predicted by linear regression from earlier visits using data only from placebo participants, adjusting for time since symptom onset, delta-FRS, baseline log-transformed serum NfL level, baseline riluzole use, baseline edaravone use, and baseline sodium phenylbutyrate/taurursodiol use.

The following SAS code specifies the imputation for an endpoint measured every 8 weeks:

```
proc mi data=work.outcomes seed=xxx nimpute=50 minmaxiter=1000
    minimum=. . . . . x x x x maximum=. . . . . y y y y ;
    class trtrnd rlz edv amx;
    fcs reg(Wk08 = sx2b1 dFRS nfl rlz edv amx Wk00);
    fcs reg(Wk16 = sx2b1 dFRS nfl rlz edv amx Wk00 Wk08);
    fcs reg(Wk24 = sx2b1 dFRS nfl rlz edv amx Wk00 Wk08 Wk16);
    mnar model( Wk08 Wk16 Wk24 / modelobs=(trtrnd="0"));
    var sx2b1 dFRS nfl rlz edv amx Wk00 Wk08 Wk16 Wk24;
run;
```

where Wk00, Wk08, Wk16, and Wk24 are the values of a given efficacy endpoint at the Baseline, Week 8, Week 16, and Week 24 Visits, respectively, trtrnd has a value of zero (0) for participants randomized to placebo, and x and y take appropriate values to specify the range of a given outcome measure (i.e., 0 and 48 for ALSFRS-R total score; 0 and . [missing value indicator] for HHD upper and lower extremity percentages and SVC).

### 6.5.8 Additional Sensitivity Analyses of Primary and Key Secondary Outcomes

To evaluate the influence of extreme values on treatment estimates from the random-slopes model, a robust linear mixed model that down-weights extreme values (Koller 2013) will be used as a sensitivity analysis of primary and key secondary outcomes using the FAS analysis set. The same fixed and random effects as specified for the primary random-slopes model in Section 6.5.3 will be used. A smoothed Huber  $\rho$ -function will be used with a tuning parameter that yields 95% asymptotic efficiency for normally distributed data.

As an alternative to CAFS for controlling the influence of potentially information loss to follow-up due to death or other causes, sensitivity of estimated treatment effects to potentially informative missingness will be analyzed using a composite endpoint of time to death and 24-week change in ALSFRS-R total score using the FAS analysis set with imputation across a range of weights spanning relative benefit or harm from the active treatment by exponential tilting (Wang et al. 2017).

### 6.5.9 Subgroup Analyses

Differences in treatment efficacy will be estimated in the following subgroups by including subgroup, subgroup  $\times$  time, and subgroup  $\times$  treatment  $\times$  time terms to random-slope models for primary and secondary efficacy endpoints using the FAS analysis set .

- Age (less than 65 years vs. 65 years or older),
- Sex (female vs. male),
- Race (white vs. any minority race with greater than 5% prevalence in the sample),
- Ethnicity (Hispanic or Latino vs. non-Hispanic or Latino),
- Baseline use of riluzole (yes vs. no),
- Baseline use of edaravone (yes vs. no),
- Baseline use of sodium phenylbutyrate/taurursodiol (yes vs. no),
- BMI (less than 25 kg/m<sup>2</sup>, 25 to less than 30 kg/m<sup>2</sup>, 30 kg/m<sup>2</sup> or more),
- Chronic kidney disease (CKD) stage (stage 1 or better [eGFR 90 mL/min/1.73m<sup>2</sup> or more], stage 2 [eGFR 60 to 89 mL/min/1.73m<sup>2</sup>], stage 3 or worse [eGFR less than 60 mL/min/1.73m<sup>2</sup>]),
- delta-FRS (fast vs. slow progression with median split),
- Time since onset of weakness (less than 18 months vs. 18 months or longer),
- El Escorial and time since onset of weakness (El Escorial definite and less than 18 months vs. not both),

- Baseline symptom severity (all ALSFRS-R questions scored 2 or greater vs. any question scored 0 or 1),
- Early disease state (all ALSFRS-R questions scored 2 or greater, SVC 80%-predicted or greater, and time since symptom onset less than 24 months vs. not meeting all three criteria),
- Site of symptom onset (bulbar vs. non-bulbar),
- Baseline plasma NfL concentration (high vs. low by median split), and
- Site (individual sites with at least 5 participants per treatment group and all participants from sites with fewer than 5 participants per treatment group pooled).

For each subgroup, unknown, not reported, and missing will be considered as one group. All individuals not included in a specified subgroup will be combined into a mixed, "other" group. The "other" group will be included in analyses if its prevalence is greater than 5%; otherwise, the "other" group will be excluded.

For each subgroup, the following estimates will be obtained:

- the treatment-group specific estimate within each subgroup level,
- the treatment difference within each subgroup level,
- the difference of differences between each pair of subgroup levels, and
- for subgroups with more than two levels, the F-test for the subgroup  $\times$  treatment  $\times$  time term will be estimated.

### 6.5.10 Comparison of Controls across Regimens

The magnitude of differences in regimen-specific rates of progression for primary and secondary efficacy endpoints will be compared by random-slopes models for functional endpoints and by Kaplan-Meier product-limit estimates, log-rank test, and Cox proportional hazards regression for survival using the placebo participants in the FAS analysis set. The random-slopes models will include fixed effects of regimen and regimen  $\times$  time interaction, adjusting for time since onset, delta-FRS, baseline log-transformed serum NfL level, baseline riluzole use, baseline edaravone use, and baseline sodium phenylbutyrate/taurursodiol and their interactions with time. Fixed terms other than regimen and month will be centered by subtracting the sample mean of a given covariate. The models will include participant-specific intercepts and slopes with unstructured covariance. Adjusted regimen-specific slope estimates, difference of the focal regimen control group vs. each other regimen and vs. the average of all other regimens, and an overall F-test of regimen-specific differences in rates of disease progression will be estimated. Regimen-specific slopes will be plotted vs. calendar year of median randomization. Cox models will include main effects for regimen, time since onset, delta-FRS, baseline log-transformed serum NfL level, baseline riluzole use, baseline edaravone use and baseline sodium phenylbutyrate/taurursodiol use. Adjusted regimen-specific hazard ratios of the focal regimen control group vs. each other regimen and vs. the average of all other regimens, and an overall F-test of regimen-specific differences in hazard of mortality will be estimated.

## 6.6 Safety Analyses

### 6.6.1 Treatment-emergent Adverse Events

TEAEs are defined as those adverse events with onset dates in the interval from treatment initiation to either the Final Safety Visit if completed, the date of death if the participant dies, the date of last study contact if the participant early terminates or is lost to follow-up, or the date of first dose of study drug during participation in the ATE, if so exposed. Adverse events with onset on the day of placebo-controlled treatment initiation and adverse events with incompletely specified onset date where the ambiguous date spans the day of double-blind treatment initiation will be assumed to be treatment emergent except for those known to precede first exposure to study drug.

The incidence of TEAEs will be summarized by the number of events of a given classification experienced by participants in each treatment group and by the number and proportion of participants experiencing such an event in each treatment group in all safety samples. The number of events and proportion of participants experiencing TEAEs will be summarized without regard to MedDRA term for any TEAE, any serious TEAE, any severe TEAE, any fatal TEAE, any related TEAE, and any TEAE leading to discontinuation of study drug. TEAEs will be summarized by MedDRA system organ class, high level term, and preferred term for all TEAEs. TEAEs, serious TEAEs, and non-serious TEAEs observed in at least 5% of participants in a given treatment group will be summarized by MedDRA system organ class and preferred term and by MedDRA preferred term alone. TEAEs will be summarized by MedDRA system organ class and preferred term for serious TEAEs, severe TEAEs, fatal TEAEs, related TEAEs, and TEAEs leading to discontinuation of study drug. TEAEs and serious TEAEs will be stratified by the following subgroups: age, sex, race, ethnicity, baseline riluzole use, baseline edaravone use, baseline sodium phenylbutyrate/taurursodiol use, weight, BMI, and CKD stage. Subgroup classifications will be the same as described in Section 6.5.9 Subgroup Analyses except that the "other" group will be retained regardless of prevalence.

For overall summaries of TEAE not classified by MedDRA term and for summaries of TEAEs, serious TEAEs, and non-serious TEAEs observed in at least 5% of participants in a given treatment group, treatment-specific TEAE incidence rates in units of number per 100 participant years and treatment-dependent differences will be estimated by Poisson regression with log link and offsets equal to the log of weeks from initiation to discontinuation of placebo-controlled treatment. Comparison-wise 95% confidence intervals will be calculated with variance estimates obtained by the delta method.

The incidence of TEAEs will be summarized without regard to MedDRA term in all safety samples for the following characteristics: (a) seriousness, (b) severity, (c) relationship to study drug, (d) action taken with study drug, (e) use of a concomitant medication, and (f) outcome. For each level of a given TEAE characteristic, summaries will include the number of events of a given classification and the number and proportion of participants for which that level of a characteristic was the worst they experienced. Any unknown characteristic of a TEAE will not be classified as worst. The level for participants who experienced no TEAE will be coded as the zero-level of each characteristic.

## 6.6.2 Safety Labs

The absolute level and the absolute change from baseline for each safety laboratory assay will be summarized as means, standard deviations, medians, and ranges at each visit by treatment group in all safety samples. The proportion of participants with abnormal levels classified to a toxicity grade based on quantitative grading using National Cancer Institute (NCI) Common Terminology Criteria for Adverse Events (CTCAE) version 5.0, the proportion with abnormal and clinically significant safety lab levels, and the proportion with possible DILI by each definition will be summarized by treatment group by visit and for any post-baseline visit in all safety samples. The proportion of participants with abnormal levels by CTCAE grade, the proportion with lab levels that are below the lower limit of normal, within normal limits, or above the upper limit of normal will be presented as shift tables vs. the status of each participant at baseline for each visit and for any post-baseline visit by treatment group in all safety samples.

## 6.6.3 ECG Results

The absolute level and the absolute change from baseline for each ECG parameter will be summarized as means, standard deviations, medians, and ranges at each visit by treatment group in all safety samples. The proportion of participants with abnormal and clinically significant ECG findings or parameter values of special concern will be summarized by finding for each treatment group by visit and at any post-baseline visit in all safety samples. The parameter values of special concern are the following: QTcF > 480 ms, QTcF > 500 ms, increase from baseline in QTcF > 60 ms, increase from baseline in PR > 25% to a value above 200 ms, increase from baseline in HR > 25% to a value below 50 bpm, increase from baseline in HR > 25% to a value above 100 bpm, and increase from baseline in QRS > 25% to a value above 110 ms.

## 6.6.4 Vital Signs and Weight

The absolute level and the absolute change from baseline for vital signs and weight will be summarized as means, standard deviations, medians, and ranges at each visit by treatment group in all safety samples. Changes in weight will also be analyzed as an efficacy endpoint by repeated-measures and random-slopes mixed models in the FAS, ECC, ERO, ECM, and EPP samples.

## 6.6.5 Suicidality

The proportion of participants who report any post-baseline suicidal ideation or any post-baseline suicidal behavior on the C-SSRS will be summarized by treatment group in all safety samples. Suicidal behaviors will include: actual attempts, non-suicidal self-injurious behavior, interrupted attempts, aborted attempts, preparatory acts or behaviors, and completed suicide. The most severe ideation, maximal frequency, maximal duration, minimal controllability, minimal deterrents, maximal reasons for ideation, maximal actual lethality or medical damage, and maximal potential lethality will be summarized by treatment group in all safety samples.

## 6.7 Other Analyses

### 6.7.1 Participant Disposition

All participants consented to the Master Protocol between the time of the first and last consent of a participant assigned to a regimen included in the FAS analysis set will be summarized as a single set for the following events: consented to the Master Protocol, failed screening for the Master Protocol, other reasons not assigned to a regimen (including timing out of the screening



window, death, withdrawal of consent, early termination, and administrative termination), and assigned to a regimen. The number of participants ever assigned and last assigned to the focal regimen, to a non-focal contributing regimen, and to a non-contributing regimen will be separately indicated. Reasons for Master Protocol screen failure will be summarized.

All participants in the above sample assigned to the focal regimen or a non-focal contributing regimen will be summarized as two sets (last assigned to the focal regimen vs. last assigned to a non-focal contributing regimen) for the following events: consented to a regimen, failed screening for a regimen, other reasons not randomized within a regimen (including timing out of the screening window, death, withdrawal of consent, early termination, and administrative termination), and randomized within a regimen. If a given individual is screened multiple times prior to randomization within a regimen, then the final screening experience of that individual will be summarized. Reasons for screen failure will be summarized separately for all participants screened for the focal regimen whether that was their final screening experience or not.

All participants in each analysis set will be summarized as two sets (randomization to active study drug vs. randomization to placebo) for the following events: initiated regimen-specific study drug, prematurely terminated study participation due to death, withdrawal of consent, early termination, loss to follow-up, or administrative termination, completed 24-week follow up, and completed a safety follow-up visit vs. continued into the ATE. Reasons for withdrawal from the study will be presented.

#### **6.7.2 Study Drug Compliance and Tolerance**

The number of days of exposure to study drug will be summarized by treatment group in the ERO and SRO analysis sets. The proportion of participants on study drug at each clinic visit will be summarized by treatment group. Study drug exposure will be calculated in two ways:

- as the number of days from dose initiation to the final safety assessment during the placebo-controlled period, inclusive, and
- as the number of days from dose initiation to drug withdrawal, inclusive, less any interval during which use of study drug was interrupted (individual missed doses will not be subtracted unless noted in the dosage management log).

The proportion of participants judged tolerant of study drug, both those who are free of intolerance due to an AE and those free of intolerance for any reason, will be summarized by treatment group.

#### **6.7.3 Concomitant Medication Use**

Concomitant medications taken during the study period will be coded using the World Health Organization (WHO) Drug Dictionary Enhanced and Default Anatomical, Therapeutic, and Chemical (ATC) class. The percentage of participants in all safety samples taking each WHO drug classified medication and each ATC class of medications will be summarized by treatment group for medications taken at baseline and those initiated after first dose of study drug.

#### **6.7.4 Medical History**

Medical histories will be summarized by MedDRA system organ class, high level term, and preferred term by treatment group for all medical conditions recorded and for medical conditions identified as ongoing at baseline in the FAS, ERO, STF, and SRO analysis sets.

### 6.7.5 Blindedness

The proportions of participants and site investigators who report on the Exit Questionnaire a guess of active vs. placebo treatment assignment, each level of surety of that guess, and each of five pre-specified reasons for making a treatment assignment will be summarized by treatment group in the FAS, ERO, and EPP analysis sets. Treatment-dependent differences in the proportion guessing active treatment assignment will be tested among all respondents and among those stating they are at least somewhat sure of their guess by Fisher's exact test, the difference in proportion guessing active treatment assignment will be estimated with confidence bounds, and the distribution of surety of guesses will be compared by exact Cochran–Armitage trend test.

### 6.7.6 Protocol Deviations

The number of major and minor protocol deviations will be summarized by type of deviation and treatment group in all analysis sets. Listings of all protocol deviations will be produced.

## 7. Validation

### 7.1 Primary Efficacy Analysis

The primary efficacy analysis will be validated according to a validation plan administered by Berry Consultants.

### 7.2 Secondary, Exploratory, and Safety Analyses

Key secondary analyses of the primary efficacy endpoint and primary analyses of secondary efficacy endpoints and summaries and analyses of serious TEAEs will be replicated from Study Data Tabulation Model (SDTM) source data by two analysts or statisticians working independently. Exact matches are required to validate results for the following integer results: treatment- and visit-specific counts of efficacy endpoints, counts of events, number of participants affected, and number of participants at risk for safety endpoints. Relative matches within 0.1% of the common mean are required to validate results for the following floating point results: estimates, standard errors, confidence bounds, and p-values.

All other analysis code and results will be reviewed by a statistician other than the initial analyst.

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## 9. Appendix I

### ALS Master Protocol Recommended Statistical Analysis, Design and Simulation Report

February 6, 2023

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## Summary of Changes

Changes since document dated May 14, 2021.

Section 1.0: Updated general descriptions of trial design, primary analysis, planned sample size, and default early stopping conditions at interim analyses.

- Section 2.1: Updated the analysis sample definition to exclude participants determined to not meet ALS diagnostic criteria. Clarified the definition of the survival endpoint. Defined “ongoing participants” and specified that data from ongoing participants would not be used in the primary analysis.
- Section 2.2: Clarified reference to the survival endpoint.
- Section 2.2.5: Added baseline log-transformed serum NfL level and baseline sodium phenylbutyrate/taurursodiol use as additional covariates. Specified that all covariates would be standardized by subtracting their means. Specified that all missing baseline covariates would be imputed by their mean. Specified that means would be calculated after transformation for any transformed baseline covariates.
- Section 3.1: Removed reference to the first three regimens specifically.
- Section 4.1: Added a table summarizing the number and percent of completed or expected ALSFRS-R assessments that do not contribute to the functional component of the primary analysis.
- Section 4.5: Added details regarding the piecewise-linear sensitivity analysis.
- Section 4.6: Added a sensitivity analysis using death alone as the survival endpoint.

## 5.0 Introduction

The ALS Master Protocol is a perpetual, multicenter, multi-regimen, randomized, placebo-controlled, adaptive platform trial. The primary objective of the trial is to evaluate the efficacy of multiple investigational products, as compared to placebo, on ALS disease progression. The master protocol will include multiple regimens where a regimen is defined as both an active treatment and the corresponding randomized placebo. Participants in the master protocol will be randomized in two stages; first to one of the actively enrolling regimens; and then 3:1 to active treatment or placebo within the regimen.

This document describes the master protocol recommended primary endpoint, primary analysis, and interim analyses. This will be the default statistical design and primary analysis for all regimens investigated in this master protocol. The recommended design is intended to support inference of substantial evidence of effectiveness when the data so warrant.

Each regimen will have a regimen-specific appendix (RSA). Deviations from the master protocol recommended statistical design are permitted. The exact statistical design along with complete operating characteristics for each regimen will be specified in the RSA.

The master protocol recommended primary efficacy endpoint is change from baseline through week 24 in disease severity as measured by the ALS Functional Rating Scale-Revised (ALSFRS-R) total score and survival. The primary analysis is a Bayesian shared parameter analysis of ALSFRS-R and survival. The primary analysis compares each regimen-specific active treatment to a shared control group.

The sample size enrolled in each regimen will be determined based on the specifics of the intervention, anticipated effect size, and number of treatment arms evaluated within that regimen. There are frequent interim analyses in order to potentially stop a regimen early for success or for futility. The default design for a regimen will include stopping early only for futility. Interim analyses will be conducted at the same time for all regimens within the master protocol, every 12 weeks of trial time. A regimen is eligible for these interim analyses once a minimum specified amount of data has been observed for that regimen.

## 2.0 Primary Efficacy Analysis

The following describes the Bayesian shared parameter primary analysis method of function and survival for each regimen, the primary analysis population, prior distributions for all model parameters, and specific posterior summaries of interest that will be used in the design. We denote the regimen being analyzed as the “analysis regimen” throughout.

### *2.1 Primary Analysis Population and Shared Control*

The primary analysis population for each analysis regimen is guided by the intent to treat (ITT) principle and includes all participants randomized to the active treatment arm within the analysis regimen, all participants concurrently randomized to the control arm within the analysis regimen, and control participants from other regimens from a shared control, excluding any participant determined to not meet ALS diagnostic criteria.

The primary analysis dataset uses longitudinal data on disease progression (ALSFRS-R measurements at 0, 4, 8, 12, 16, 20 and 24 weeks, at early termination visits, and assessments during double-blind follow-up that substitute for missed scheduled visits) for those who have not been censored due to reaching the survival endpoint (a composite of death or death-equivalent) as well as occurrence of the survival endpoint during planned double-blind follow-up on participants within the primary analysis population. Follow-up for primary analysis of survival among participants who do not die or reach a death equivalent prior to the Week 24 Visit will be censored at the Week 24 Visit date, if completed, or at the earlier of the end of the Week 24 Visit window or date last known alive if a Week 24 visit is not completed. Off-schedule results, e.g., those collected as part of an Early Termination Visit, will be used in place of the closest missing scheduled visit (while preserving visit sequence) for analyses that depend on visit-specific data. Off-schedule results that do not substitute for a missed schedule visit will not be used.

The shared control group will include participants randomized to the control arm across other actively enrolling regimens (concurrent shared controls) and previously completed regimens (non-concurrent shared controls) within the master protocol as specified in the RSA. Concurrent shared controls are defined as participants on a control arm who were randomized within a 6-month window of the start and stop of randomization of the analysis regimen. For the purposes of analysis, 6 months is defined as 180 days. For example, if the analysis regimen enrolls from Month 12 – Month 24 in the platform, the

concurrent shared controls will include all available data from all participants randomized between Month 6 and Month 30 within the platform across all regimens.

For interim analyses, all available data at the time of the analysis will be used including data on participants who have not had the opportunity to complete 24 weeks of follow-up (denoted ongoing). The phrase “opportunity to complete 24 weeks” is meant to describe participants who have at least 24 weeks of time elapse since their baseline visit.

For final primary analyses, data from ongoing participants will not be used, but participants from open regimens who have had the opportunity to complete 24 weeks of follow-up may be used. To minimize risk of potential unblinding of data for regimens outside of the analysis regimen that have not yet had last patient last visit (open regimens) if shared control data from open regimens are used within the primary analysis, no participant-level data will be shared and no summaries of the data will be provided from which individual participants could be identified until all regimens that are included in the shared control for the analysis regimen have been closed (last patient last visit) and locked. For this reason, regimens may choose to use shared control data only from regimens that have been closed (last patient last visit) to avoid delay in release of participant-level data. The shared control population that will be included in the final primary analysis for each regimen will be prespecified in each regimen-specific appendix.

Finally, the master protocol allows for additional and more restrictive inclusion/exclusion criteria within a regimen specific appendix (RSA). However, these additional inclusion/exclusion criteria will be limited and will be allowed based only on restrictions due to safety or mechanism of action. Major additional RSA inclusion/exclusion criteria that can be assessed for all participants entering the platform will be used to subset the shared control to include only shared control participants that meet these major additional RSA inclusion/exclusion criteria. Major additional RSA inclusion/exclusion criteria that will be used to subset the shared control will be limited to those criteria that have the potential to lead to significant differences in the patient population in terms of the rate of disease progression within the primary analysis population and will be prespecified within each RSA. The first three regimens enrolling within the platform did not include any major additional inclusion/exclusion criteria.

## *2.2 Bayesian Shared Parameter Model of Function and Mortality*

The primary analysis for each analysis regimen is a Bayesian shared parameter model of function and survival with the following components:

- **Function:** Bayesian repeated measures model of ALSFRS-R at baseline through 24 weeks for all participants who have not experienced mortality (per definition of the composite survival endpoint). Participants who have experienced mortality will not contribute to the functional component of the analysis. The Bayesian repeated measures model of ALSFRS-R measures the

slowing in the rate of progression for ALSFRS-R in treated participants relative to control for survivors.

- **Mortality:** Exponential proportional hazards model of time to the composite survival endpoint of death or death-equivalent through 24 weeks that measures the mortality hazard ratio in treated participants relative to control.
- **Shared Treatment Effect:** A shared treatment effect parameter (slowing in time of the progression of the disease) is common between the function and mortality components and provides an overall estimate of the treatment effect that is averaged across the two endpoints. This shared parameter corresponds to a slowing in the rate of ALSFRS-R progression as well as a slowing in the time to mortality.

The ALSFRS-R repeated measures component of survivors is based on a linear rate of progression in ALSFRS-R for placebo participants and a proportional slowing in the rate of progression for treated participants at each time point. The model incorporates participant-level random effects in the baseline value of the ALSFRS-R (intercept) and in the rate of progression (slope); covariate effects to account for covariate-explainable differences in rates of progression based on participant-specific baseline covariates; regimen-specific differences in baseline values, rates of progression and measurement error, and a time-trend effect to accommodate non-concurrently randomized controls. Assuming there are  $N_{func}$  total participants who have survived, let the ALSFRS-R at timepoint  $j$  for participant  $i$  be labelled  $Y_{ij}$ , for  $j =$  the number of weeks at the visit with expected values around 0, 4, 8, 12, 16, 20, and 24 weeks and  $i = 1, \dots, N_{func}$ . We note that the value for  $j$  will be the actual week that the visit took place rounded to the nearest day. The Bayesian repeated measures model for ALSFRS-R is

$$Y_{ij} = (\alpha_{r(i),t(i)} + \gamma_i) - \beta_i * \exp(\theta_{t(i)} + \delta X_i + \rho_{d(i)} + \eta_{r(i)}) * \left(\frac{j * 12}{52}\right) + \epsilon_{ij};$$

$$\epsilon_{ij} \sim N(0, \sigma_{r(i),t(i)}^2).$$

The mortality component is an exponential proportional hazards model that assumes a constant baseline hazard rate for all control participants. Given the low expected rate of mortality within this study (5%) and to avoid over-parameterization, we do not include covariate, regimen-specific, or time-trend effects in the mortality component. Assuming there are  $N_{mort}$  total participants, the hazard rate of mortality per month for participants  $i = 1, \dots, N_{mort}$  is denoted  $\lambda_i$  and is

$$\lambda_i = \lambda_0 \exp(\theta_{t(i)}); i = 1, \dots, N_{mort}.$$

The shared parameter model makes use of the following additional notation:

- Treatment arm:  $t(i)$  denotes the treatment arm,  $T$ , to which participant  $i$  is randomized where  $T = 0,1$  corresponds to placebo and active arm, respectively.
- Regimen:  $r(i)$  denotes the regimen,  $R$ , to which participant  $i$  is randomized where  $R = 1, \dots, N_r$  corresponds to the analysis regimen ( $R=1$ ) and all other  $N_r$ , enrolling regimens respectively.



- Time of Randomization:  $d(i)$  denotes the time of randomization,  $d$ , for participant  $i$ . The time of randomization is expressed as an ordinal discrete outcome that where  $d = 0$  is the current time period for concurrent controls (defined as in Section 2.1 with those participants randomized within 6 months (180 days) from the start or finish of randomization within the analysis regimen) and  $d = 1, \dots, D$  are the number of additional 6-month intervals preceding the start of the current time period.
- Baseline Covariates:  $X_i$  is a vector of dimension  $C$  of baseline covariates for participant  $i$  that have been standardized.

Each parameter in the above model and the prior distribution assumed for each parameter is described below in Sections 2.2.1 – 2.2.7. The majority of the prior distributions are chosen to be non-informative (large amount of variability and uncertainty) to allow the posterior distributions to reflect what is observed in the data.

### 2.2.1 Treatment Effect ALSFRS-R and Mortality

The disease rate ratio (DRR) under treatment  $T$  is the parameter  $\exp(\theta_T)$ . For a placebo patient, the DRR is set equal to 1 ( $\theta_0 = 0$ ). Thus, the DRR parameter for active treatment represents the relative change to the rate of decline of ALSFRS-R and the rate of mortality of treated participant relative to a placebo participant. The estimated DRR can also be interpreted as the average rate of decline in function and mortality, with the weights across function and mortality depending on the proportion of patients with observed mortality. Appropriately, more weight is given to mortality relative to function as the proportion of observed mortality increases. If the DRR is less than 1 then the rate of disease progression on function and mortality for actively treated participants is slower than that for placebo. The DRR value represents the proportional slowing in time (or increase) of the disease. A DRR = 0.75 corresponds to a 25% slowing in the rate of functional progression and time to mortality and a value of 0.25 represents a 75% slowing in the rate of functional progression and time to mortality.

The prior distribution for the DRR for the regimen-specific active treatment,  $T=1$ , is uniform between 0 and 2; placing equal weight on a DRR<1 and DRR>1:

$$\exp(\theta_1) \sim \text{Unif}(0,2).$$

### 2.2.2 Random Intercept and Slope Effects for ALSFRS-R

Participant-specific random intercept effects,  $\gamma_i, i = 1, \dots, N$ , represent the increase or decrease in the participant-specific expected ALSFRS-R at the time of randomization compared to the overall mean for the treatment arm and regimen. Participant-specific

random slope effects,  $I, i = 1, \dots, N$ , represent the participant-specific rate of decline per month in ALSFRS-R expected under placebo.

The overall mean estimate per treatment arm and regimen is assumed to have a non-informative prior normal distribution with a mean of 38 (mean value observed in historical trials within the PRO-ACT database) and SD of 10:

$$\alpha_{R,T} \sim N(38, 10^2); T = 0, 1; R = 1, \dots, N_R$$

Participant-level random intercept effects are modelled using a hierarchical normal distribution with a population mean of 0 and population standard deviation,  $\sigma_\gamma$ , with a uniform hyper-prior:

$$\begin{aligned} \gamma_i &\sim N(0, \sigma_\gamma^2); i = 1, \dots, N; \\ \sigma_\gamma &\sim Unif(0, 10). \end{aligned}$$

The uniform hyper-prior on the standard deviation is non-informative with a range from 0 to 10 (twice observed SD of the participant-level intercepts in historical trials within the PRO-ACT database).

Participant-level random slope effects are modelled using a hierarchical normal distribution with overall population mean,  $\mu_\beta$ , with a normal hyper-prior; and an arm-specific population standard deviation,  $\sigma_{\beta,T}$ ;  $T = 0, 1$ , with a uniform hyper-prior:

$$\begin{aligned} \beta_i &\sim N(\mu_\beta, \sigma_{\beta,t(i)}^2); i = 1, \dots, N; \\ \mu_\beta &\sim N(1, 1^2); \\ \sigma_{\beta,T} &\sim Unif(0, 2); T = 0, 1. \end{aligned}$$

The mean and variance of the participant-level random effects are estimated from the data. This is done by placing hyperprior distributions on the relevant parameters in the random effects distribution and using the observed data to estimate the hyper-parameters. Non-informative distributions are used for these hyper-priors to ensure that the data determine the corresponding hyper-prior estimates and the estimated random effect distributions. In particular, the normal hyper-prior on the mean of the participant-level random slope effects is non-informative with a mean of 1 (mean value observed in historical trials within the PRO-ACT database) with a standard deviation of 1. The uniform hyper-prior on the standard deviation is non-informative with a range from 0 to 2 (over twice observed SD of the participant-level slopes in historical trials within the PRO-ACT database).

### 2.2.3 Regimen-Specific Random Slope Effects ALSFRS-R

The increase or decrease in the rate of progression for placebo participants for all regimens compared to that of the analysis regimen is the parameter  $\exp(\eta_r)$ . For the analysis regimen ( $r=1$ ), the effect is assumed to be 1 ( $\eta_1 = 0$ ).

The log of the regimen-specific random effects,  $\eta_r$ , will be modelled using a hierarchical model to allow for dynamic borrowing within the shared control across different regimens:

$$\begin{aligned}\eta_r &\sim N(0, \sigma_\eta^2); r = 2, \dots, N_r; \\ 1/\sigma_\eta^2 &\sim \text{Gamma}(1, .05^2).\end{aligned}$$

The gamma (shape and rate parameterization) hyper-prior is informed based on estimated between-trial variability in rates of progression for ALSFRS-R from historical trials.

#### 2.2.4 Time-Trend Effects ALSFRS-R

The log of the time-trend effects will be modelled using a simple normal dynamic linear model where the distribution of each time-trend effect is normal with a mean equal to the effect at the previous time period and a standard deviation that has a uniform hyper-prior:

$$\begin{aligned}\rho_0 &= 0; \\ \rho_d &\sim N(\rho_{d-1}, \sigma_\rho^2); d = 1, \dots, D; \\ \sigma_\rho &\sim \text{Unif}(0, 2).\end{aligned}$$

#### 2.2.5 Baseline Covariate Effects ALSFRS-R

Covariates included in the primary efficacy analysis are: time since onset, pre-baseline slope of ALSFRS-R, baseline log-transformed serum NfL level, baseline riluzole use, baseline edaravone use, and baseline sodium phenylbutyrate/taurursodiol use as indicated in participants' concomitant medication logs. All covariates will be standardized by subtracting their means using all participants in the regimen-specific analysis data set (regimen-specific treatment and control as well as additional shared control). Missing baseline covariates for a participant will be imputed using the mean value across all participants in the applicable analysis dataset (regimen-specific treatment and control as well as additional shared control for the primary analysis). For any baseline covariate that is transformed for analysis, means will be calculated after transformation. Covariate effects are modelled as  $\exp(\delta_c)$ ,  $c = 1, \dots, C$ , accounting for covariate-explainable variability in the rate of ALSFRS-R progression due to participant-specific baseline covariates.

Prior distributions for the log of the covariate effects are independent and identically distributed as:

$$\delta_c \sim N(0, 1^2); c = 1, \dots, C.$$

### 2.2.6 Residual Errors ALSFRS-R

The residual errors for the individual observations,  $\epsilon_{ij}$ ,  $i = 1, \dots, N$  and  $j = 0, \dots, 24$  are modelled as independent normal distributions with a mean of 0 and with an treatment arm and regimen-specific standard deviation of  $\sigma_{R,T}$ ;  $T = 0, 1$ ;  $R = 1, \dots, N_R$ .

The prior distribution for the standard deviation of the residual error term in each arm are independent and identically distributed as:

$$\sigma_{R,T} \sim \text{Unif}(0, 4); T = 0, 1; R = 1, \dots, N_R.$$

The uniform prior distribution is non-informative, ranging from 0 to 4 (over twice that observed in historical trials within the PRO-ACT database).

### 2.2.7 Baseline Mortality Hazard Rate

The baseline mortality hazard rate per month common across all placebo participants is  $\lambda_0$ . The prior distribution for the placebo mortality hazard rate is distributed as Gamma (shape and rate parametrization):

$$\lambda_0 \sim \text{Gamma}(.01, 1);$$

with a mean of 0.01 and a standard deviation of 0.1. This prior distribution is centered on the value observed within the PRO-ACT database.

## 2.3 Posterior Summaries

The Bayesian posterior distribution of all model parameters is calculated using Markov chain Monte Carlo (MCMC). We will draw at least 500,000 samples from the posterior distribution with a thinning of 10. The exact number of samples from the MCMC will be determined based on convergence diagnostics to ensure an adequate effective sample size of the posterior distribution.

Summaries of the Bayesian posterior distribution for the DRR under the regimen-specific active treatment,  $\exp(\theta_1)$ , will be provided including the mean and 95% credible intervals; the probability that the treatment is superior to placebo,  $Pr(\exp(\theta_1) < 1)$ ; and the probability the treatment slows the rate of progression by at least 10%,  $Pr(\exp(\theta_1) < .9)$ .

## 3.0 Success Thresholds, Interim Analyses and Trial Adaptations

Interim analyses will occur simultaneously for all actively enrolling regimens. Interims will begin after at least one regimen within the master protocol has 40 randomized participants with the opportunity to complete at least 24 weeks of follow-up. Subsequent interims will occur approximately every 12 weeks. At an interim analysis, a regimen can either stop

early for success or for futility. A regimen is eligible to stop early for success or for futility once there are 40 randomized participants within that regimen with the opportunity to complete at least 24 weeks of follow-up. If a regimen has not stopped early for success or futility at an interim analysis, the regimen-specific final analysis will take place after all participants randomized to that regimen (typically 120 active and 40 control) have had the opportunity to complete 24 weeks of follow-up.

### 3.1 Regimen Success

In the default design, early success stopping at interim analyses will not be conducted. Success will be declared for a regimen at the final analysis if the posterior probability that the treatment,  $T=1$ , is superior to placebo group is greater than .979:

$$Pr(\exp(\theta_1) < 1) > .979.$$

The threshold for the final analysis of .979 was chosen to control type I error at 2.5% across all Null simulation scenarios (except for when the analysis regimen has a slower rate of progression on ALSFRS-R compared to all other regimens) without futility stopping.

### 3.2 Regimen Futility

Futility will be declared for a regimen at an interim analysis if the posterior probability that the treatment,  $T=1$ , slows disease progression by at least 10% is less than 5%:

## 5.0

### $Pr(\exp(\theta_1) < .9) < .05$ Goodness of Fit Diagnostics and Sensitivity Analyses of Primary Efficacy Analysis

**5.0** The following goodness-of-fit diagnostics, supportive analyses, and sensitivity analyses will be performed for the primary efficacy analysis and will be conducted on the primary analysis population unless otherwise specified.

#### 1 Goodness-of-fit Diagnostics

The following goodness-of-fit diagnostics will be performed to assess the goodness-of-fit of the proposed Bayesian shared parameter analysis model:

- Residual plots for ALSFRS-R: residuals vs. fitted values of ALSFRS-R at each 4-week visit and for each treatment group.

- Mean and SD in change from baseline of ALSFRS-R at each 4-week visit for each treatment group compared to model fitted expected values from primary analysis model.
- Kaplan-Meier curve for each treatment group compared to fitted exponential curves from primary analysis model.

**5.0** A table summarizing the number and percent of completed or expected ALSFRS-R assessments that do not contribute to the functional component of the primary analysis separated by visit and categorized by a) missing due to death or death equivalent, and b) missing for other reasons.*2 Key Supportive Joint Rank Analysis of ALSFRS-R and Mortality*

**5.0** As a key supportive analysis, a joint rank test of survival and function will be performed. The analysis will rank all participants based first on survival time and second, if they survived, the change in ALSFRS-R from 0 to 24-weeks as described in Berry et al. 2013 (Berry, 2013). Given that participants may have different lengths of follow-up, comparisons will be made between any two participants based on the shortest duration of follow-up between the two. For example, if participant 1 has not experienced mortality and has ALSFRS-R values through 12 weeks and participant 2 has not experienced mortality and has ALSFRS-R values through 24 weeks, the comparison of the change from baseline at 12 weeks in ALSFRS-R will be made. Additionally, if participant 1 has ALSFRS-R values through 36 weeks and then experienced mortality at 36 weeks and participant 2 has ALSFRS-R values only through 24 weeks, a comparison is made based on ALSFRS-R values through 24 weeks. A linear regression analysis will be performed on the ranks to adjust for important covariates described in Section 2.2.5 using a fixed effect for each covariate as well as a treatment effect. This secondary analysis will be performed on the primary analysis population as well as the sensitivity analysis populations described in Section 4.3.3 *Sensitivity of Analysis Population*

As sensitivity analyses, the Bayesian primary analysis model will be applied to the following analysis populations:

- **Concurrent Shared Control Population:** All participants randomized to the active treatment arm or placebo arm within the analysis regimen and all participants randomized to the placebo arm across other regimens within the master protocol who have been randomized within a 6-month window of when the analysis regimen has been enrolling.
- **Common Mode of Administration Population:** All participants randomized to the active treatment arm or placebo arm within the analysis regimen and all participants randomized to the placebo arm across other regimens within the master protocol that have the same mode of administration.



**5.0 Regimen Only Population:** All participants randomized to the active treatment arm or placebo arm within the analysis regimen only. Because the population only involves one regimen, certain aspects of the analysis model will require modification (e.g. no regimen-specific effects).*4 Sensitivity Analysis for Handling Missing Data*

The Bayesian repeated measures component of ALSFRS-R naturally accommodates differential length of follow-up and utilizes all available longitudinal data for each participant who has not been censored due to mortality. Additionally, we provide the following sensitivity analyses to missing (not due to mortality) ALSFRS-R data within the Bayesian repeated measures component of ALSFRS-R.

- Impute all missing ALSFRS-R based on last observation carried forward.
- Impute all missing ALSFRS-R as 0 for both treatment and control arms.
- Impute all missing ALSFRS-R as the worst value observed within the combined treatment and control arms for both treatment and control arms.

**5.0** Impute all missing ALSFRS-R as the worst value observed within the respective treatment/control arms.*5 Shared Parameter Model Sensitivity Analyses*

The following analyses will be conducted to test the sensitivity of the results to the primary efficacy model assumptions.

- The functional component of the primary analysis model, the Bayesian repeated measures model of ALSFRS-R, will be updated to allow for different rates of progression within each 4-week interval ( $k$ ) to assess the sensitivity of the results to the linearity assumption in the primary efficacy analysis. Let  $Y_{i,j,k}$  denote the ALSFRS-R at visit  $j$  for participant  $i$ . The ALSFRS-R component is then defined as:

$$Y_{i,j} = (\alpha_{r(i),t(i)} + \gamma_i) - \sum_{k=1:6} \mu_{\beta,k} * \exp(\beta_i + \theta_{t(i)} + \delta X_i + \rho_{d(i)} + \eta_{r(i)}) * \left( \frac{E_{i,j,k} * 12}{52} \right) + \epsilon_{i,j};$$

$$\epsilon_{ij} \sim N(0, \sigma_{r(i),t(i)}^2).$$

Where  $E_{i,j,k}$  is the number of weeks of exposure for subject  $i$  in visit  $j$  within each 4-week interval  $k=1,2,...,6$ ;  $\mu_{\beta,k}$  is the overall slope for time interval  $k$ ; and  $\beta_i$  is a common subject-level effect to the slope over time periods. The following prior distributions are placed on the overall mean slope per time period and the subject-level effect to the slope:

$$\begin{aligned} \mu_{\beta,k} &\sim N(1, 1^2); k = 1:6; \\ \beta_i &\sim N(0, \sigma_{\beta,t(i)}^2); i = 1, \dots, N; \\ \sigma_{\beta,T} &\sim Unif(0, 2); T = 0, 1. \end{aligned}$$

Prior distributions for all other parameters remain the same as the primary analysis.

- The shared parameter primary analysis model will be updated to allow for different treatment effects across time (4-week intervals) for the functional component and a different treatment effect between function and mortality.
- The shared parameter primary analysis model will be updated to allow for different treatment effects between function and mortality.
- The shared parameter primary analysis model will be updated to explore a range of degrees of fixed borrowing of controls across regimens. In particular, the variance of the regimen-specific random slope effects (Section 2.2.3) will be fixed across a set of values ranging from  $0.001^2$  (full borrowing) to  $10^2$  (no borrowing):

$$\eta_r \sim N(0, \sigma_\eta^2); r = 2, \dots, N_r;$$

$$\sigma_\eta^2 = .001^2, .01^2, .05^2, .1^2, .5^2, 1^2, 5^2, 10^2.$$

## 5.0 Summaries of the estimated treatment effect and estimated differences across regimens within each sensitivity analysis will be provided to assess the robustness of the primary analysis results to the degree of borrowing within the shared control. .6

### *Sensitivity of Survival Endpoint*

As a sensitivity analysis, the Bayesian primary analysis model will be applied to the primary analysis population with the composite survival endpoint of death or death-equivalent replaced by a survival endpoint of death alone.

## 5.0 Clinical Trial Simulations

Clinical trial simulation is used to provide simulated example trials and to quantify operating characteristics for each regimen. In particular, virtual patient outcomes are created under different assumptions for key design parameters including placebo rates of progression for ALSFRS-R, measurement error for ALSFRS-R, mortality rates, treatment effects on ALSFRS-R, treatment effects on mortality, dropout rates and accrual rates. For each set of simulation assumptions (i.e., a scenario), many trials are simulated and virtually executed, including all interim analyses for early success and early futility. Trial operating characteristics are summarized across all simulated trials for each scenario.

### 5.1 Virtual Patient ALSFRS-R Simulation

Participant-level ALSFRS-R data at each timepoint is simulated using placebo data from the Pooled Resource Open-Access ALS Clinical Trials Database (PRO-ACT) with a bootstrap approach. In particular, a simulation database is created from the PRO-ACT database using each placebo participant with at least 3 ALSFRS-R measurements over the course of at least 24 weeks. For each of these participants, an intercept, slope and residual standard-error is calculated from a participant-specific linear regression of the measured ALSFRS-R within the PRO-ACT database as a function of the month of each measured visit. These linear regression values as well as baseline covariates are recorded for each participant within the simulation database.

Within the current simulations, a subset of the PRO-ACT simulation database is sampled based on those participants that have a time since onset less than 3 years and have a slow vital capacity or forced vital capacity (SVC / FVC) greater than 50% (N=2175). Within this subset, the average subject-specific slope (ALSFRS-R decline per month) is -1.03 with a SD of 0.76. The average residual standard error is 1.77.

To create a dataset of participant-level ALSFRS-R scores at each 4-week visit for each simulated clinical trial from the PRO-ACT simulation database, the following steps are taken:

- Choose a participant,  $i$ , from the PRO-ACT simulation database and record values of the participant-specific intercept ( $\gamma_i$ ), slope ( $\beta_i$ ), residual standard error ( $\sigma_i^2$ ), and baseline covariates.
- Simulate participant-specific ALSFRS-R at 0, 4, 8, 12, 16, 20 and 24 weeks ( $j$ ) from:

$$Y_{i,j} \sim N(\gamma_i + (\beta_i + \theta_{t(i)}) * \frac{j*12}{52}, \sigma_i^2);$$

assuming  $\theta_0 = 0$  for placebo and  $\theta_1$  is found such that  $(\hat{\beta} + \theta_1)/\hat{\beta} = 1 - \text{assumed \% slowing of treatment}$ ; where  $\hat{\beta}$  is the average slope across all subjects.

The treatment effect within the simulations is introduced as an additive effect to avoid introducing differences in the variance of the participant-specific rates of progression between the treatment and control arms. This is to provide a conservative estimate of

power as the multiplicative effect of treatment on the rate of progression decreases the variance of participant-level slopes in the treatment arm as the treatment becomes more effective.

## 5.2 Base Simulation Scenarios and Sensitivity Scenarios

All trial simulations assume there are 3 regimens within the platform; all are enrolling within the same population and randomizing concurrently (as defined in Section 2.1). Treatment effect assumptions will vary across the first regimen and operating characteristics will be reported for this regimen. The other two regimens will be assumed to be Null (no benefit on ALSFRS-R or mortality).

The trial simulations are performed under a base scenario (set of simulation assumptions) as well as numerous sensitivity scenarios. In particular, a simulation scenario is defined by 6 different parameters (Table 5.2.1). This set of 6 parameters characterizes various nuisance parameters, but also key assumptions on the endpoints such as their variability and the behavior of the control arm and treatment effect. We define a “base case” set of values for these 6 parameters as well as two sensitivity analysis values. The table below lists each of these 6 parameters along with the base case value and the two sensitivity analysis values.

Table 5.2.2 describes treatment effect assumptions. We simulate the base case of parameter values from Table 5.2.1 under all treatment effect scenarios in Table 5.2.2 and consider this as the primary set of operating characteristics for this trial (6 scenarios). Then, for each of 6 parameters that define a scenario, we vary one sensitivity analysis value at a time from the base case assumptions. For the first 5 parameter values we simulate under the null treatment effect and the common treatment effect with 30% slowing and a HR of .7 (10 sensitivity analyses across 2 treatment effect assumptions). For parameter 6 (mortality rate) we simulate under 4 treatment effect scenarios (Null, Common Effect with HR=.7 and Slowing=30%, No Mortality Benefit, and Worse Mortality). All sensitivity scenarios will be simulated with and without futility stopping.

| <b>Table 5.2.1: Simulation Scenario Parameter Assumptions</b> |                  |                                          |                                          |
|---------------------------------------------------------------|------------------|------------------------------------------|------------------------------------------|
| <b>Parameter</b>                                              | <b>Base Case</b> | <b>Sensitivity Analysis Values</b>       |                                          |
|                                                               |                  | <b>a</b>                                 | <b>b</b>                                 |
| 1. Accrual rate (pts/mo)                                      | 40               | 20                                       | 60                                       |
| 2. Non-mortality dropout rate per month                       | 2%               | 4%                                       | 6%                                       |
| 3. Timing of Start of Enrollment of Other Regimens            | 1 Month apart    | 3 Months apart                           | 6 Months apart                           |
| 4. ALSFRS-R Slope Distribution All Regimens                   | Same as PRO-ACT  | 10% Slower Progression and less variable | 10% Faster Progression and more variable |
| 5. ALSFRS-R Measurement Error All Regimens                    | Same as PRO-ACT  | 10% less                                 | 10% more                                 |

|                                                               |      |                                            |                                            |
|---------------------------------------------------------------|------|--------------------------------------------|--------------------------------------------|
| 6. ALSFRS-R Differences in Slope Distribution Across Regimens | None | 10% Slower Progression of Analysis Regimen | 10% Faster Progression of Analysis Regimen |
| 7. ALSFRS-R Differences in Measurement Error Across Regimens  | None | 10% less for Analysis Regimen              | 10% more for Analysis Regimen              |
| 8. 24-Week Mortality Rate                                     | 5%   | 10%                                        | 20%                                        |

**Table 5.2.2: Treatment Effect Assumptions Mortality and Function**

| Scenario                              | ALSFRS-R Slowing (1-DRR) | Mortality HR |
|---------------------------------------|--------------------------|--------------|
| Null                                  | 0%                       | 1.0          |
| Common Mortality and Function Benefit | 25%                      | .75          |
|                                       | 30%                      | .70          |
|                                       | 35%                      | .65          |
| No Mortality Benefit                  | 30%                      | 1.0          |
| Worse Mortality                       | 30%                      | 1.3          |

## 6.0 Example Trials and Operating Characteristics

For each scenario and treatment effect we simulate 10000 clinical trials. We present example simulated trials as well as overall average operating characteristics across all simulated trials. For each simulation scenario and treatment effect, we report the following regimen-specific operating characteristics for the first regimen:

- Mean duration (months)
- Mean sample size randomized to the regimen
- Probability of early success, total probability of success, and cumulative probability of stopping for success as a function of trial duration.
- Probability of early futility and cumulative probability of stopping for futility as a function of trial duration.
- Average estimate of the treatment effect and estimate of the treatment effect as a function of total trial duration.

### 6.1 Operating Characteristics Base Scenarios

Table 6.1.1 shows the overall operating characteristics for a single regimen under the master protocol recommended design for the Base simulation scenarios with futility stopping. Additionally, Table 6.1.2 summarizes the observed treatment effect estimates in each simulated trial as a function of the trial duration and outcome.

Under the null scenario there is an overall one-sided Type I error rate of 2.4% for the regimen. The average duration of the regimen is 14 months. There is a 28.0% chance that the regimen will be stopped early for futility.

The master protocol recommended design has 61%, 77%, and 88% power to detect a slowing in the rate of disease progression of 25%, 30% and 35% respectively when mortality and function have a common treatment effect. When there is no benefit in mortality and a 30% slowing in ALSFRS-R progression the power drops from 77% to 72%. When there is a negative effect on mortality and a 30% slowing in ALSFRS-R progression, the power drops further to 68%. Finally, there is less than 1% chance in these scenarios that the regimen will be stopped early for futility.

Given the expected accrual rate, interim analyses will begin approximately 7 months into the trial. Therefore, the final analysis for a regimen is expected to occur around 16 months after that regimen is initiated. Table 6.1.2 summarizes the observed treatment effect estimates that would lead to early stopping for futility or final success. If a regimen stops at the first interim analysis for futility, the average treatment effect estimate (DRR) is 1.35. The smallest observed DRR stopping for futility at this look across all simulated trials was 1.13, corresponding to a 13-35% increase in the rate of disease progression for treatment relative to placebo. At later analyses, 10-13 months from the start of the regimen, the average treatment effect estimate (DRR) observed among regimens that stopped for futility was 1.16-1.21 and the smallest observed DRR across all simulated trials observed to stop



for futility at these looks was 1.05-1.08, corresponding to a 5-8% increase in the rate of disease progression for treatment relative to placebo. Regimens never stopped early for futility if observed treatment effect was less than 1 (signaling a slowing in the rate of progression) and only stopped for futility if there was at least a 5% estimated increase in the rate of progression under treatment.

| Table 6.1.1: Base Scenarios Operating Characteristics with Early stopping for Futility |             |                              |                  |                         |                          |
|----------------------------------------------------------------------------------------|-------------|------------------------------|------------------|-------------------------|--------------------------|
| Scenario<br>(% Slowing<br>ALSFRS-R)                                                    | HR<br>Mort. | Mean<br>Duration<br>(Months) | Prob.<br>Success | Prob.<br>Early Futility | Mean<br>Estimated<br>DRR |
| 0%                                                                                     | 1           | 14                           | 0.024            | 0.28                    | 1.05                     |
| 25%                                                                                    | .75         | 15                           | 0.61             | 0.01                    | 0.75                     |
| 30%                                                                                    | .7          | 15                           | 0.77             | 0.00                    | 0.69                     |
| 35%                                                                                    | .65         | 15                           | 0.88             | 0.00                    | 0.64                     |
| 30%                                                                                    | 1           | 15                           | 0.72             | 0.00                    | 0.71                     |
| 30%                                                                                    | 1.3         | 15                           | 0.68             | 0.01                    | 0.72                     |

| Table 6.1.2: Base Scenarios Observed Treatment Effect (DRR = 1 - % Slowing) Estimate per Trial Duration and Outcome |                                          |                |      |         |      |
|---------------------------------------------------------------------------------------------------------------------|------------------------------------------|----------------|------|---------|------|
| Interim<br>/Final                                                                                                   | Approx.<br>Trial<br>Duration<br>(Months) | Early Futility |      | Success |      |
|                                                                                                                     |                                          | Mean           | Min  | Mean    | Max  |
| 1                                                                                                                   | 7                                        | 1.35           | 1.13 | -       | -    |
| 2                                                                                                                   | 10                                       | 1.21           | 1.08 | -       | -    |
| 3                                                                                                                   | 13                                       | 1.16           | 1.05 | -       | -    |
| Final                                                                                                               | 16                                       | -              | -    | 0.65    | 0.82 |

## 6.2 Operating Characteristics and Example Trials Integration of Mortality and Function

This section provides sensitivity of the overall operating characteristics to the assumptions of an overall 5% 6-month mortality rate as well as the assumption that mortality and function have a common treatment effect. All operating characteristics in this section assume no futility stopping. Additionally, we show results from two example trials to demonstrate how the proposed Bayesian shared parameter primary analysis accounts for mortality.

Table 6.2.1 provides Type I error, power and DRR estimates for the base scenario (5% 6-month mortality) as well as scenarios when the mortality rate is increased (10-20% 6-month mortality). Power and average DRR estimates are provided for treatment effect scenarios where the true DRR for mortality and function is the same and equal to .70 and when the true DRR for mortality and function is different (null or negative effect on mortality).

Under the null scenario when the true DRR equals 1.0 for function and true HR=1.0 for mortality, the type I error and the estimated DRR are not sensitive to the assumption of the percentage of 6-month mortality.

Under scenarios when the treatment effect for mortality is null or worse (true HR=1.0 or HR=1.3) and there is a 30% slowing in ALSFRS-R (true DRR=0.70), the power and estimated DRR decrease as the rate of mortality increases. This is due to the overall treatment effect being more heavily weighted by the mortality effect. The average estimated DRR equals 0.69 when the true DRR and HR parameters equal 0.70. Under a 20% 6-month mortality rate and a true DRR equal to 0.70, the average estimated DRR equals 0.75 and 0.81 when the true mortality HR equals 1.0 and 1.3, respectively, thereby decreasing the estimated functional treatment effect by 6-12% (absolute) compared to the scenario with both true DRR and true HR equal 0.70.

| <b>Table 6.2.1: Base and Sensitivity Scenarios on Combining Mortality and Function; No Early Futility Stopping</b> |                          |                      |                           |                      |                        |                      |                        |                      |
|--------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------|---------------------------|----------------------|------------------------|----------------------|------------------------|----------------------|
| <b>Scenario</b>                                                                                                    | <b>DRR ALSFRS-R= 1.0</b> |                      | <b>DRR ALSFRS-R = .70</b> |                      |                        |                      |                        |                      |
|                                                                                                                    | <b>DRR Mort. = 1.0</b>   |                      | <b>DRR Mort. = .70</b>    |                      | <b>DRR Mort. = 1.0</b> |                      | <b>DRR Mort. = 1.3</b> |                      |
|                                                                                                                    | <b>Type I Error</b>      | <b>Mean Est. DRR</b> | <b>Power</b>              | <b>Mean Est. DRR</b> | <b>Power</b>           | <b>Mean Est. DRR</b> | <b>Power</b>           | <b>Mean Est. DRR</b> |
| Base (5% Mort. Rate)                                                                                               | 0.024                    | 1.01                 | 0.77                      | 0.69                 | 0.72                   | 0.71                 | 0.68                   | 0.72                 |
| 10% Mort. Rate                                                                                                     | 0.023                    | 1.01                 | 0.77                      | 0.69                 | 0.67                   | 0.72                 | 0.57                   | 0.75                 |
| 20% Mort. Rate                                                                                                     | 0.024                    | 1.01                 | 0.77                      | 0.69                 | 0.56                   | 0.75                 | 0.35                   | 0.81                 |

Additionally, to demonstrate that our model properly accounts for mortality and to show the sensitivity of the model to treatment-dependent differences in mortality, two simulated trials have been explored (Tables 6.2.2 and 6.2.3 and Figures 6.2.1-6.2.2). Within each simulated trial, we fix the ALSFRS-R data for each participant as well as the mortality data of participants in the placebo arm. We then let the mortality data on the active treatment arm vary to explore how the key supportive joint rank analysis (CAFS) and the proposed Bayesian shared-parameter primary analysis respond to an increasing number of deaths on the active treatment arm. The point at which inference about benefit from treatment switches from significant to non-significant, the tipping point, reveals how sensitive a given analysis is to an excess of deaths in the active arm. An analysis with a smaller tipping point is more sensitive and would be less likely to support a claim of benefit from the active treatment when the active treatment slows functional progression does not also reduce mortality rates.

The base simulations assume a substantial slowing in rate of functional decline of 30%. Within these example trials, as more participants on the active treatment arm experience mortality, the estimated disease rate ratio (DRR), the common treatment effect estimate between mortality and function, increases towards 1 (no effect of treatment) and the posterior probability that the treatment slows the rate of progression ( $DRR < 1$ ) decreases. This is similar to the change seen in the CAFS joint rank analysis: as mortality in the active treatment arm increases, its p-value becomes less significant (see details below). The two example trials differ in the overall mortality rate (Example 1 has a higher rate than Example 2). Comparison of the two example trials shows that the sensitivity of the results to the mortality hazard ratio depends on the overall mortality rate (the mortality hazard ratio is smaller at the tipping point for Example 1 than for Example 2). These examples demonstrate that the proposed Bayesian shared-parameter model is more sensitive to discrepancies in the effect of treatment on mortality vs. the effect of treatment on function relative to the CAFS joint rank analysis.

For each example, we tabulate the following parameters across a range of mortality rates in the active treatment arm (Tables 6.2.2 and 6.2.3): the number of deaths on active treatment and shared placebo, the posterior DRR parameter estimates from the shared-parameter analysis vs. the sensitivity analysis that allows different DRR treatment effects between function and mortality, CAFS one-sided p-value, and posterior probability that the DRR is less than 1 from the shared parameter model. For Example 1, we present several graphs from specific representative trial simulations (the highlighted rows from Table 6.2.2; Plots 6.2.1-6.2.2). The top left plot shows estimates of change from baseline in ALSFRS-R through 24-weeks. The results include fitted values for the active treatment and the shared placebo from the Bayesian shared-parameter model (solid lines), fitted values for the active treatment and the shared placebo from the sensitivity analysis with different treatment effects on mortality and function (dotted lines), and raw mean values of the change from baseline at each visit (circles). The top right plot is a residual plot of fitted ALSFRS-R value (x-axis) at each visit vs. the residual ALSFRS-R value (y-axis) from the Bayesian shared-parameter model. The color of the circle represents active treatment vs. shared placebo. The size of the circle represents visit time (24-week visits are larger circles than 4-week visits). The bottom left plot shows Kaplan-Meier survival curves for mortality for the active

treatment and the shared placebo as well as the fitted exponential curves from the Bayesian shared-parameter model. The bottom right plot shows the DRR parameter estimates from the shared-parameter analysis vs. the sensitivity analysis that allows different DRR treatment effects between function and mortality.

Within Example Trial 1 (Table 6.2.2 and Figures 6.2.1-6.2.2), there are 18 deaths on the placebo arm (15%). Results are shown for 11-29 total deaths on the treatment arm with mortality hazard ratios from the mortality component alone ranging from 0.68 to 1.58. Simulations representative of rows highlighted in Table 6.2.2 are presented in Figures 6.2.1-6.2.2. The treatment effect estimated from the functional component alone for those participants who did not experience mortality range from 0.71-0.73. As the number of deaths increases on the active treatment arm, the posterior estimate for the shared treatment effect between mortality and function from the Bayesian shared-parameter model increases (0.73 to 0.88) and the posterior probability that the DRR is less than one decreases (0.994 to 0.853). Similarly, the CAFS one-sided p-value increases (0.002 to 0.233). For all mortality hazard ratios less than or equal to one, both CAFS and the Bayesian shared-parameter model indicate a significant treatment benefit. For the Bayesian shared-parameter model, the tipping point at which results become non-significant occurs between mortality hazard ratios of 1.04 and 1.10 (17 to 18 deaths in the active treatment arm). Under the CAFS analysis, the tipping point at which results become non-significant occurs between mortality hazard ratios of 1.15 to 1.27 (19 to 21 deaths on the active treatment arm). Under the simulated conditions, the Bayesian shared-parameter model was more sensitive to a discrepancy between the effect of treatment on function and mortality than the CAFS analysis.

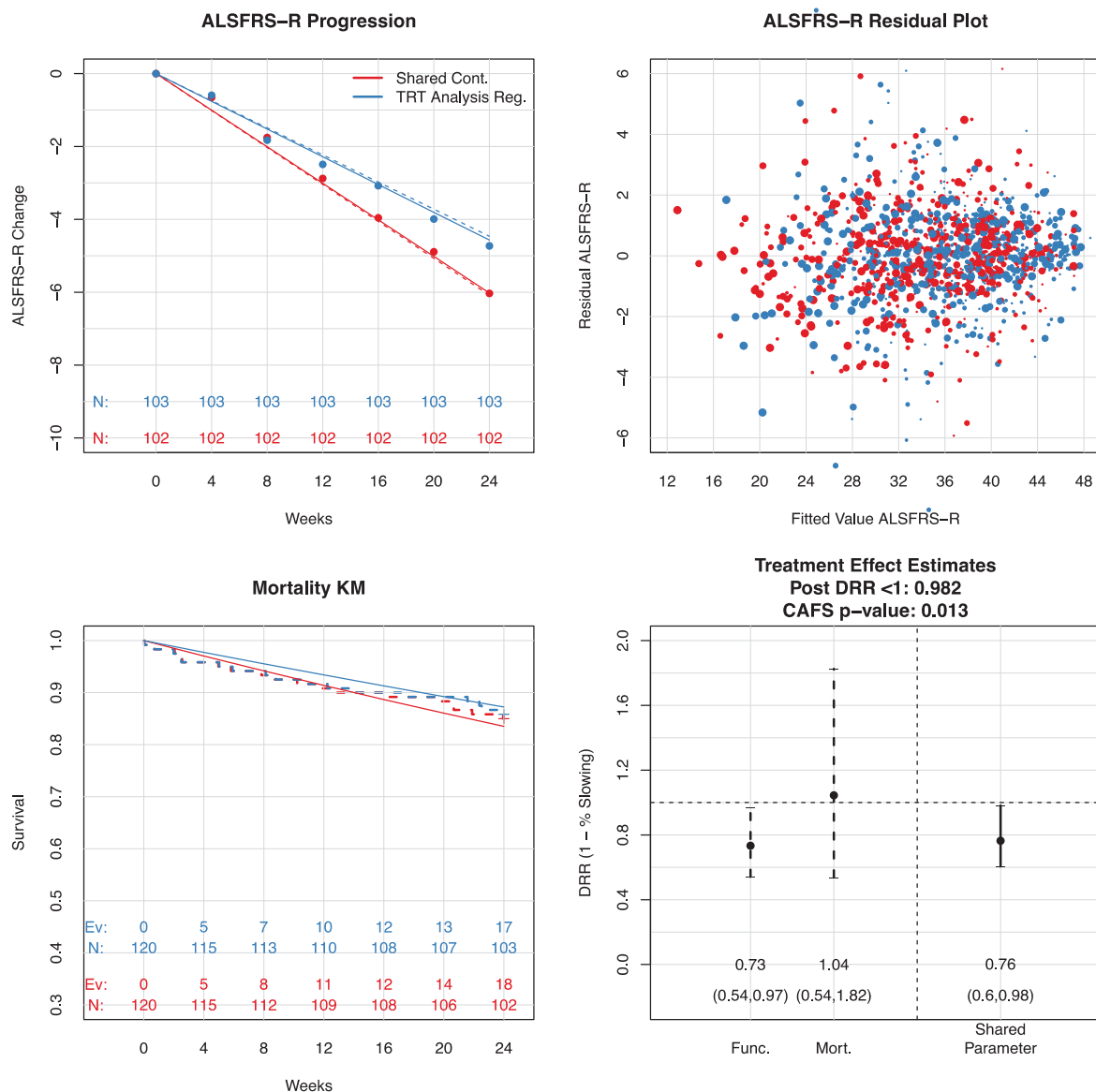
Within Example Trial 2 (Table 6.2.3), there are 11 deaths on the placebo arm (9%). Results are shown for 6-19 total deaths on the treatment arm with mortality hazard ratios from the mortality component alone ranging from 0.66 to 1.57. The treatment effect estimated from the functional component alone for those participants who did not experience mortality ranges from 0.72-0.75. As the number of deaths increases on the active treatment arm, the posterior estimate for the shared treatment effect between mortality and function from the Bayesian shared-parameter model increases (0.71 to 0.85) and the posterior probability that the DRR is less than one decreases (0.998 to 0.917). Similarly, the CAFS one-sided p-value increases (0.001 to 0.085). For all mortality hazard ratios less than or equal to 1.13 both CAFS and the Bayesian shared-parameter model indicate a significant treatment benefit (CAFS p-value less than .025 and Bayesian posterior greater than 0.979). For the Bayesian shared-parameter model, the tipping point at which results become non-significant occurs between mortality hazard ratios of 1.13 to 1.21 (11 to 12 deaths on the active treatment arm). Under the CAFS analysis, the tipping point at which results become non-significant occurs between mortality hazard ratios of 1.27 to 1.33 (13 to 14 deaths on the active treatment arm). As in Example Trial 1, the Bayesian shared-parameter model was more sensitive to a discrepancy between the effect of treatment on function and mortality than the CAFS analysis. We also note that with the lower mortality rate, both the Bayesian shared-parameter analysis and the CAFS analysis were less sensitive to a discrepancy between the effect of treatment on function and mortality.

**Table 6.2.2: Example Trial 1 demonstrating integration of mortality and function in primary analysis. Highlighted rows correspond to Figures 6.2.1 and 6.2.2**

| Deaths PBO | Deaths TRT | Posterior Mean Estimate |              |            | CAFS P-Value | Posterior Probability of DRR <1 |
|------------|------------|-------------------------|--------------|------------|--------------|---------------------------------|
|            |            | Mortality DRR (HR)      | Function DRR | Common DRR |              |                                 |
| 18         | 11         | 0.68                    | 0.73         | 0.73       | 0.002*       | 0.994*                          |
| 18         | 13         | 0.81                    | 0.72         | 0.73       | 0.003*       | 0.991*                          |
| 18         | 14         | 0.87                    | 0.73         | 0.74       | 0.005*       | 0.988*                          |
| 18         | 17         | 1.04                    | 0.73         | 0.76       | 0.013*       | 0.982*                          |
| 18         | 18         | 1.10                    | 0.73         | 0.77       | 0.015*       | 0.975                           |
| 18         | 19         | 1.15                    | 0.73         | 0.78       | 0.022*       | 0.973                           |
| 18         | 21         | 1.27                    | 0.73         | 0.79       | 0.028        | 0.972                           |
| 18         | 23         | 1.36                    | 0.71         | 0.80       | 0.035        | 0.956                           |
| 18         | 24         | 1.41                    | 0.71         | 0.81       | 0.056        | 0.950                           |
| 18         | 26         | 1.48                    | 0.71         | 0.82       | 0.091        | 0.948                           |
| 18         | 27         | 1.52                    | 0.71         | 0.84       | 0.135        | 0.908                           |
| 18         | 29         | 1.58                    | 0.73         | 0.88       | 0.233        | 0.853                           |

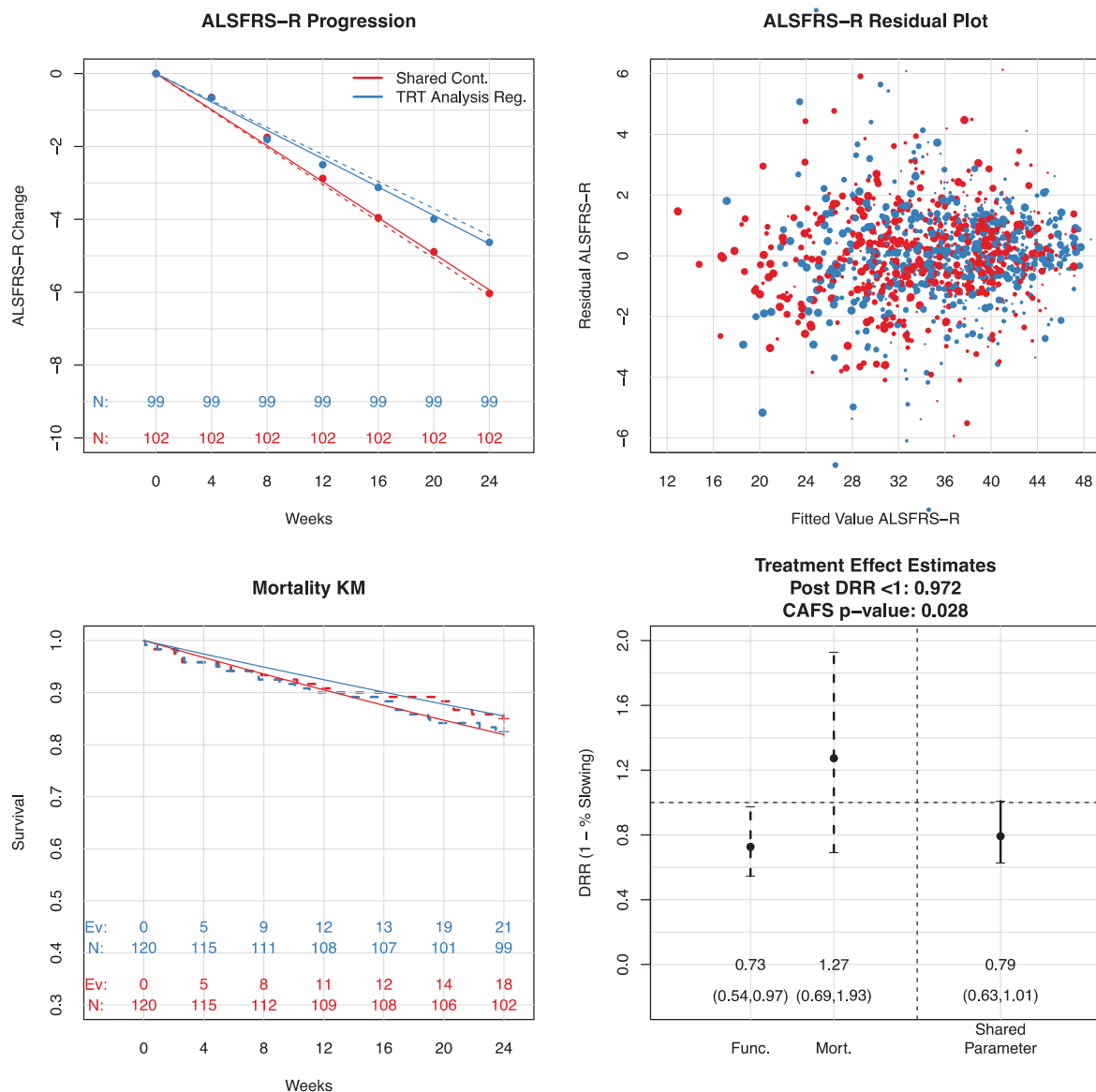
**Table 6.2.3: Example Trial 2 demonstrating integration of mortality and function in primary analysis**

| Deaths PBO | Deaths TRT | Posterior Mean Estimate |              |            | CAFS P-Value | Posterior Probability of DRR <1 |
|------------|------------|-------------------------|--------------|------------|--------------|---------------------------------|
|            |            | Mortality DRR (HR)      | Function DRR | Common DRR |              |                                 |
| 11         | 6          | 0.66                    | 0.72         | 0.71       | 0.001*       | 0.998*                          |
| 11         | 7          | 0.77                    | 0.72         | 0.73       | 0.003*       | 0.993*                          |
| 11         | 8          | 0.87                    | 0.74         | 0.74       | 0.004*       | 0.992*                          |
| 11         | 9          | 0.96                    | 0.74         | 0.75       | 0.006*       | 0.988*                          |
| 11         | 10         | 1.06                    | 0.73         | 0.75       | 0.009*       | 0.982*                          |
| 11         | 11         | 1.13                    | 0.74         | 0.78       | 0.013*       | 0.979*                          |
| 11         | 12         | 1.21                    | 0.74         | 0.78       | 0.014*       | 0.976                           |
| 11         | 13         | 1.27                    | 0.73         | 0.78       | 0.017*       | 0.975                           |
| 11         | 14         | 1.33                    | 0.74         | 0.80       | 0.028        | 0.961                           |
| 11         | 16         | 1.45                    | 0.73         | 0.80       | 0.036        | 0.959                           |
| 11         | 17         | 1.49                    | 0.74         | 0.82       | 0.058        | 0.939                           |
| 11         | 19         | 1.57                    | 0.75         | 0.85       | 0.085        | 0.917                           |



**Figure 6.2.1: Example Trial 1 with 18 Deaths on Placebo and 17 Deaths on Treatment demonstrating integration of mortality and function in primary analysis.**





**Figure 6.2.2: Example Trial 1 with 18 Deaths on Placebo and 21 Deaths on Treatment demonstrating integration of mortality and function in primary analysis.**

### 6.3 Operating Characteristics and Example Trials Differences Across Regimens

This section provides sensitivity of the overall operating characteristics to the assumption that regimens within the shared control have the same ALSFRS-R distributions. In particular, we look at the sensitivity of operating characteristics to differences across the regimens in the rate of progression of ALSFRS-R as well as differences in the residual error of ALSFRS-R. All operating characteristics in this section assume no futility stopping.

Table 6.3 provides Type I error, power and DRR estimates for the base scenario (all regimens have the same ALSFRS-R progression rate and residual error) as well as scenarios when the analysis regimen differs from all other regimens in terms of the ALSFRS-R progression rate and ALSFRS-R residual error. Power and DRR estimates are provided for treatment effect scenarios where the DRR for mortality and function is the same and equal to .70 and when the DRR for mortality and function is different (null or negative effect on mortality).

| <b>Table 6.3: Base and Sensitivity Scenarios on Differences Across Regimens; No Early Futility Stopping; Common Mortality and Function Effect</b> |                 |                  |                         |                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------|-------------------------|------------------|
| Scenario                                                                                                                                          | Null<br>DRR 1.0 |                  | Alternative<br>DRR 0.70 |                  |
|                                                                                                                                                   | Type I<br>Error | Mean Est.<br>DRR | Power                   | Mean Est.<br>DRR |
| Base                                                                                                                                              | 0.024           | 1.01             | 0.77                    | 0.69             |
| ALSFRS-R Slower Progress Analysis Reg.                                                                                                            | 0.060           | 0.96             | 0.85                    | 0.66             |
| ALSFRS-R Faster Progress Analysis Reg.                                                                                                            | 0.010           | 1.05             | 0.67                    | 0.72             |
| ALSFRS-R Lower Resid. Error Analysis Reg.                                                                                                         | 0.024           | 1.01             | 0.79                    | 0.69             |
| ALSFRS-R Higher Resid. Error Analysis Reg.                                                                                                        | 0.023           | 1.01             | 0.75                    | 0.69             |

Under the extreme assumption that participants in the analysis regimen progress at a 10% slower rate of progression on ALSFRS-R than those in all other regimens, there is an increased Type I error of 6% and a small positive bias in the treatment effect estimate of .96 (4% slowing in progression). Under the assumption that participants in the analysis regimen progress at a 10% faster rate of progression on ALSFRS-R than those in all other regimens, the Type I error decreases to 1% with a small negative bias in the treatment effect estimate equal to 1.05 (5% increase in progression). The sensitivity of these results is due to the borrowing across regimens for the shared control.

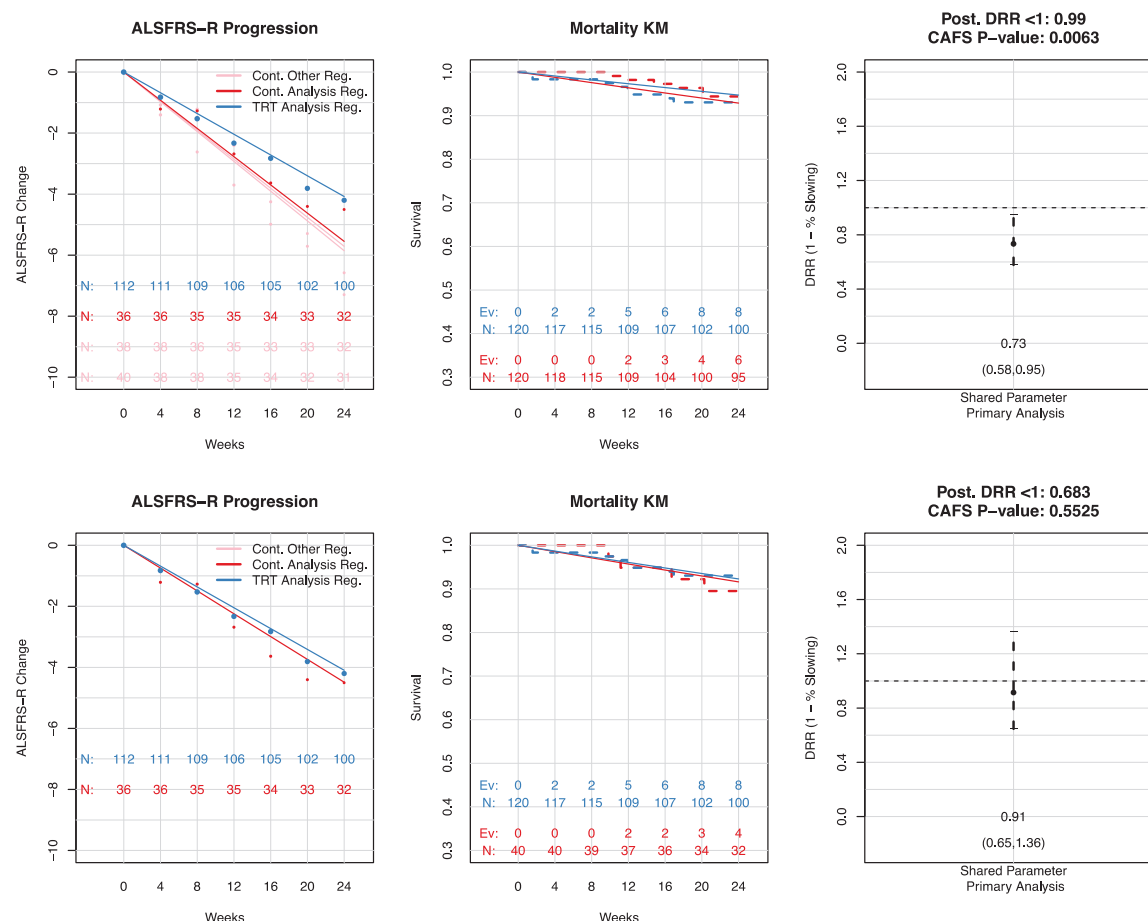
Operating characteristics are not sensitive to differences in the ALSFRS-R residual error for the analysis regimen compared to other regimens other than having an increase or decrease in power when ALSFRS-R under the analysis regimen has more or less residual error.

In addition to the primary analysis, sensitivity analyses will be provided with regards to the shared control (Section 4.2) to assess differences in rates of progression across regimens and how treatment effect estimates differ under the analysis regimen when the treated

group is compared to only the regimen-specific controls and compared to only the controls with the same mode of administration. We provide examples from simulated trials (Figures 6.3.1-6.3.3) to demonstrate how discrepancies between the primary analysis and proposed sensitivity analyses will address the concern of a potential type I error and bias due to sharing of controls that are heterogenous.

For each simulated example trial, the left plots show results from the primary shared-parameter model for the change from baseline in ALSFRS-R through 24 weeks using all shared controls (top) and for the sensitivity analysis using only regimen-specific controls (bottom). The results include fitted values for the active treatment and for each regimen-specific control (blue line for active treatment; red line for matching control from analysis regimen; pink for additional shared controls when included in the analysis) and raw mean values in the change from baseline at each visit (circles). The middle plots show mortality KM plots as well as exponential fitted curves for the primary analysis using all shared controls (top) vs. the sensitivity analysis with only regimen-specific controls (bottom). The right plots show the DRR parameter estimates, i.e., the posterior probability of slowing progression ( $DRR < 1$ ), from the primary analysis method using all shared controls (top) vs. the sensitivity analysis using only regimen-specific controls (bottom). Additionally, we note the one-sided p-value from the key supportive CAFS analysis using all shared controls (top right) compared to only regimen-specific controls (bottom right).

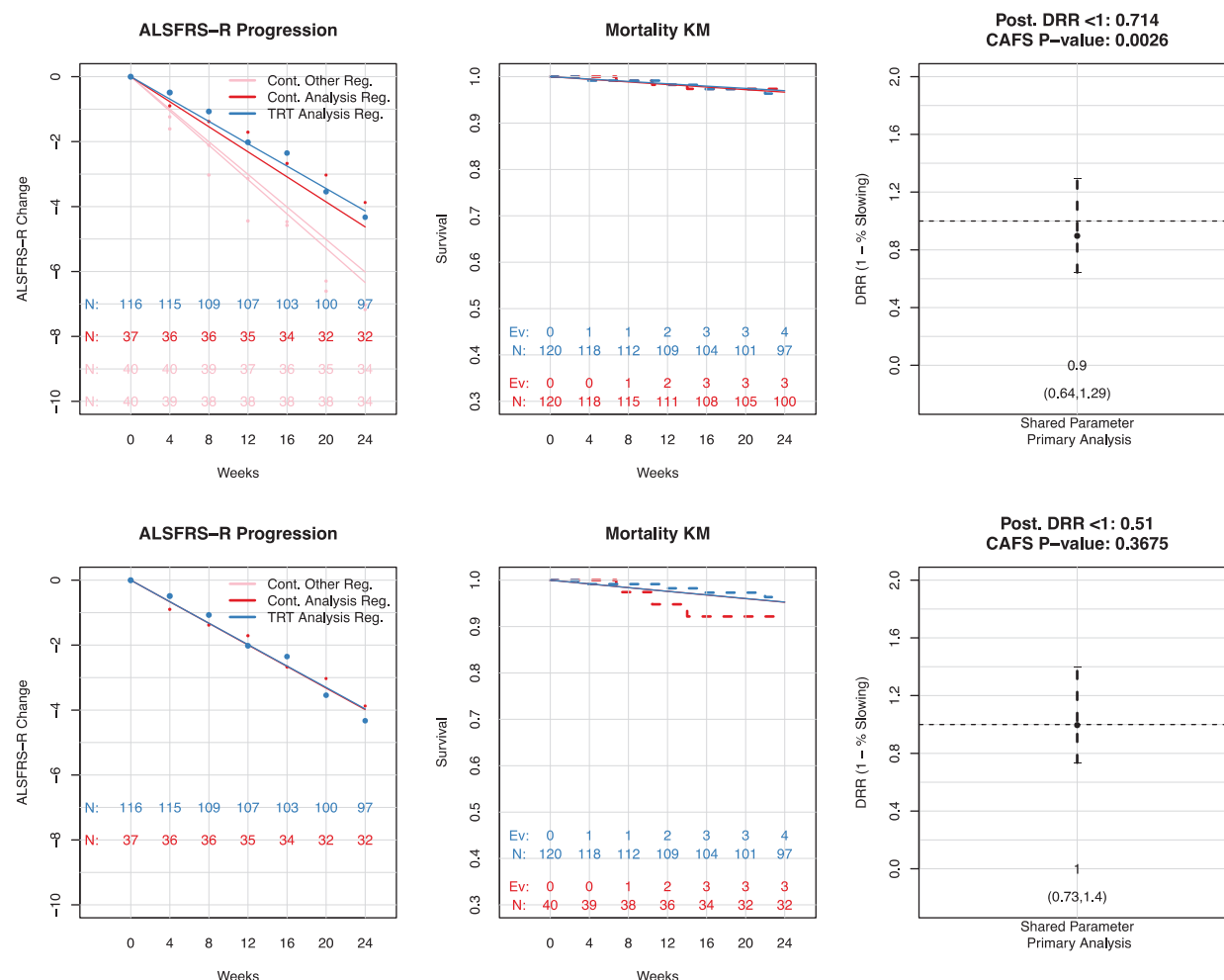
Figure 6.3.1 shows results from a null simulated trial (no treatment effect) where there are differences across regimens in the shared control. In particular, the matched regimen-specific control from the analysis regimen has a slower rate of progression on ALSFRS-R (red circles) than the controls from the other regimens in the shared control (pink circles). The raw mean change from baseline at 24 weeks in the matched regimen-specific controls is approximately -4.5 and the mean change from baseline at 24 weeks in the other shared controls is approximately -6.5 and -7.2. These differences across regimens within the shared control lead to a fitted ALSFRS-R slope for the matched regimen-specific control to be pulled down and estimated to be larger (faster progression) than what is observed in that regimen only (red line falls below red circles). This causes a positive treatment effect bias in the DRR (estimated to be 0.73) under the primary analysis method. The posterior probability that the DRR is less than 1 is 0.990 and the corresponding one-sided p-value from the CAFS is 0.0063. Both analysis methods using shared controls would lead to a type I error. In comparison, the sensitivity analysis using only matched regimen-specific controls leads to a fitted value for the regimen-only controls ALSFRS-R slope that estimates slower decline than what was estimated using all shared controls. This results in a smaller treatment effect estimate of 0.91 under the primary shared-parameter analysis. The posterior probability that the DRR is less than 1 is 0.683 and the corresponding one-sided p-value from the CAFS is 0.5525. The clear discrepancies between the primary analysis and the proposed sensitivity analysis using only regimen-specific controls in the Bayesian shared parameter analysis as well as the CAFS analysis demonstrates that the positive result in the primary analysis may be a type I error due to the sharing of heterogenous controls.



**Figure 6.3.1:** Example Trial 3. Null Treatment Effect ( $DRR = 1.0$ ). Differences across regimens in shared control. Top row shows results from the primary analysis using all shared controls. Bottom row shows results from sensitivity analyses using only matched regimen-specific controls.

Figure 6.3.2 shows results from a null simulated trial (no treatment effect) where there are more substantial differences across regimens in the shared control than in the previous example. The raw mean change from baseline at 24 weeks in the matched regimen-specific controls is approximately -4.0 and the mean change from baseline at 24 weeks in the other shared controls is approximately -7.0 for both. Because the Bayesian primary analysis allows for dynamic sharing of controls (more sharing if they look similar vs. less sharing if they look different) these substantial differences across regimens allows the model to learn the controls are likely very different across regimens and leads to a fitted ALSFRS-R slope for the matched regimen-specific control that is less influenced and pulled down by the controls of the other two regimens than in the previous example (red line falls slightly below red circles). This causes a slight positive treatment effect bias in the DRR (estimated to be 0.90) under the primary analysis method. The posterior probability that the DRR is less than 1 is 0.714 and the Bayesian shared parameter analysis leads to a null result. However, because the CAFS analysis uses all shared controls and the amount of sharing is not dynamic, the results of this analysis are more sensitive to the increased rate of progression in the other two control groups and the corresponding one-sided p-value from the CAFS is 0.0026. Leading to a Type I error in the CAFS analysis but not the Bayesian

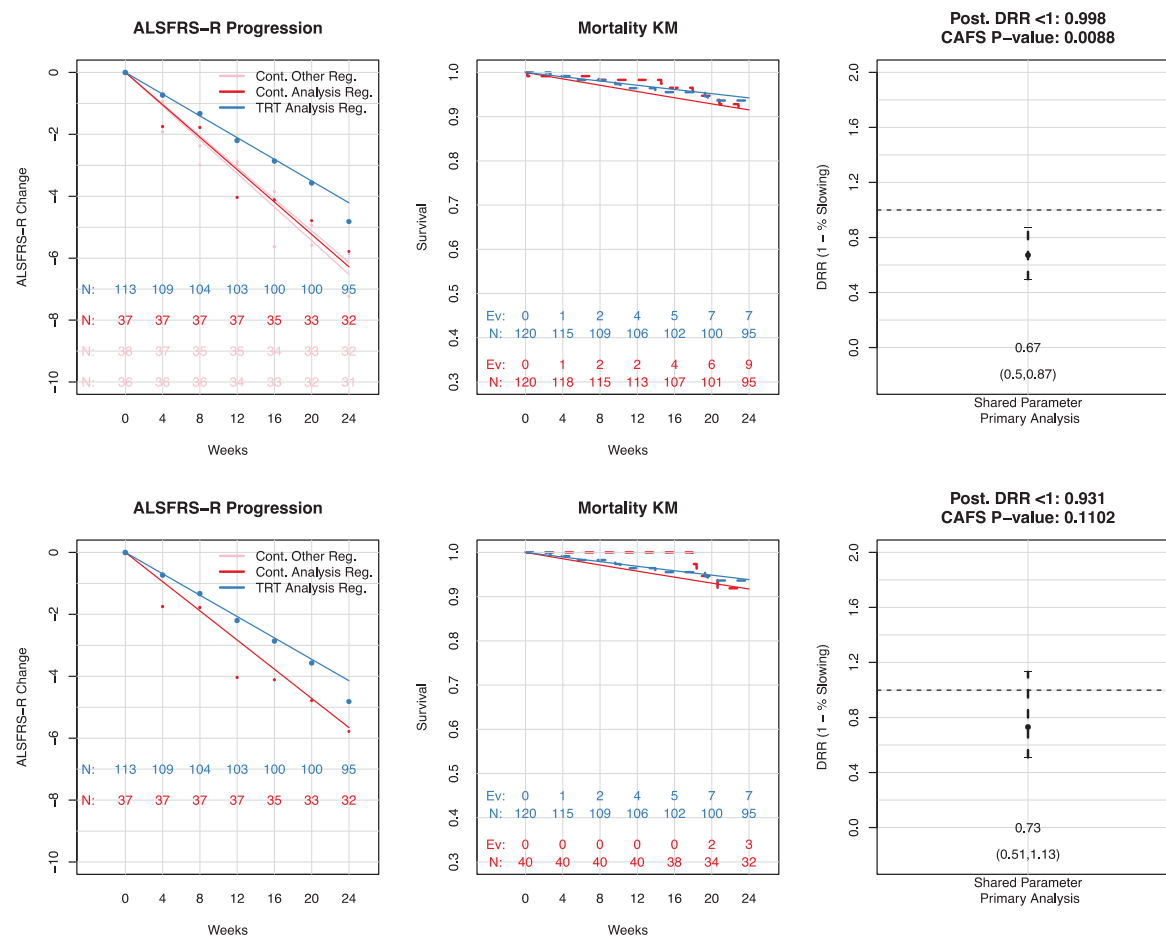
shared parameter analysis. When matched regimen-specific controls are only used, both the Bayesian shared parameter analysis and the CAFS analysis lead to a null result, (treatment effect estimate of 1.0, posterior probability  $DRR < 1$  of 0.51 and CAFS p-value of 0.3675).



**Figure 6.3.2:** Example Trial 4. Null Treatment Effect ( $DRR = 1.00$ ). Differences across regimens in shared control. Top row shows results from the primary analysis using all shared controls. Bottom row shows results from sensitivity analyses using only matched regimen-specific controls.

Figure 6.3.3 shows results from a positive simulated trial (30% slowing in disease progression) where there are no simulated differences across regimens in the shared control. Within this example, the matched regimen-specific control from the analysis regimen has a similar rate of progression on ALSFRS-R (red circles) as the controls from the other regimens in the shared control (pink circles). In particular, the raw mean change from baseline at 24 weeks in the matched regimen-specific controls is approximately -5.9 and the mean change from baseline at 24 weeks in the other shared controls is approximately -5.9 and -7.2. The similarities across regimens within the shared control leads to a fitted ALSFRS-R slope for the matched regimen-specific control that provides a

good fit to the observed data (red line falls within red circles). The treatment effect is estimated to be 0.67, the posterior probability that the DRR is less than 1 is 0.998, and the corresponding one-sided p-value from the CAFS is 0.0088. The sensitivity analysis using only matched regimen-specific controls leads to a similar treatment effect estimate of 0.73, but with more uncertainty. The posterior probability that the DRR is less than 1 is 0.931 and the corresponding one-sided p-value from the CAFS is 0.1102. The consistency in results between the primary analysis using all shared controls as well as the sensitivity analysis using only matched regimen-specific controls provides confidence that the positive primary analysis results are not biased by the shared controls.



**Figure 6.3.3:** Example Trial 5. Positive Treatment Effect (DRR = 0.70). No differences across regimens in shared control. Top row shows results from the primary analysis using all shared controls. Bottom row shows results from sensitivity analyses using only matched regimen-specific controls.

### 6.4 Operating Characteristics Additional Sensitivity Scenarios

This section provides additional sensitivity results of the overall operating characteristics to the assumptions of accrual rates, dropout rates, timing of enrolling additional regimens, and distributional assumptions of ALSFRS-R common to all regimens. All operating characteristics in this section assume no futility stopping.

Table 6.4 shows the overall power (under a DRR of .70 for Mortality and Function) and Type I error for the base scenarios as well as all additional sensitivity scenarios described in Section 5.2 without futility stopping.

Under all Null scenarios, Type I error is controlled at 2.5% or less without futility stopping. Type I error decreases as there is less shared controls in the final analysis (faster accrual and regimens enrolling further apart).

Under all scenarios where the treatment effect is common between ALSFRS-R slowing and mortality (DRR is 0.70) the power of the trial ranges between 56-79%. Power is the lowest when the regimens start six months apart since the first regimen will have less shared control. Mean DRR estimates range from 0.69 – 0.71 (29-31% slowing).

| <b>Table 6.4: Base and Sensitivity Scenarios with No Early Futility Stopping; Common Mortality and Function Effect</b> |                         |                          |                                 |                          |
|------------------------------------------------------------------------------------------------------------------------|-------------------------|--------------------------|---------------------------------|--------------------------|
| <b>Scenario</b>                                                                                                        | <b>Null<br/>DRR 1.0</b> |                          | <b>Alternative<br/>DRR 0.70</b> |                          |
|                                                                                                                        | <b>Type I Error</b>     | <b>Mean Est.<br/>DRR</b> | <b>Power</b>                    | <b>Mean Est.<br/>DRR</b> |
| Base                                                                                                                   | 0.024                   | 1.01                     | 0.77                            | 0.69                     |
| Slower Accrual                                                                                                         | 0.025                   | 1.01                     | 0.77                            | 0.69                     |
| Faster Accrual                                                                                                         | 0.020                   | 1.01                     | 0.76                            | 0.69                     |
| 4% Dropout                                                                                                             | 0.024                   | 1.01                     | 0.74                            | 0.69                     |
| 6% Dropout                                                                                                             | 0.024                   | 1.01                     | 0.71                            | 0.69                     |
| Reg. Start 3 Months Apart                                                                                              | 0.023                   | 1.02                     | 0.67                            | 0.70                     |
| Reg. Start 6 Months Apart                                                                                              | 0.021                   | 1.03                     | 0.56                            | 0.71                     |
| ALSFRS-R Slower Progress All Reg.                                                                                      | 0.023                   | 1.01                     | 0.75                            | 0.69                     |
| ALSFRS-R Faster Progress All Reg.                                                                                      | 0.023                   | 1.01                     | 0.78                            | 0.69                     |
| ALSFRS-R Lower Resid. Error All Reg.                                                                                   | 0.023                   | 1.01                     | 0.79                            | 0.69                     |
| ALSFRS-R Higher Resid. Error All Reg.                                                                                  | 0.023                   | 1.01                     | 0.75                            | 0.69                     |



## 7.0 References

Berry, J. D., et al., 2013. The Combined Assessment of Function and Survival (CAFS): a new endpoint for ALS clinical trials. *Amyotrophic Lateral Sclerosis and Frontotemporal Degeneration*, Volume 14.3, pp. 162-168.

**RGC REGIMEN-SPECIFIC STATISTICAL ANALYSIS PLAN (R-SAP)**

|                                |                                                                                                                                                                                                                          |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Master Protocol</b>         | Platform Trial for the Treatment of Amyotrophic Lateral Sclerosis (ALS): A perpetual multi-center, multi-regimen, clinical trial evaluating the safety and efficacy of investigational products for the treatment of ALS |
| <b>Regimen</b>                 | RGC: CNM-Au8                                                                                                                                                                                                             |
| <b>Regimen Partner</b>         | Clene Nanomedicine, Inc.                                                                                                                                                                                                 |
| <b>Regulatory Sponsor</b>      | Merit E. Cudkowicz, MD                                                                                                                                                                                                   |
| <b>Master Protocol Version</b> | 4.0, 31 Aug 2020                                                                                                                                                                                                         |
| <b>RSA Version</b>             | 5.0, 22 Nov 2021                                                                                                                                                                                                         |
| <b>Master SAP Version</b>      | 1.0, 24 Jun 2020                                                                                                                                                                                                         |
| <b>R-SAP Version</b>           | 1.0, 17 Mar 2022                                                                                                                                                                                                         |

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## SAP REVISION HISTORY

| Version | Date        | Description of Changes |
|---------|-------------|------------------------|
| 1.0     | 17 Mar 2022 | Initial version        |

## ABBREVIATIONS

|           |                                                                         |
|-----------|-------------------------------------------------------------------------|
| ALP       | Alkaline Phosphatase                                                    |
| ALS       | Amyotrophic Lateral Sclerosis                                           |
| ALSAQ-40  | Amyotrophic Lateral Sclerosis Assessment Questionnaire, 40-item version |
| ALSFRS-R  | Amyotrophic Lateral Sclerosis Functional Rating Scale, Revised          |
| ALT       | Alanine Transaminase                                                    |
| AST       | Aspartate Transaminase                                                  |
| ATC       | WHODrug Anatomical, Therapeutic, and Chemical class                     |
| ATS       | American Thoracic Society                                               |
| BLQ       | Below the Limit of Quantitation                                         |
| BMI       | Body Mass Index                                                         |
| C-SSRS    | Columbia Suicide Severity Rating Scale                                  |
| CAFS      | Combined Assessment of Function and Survival                            |
| CBC       | Complete Blood Count                                                    |
| CKD       | Chronic Kidney Disease                                                  |
| COVID-19  | Coronavirus Disease 2019                                                |
| CNS-BFS   | Center for Neurologic Study Bulbar Function Scale                       |
| CSF       | Cerebrospinal Fluid                                                     |
| CSR       | Clinical Study Report                                                   |
| CTCAE     | Common Terminology Criteria for Adverse Events                          |
| delta-FRS | Pre-baseline Slope in ALSFRS-R                                          |
| DAP       | Data Analysis Plan                                                      |
| DILI      | Drug-induced Liver Injury                                               |
| DNA       | Deoxyribonucleic Acid                                                   |
| DRR       | Disease Rate Ratio                                                      |
| ECC       | Efficacy Concurrent Control                                             |
| ECG       | Electrocardiography or Electrocardiogram                                |
| eGFR      | Estimated Glomerular Filtration Rate                                    |
| EPP       | Efficacy Per-protocol                                                   |
| ERO       | Efficacy Regimen-only                                                   |
| ELISA     | Enzyme-linked Immunosorbent Assay                                       |
| FAS       | Full Analysis Set                                                       |

## ABBREVIATIONS (continued)

|        |                                                                                    |
|--------|------------------------------------------------------------------------------------|
| FVC    | Forced Vital Capacity                                                              |
| GLI    | Global Lung Initiative                                                             |
| hCG    | Human Chorionic Gonadotropin                                                       |
| HHD    | Hand-held Dynamometry                                                              |
| HLT    | MedDRA High Level Term                                                             |
| ICF    | Informed Consent Form                                                              |
| ITT    | Intention-to-treat Principle                                                       |
| M-SAP  | Master Statistical Analysis Plan                                                   |
| MDRD   | Modification of Diet in Renal Disease                                              |
| MedDRA | Medical Dictionary for Regulatory Activities                                       |
| MP     | Master Protocol                                                                    |
| MPRDR  | ALS Master Protocol Recommended Statistical Analysis, Design and Simulation Report |
| NCI    | National Cancer Institute                                                          |
| NEALS  | Northeast ALS                                                                      |
| NfL    | Neurofilament Light Chain                                                          |
| NIV    | Noninvasive Ventilation                                                            |
| OLE    | Open-label Extension                                                               |
| PAV    | Permanent Assisted Ventilation                                                     |
| PD     | Pharmacodynamics                                                                   |
| PK     | Pharmacokinetics                                                                   |
| PT     | MedDRA Preferred Term                                                              |
| RBC    | Red Blood Cell                                                                     |
| RDW    | RBC Distribution Width                                                             |
| RGC    | Regimen C (CNM-Au8, 30 mg/d and 60 mg/d)                                           |
| RSA    | Regimen-specific Appendix                                                          |
| R-SAP  | Regimen-specific Statistical Analysis Plan                                         |
| SAE    | Serious Adverse Event                                                              |
| SAP    | Statistical Analysis Plan                                                          |
| SGOT   | Serum Glutamic Oxaloacetic Transaminase                                            |
| SGPT   | Serum Glutamic Pyruvic Transaminase                                                |

## ABBREVIATIONS (continued)

|         |                                                    |
|---------|----------------------------------------------------|
| SoA     | Schedule of Activities                             |
| SOC     | MedDRA System Organ Class                          |
| SRO     | Safety Regimen-only                                |
| STF     | Safety and Tolerability Full                       |
| STN     | Safety and Tolerability Narrow                     |
| SVC     | Slow Vital Capacity                                |
| TBL     | Total Bilirubin                                    |
| TEAE    | Treatment-emergent Adverse Event                   |
| TSH     | Thyroid Stimulating Hormone                        |
| ULN     | Upper Limit of Normal                              |
| WBC     | White Blood Cell                                   |
| WHODrug | World Health Organization Drug Dictionary Enhanced |



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## 1. Governing Documents

This Regimen-specific Statistical Analysis Plan (R-SAP) for the CNM-Au8 regimen (RGC) specifies any modification from the default outcome measures, analysis samples, and planned analyses for the placebo-controlled period of the HEALEY ALS Platform Trial as specified in the Master SAP (M-SAP). The M-SAP and this R-SAP supplement the Master Protocol, the "ALS Master Protocol Recommended Statistical Analysis, Design and Simulation Report" (Appendix 1 to the Master Protocol), and the RGC Regimen-specific Appendix (RGC RSA). Please refer to the Master Protocol and the RGC RSA for details on the rationale for the study design, eligibility criteria, conduct of the trial, clinical assessments and schedule of assessments, definitions and reporting of adverse events, data management conventions, and regulatory oversight and compliance procedures. The "ALS Master Protocol Recommended Statistical Analysis, Design and Simulation Report" (MPRDR) and any regimen-specific deviations described in the RGC RSA, and this R-SAP are authoritative in defining the primary and interim analyses. In case of discrepancies between the RGC RSA and this R-SAP concerning use of shared placebos, this R-SAP is authoritative. In case of discrepancies between either SAP and the Master Protocol and the RGC RSA concerning matters of analysis other than the primary and interim analyses and use of shared placebos, the M-SAP and this R-SAP are authoritative. In case of discrepancies between the M-SAP and this R-SAP, this R-SAP is authoritative. On all matters not related to analysis, the Master Protocol and the RGC RSA are authoritative. The following table describes relationships among the relevant documents in adjudicating possible discrepancies with higher numbers indicating greater authority.

| Issues potentially requiring adjudication                                                                | Master Protocol | RGA RSA | MPRDR | M-SAP | RGA R-SAP |
|----------------------------------------------------------------------------------------------------------|-----------------|---------|-------|-------|-----------|
| Use of shared placebos                                                                                   | 1               | 4       | 2     | 3     | 5         |
| Primary and interim analysis specifications not related to use of shared placebo                         | 1               | 5       | 4     | 2     | 3         |
| Statistical analysis specifications not related to use of shared placebo or primary and interim analyses | 1               | 3       | 2     | 4     | 5         |
| All matters not related to statistical analysis                                                          | 4               | 5       | 1     | 2     | 3         |

## 2. Study Design

### 2.1 Overview

The HEALEY ALS Platform Trial is a perpetual multi-center, multi-regimen clinical trial evaluating the safety and efficacy of investigational products for the treatment of ALS. RGC evaluates the safety and efficacy of CNM-Au8 administered orally at dosages of 30 mg/d and 60 mg/d vs. placebo. The RGC RSA describes the nature of the intervention and its mechanism of action, the mode and frequency of administration, additional eligibility criteria beyond those specified in the Master Protocol, additional enrollment procedures, and additions and modifications of safety and efficacy assessments relative to those outlined in the Master Protocol.

## 2.2 Study Objectives

Primary Efficacy Objective:

- To evaluate the efficacy of CNM-Au8 as compared to placebo on ALS disease progression.

Secondary Efficacy Objectives:

- To evaluate the effect of CNM-Au8 on selected secondary measures of disease progression, including survival.

Safety Objectives:

- To evaluate the safety of CNM-Au8 for ALS.

Exploratory Efficacy Objectives:

- To evaluate the effect of CNM-Au8 on selected biomarkers and endpoints.
- To explore CNM-Au8 pharmacokinetics (PK) and pharmacodynamic (PD) effects.

## 2.3 Study Population

In addition to eligibility criteria specified in the Master Protocol, participants in RGC must not have a history of allergy to gold, gold salts, or colloidal gold preparations.

Participants will be recruited from approximately 60 centers located throughout the US that are part of the Northeast ALS (NEALS) Consortium.

## 2.4 Participant Flow

Participants in RGC follow the consenting, Master screening, regimen assignment, regimen-specific screening, randomization to active or placebo treatment, and follow-up procedures and timing described in the M-SAP. Detailed descriptions of study procedures and timing are specified in the Master Protocol and the RGC RSA.

## 2.5 Regimen Allocation

Participants in RGC are those determined eligible for Master Protocol-level inclusion and exclusion criteria and randomly assigned to RGC, stratified by use of: (i) riluzole, (ii) edaravone, (iii) both, or (iv) neither at the time of screening for the Master Protocol. Details of regimen assignment are described in the Platform Trial Regimen Assignment Plan.

## 2.6 Treatment Allocation

Participants in RGC are randomly allocated in a 3:3:2 ratio to 30 mg/d CNM-Au8, 60 mg/d CNM-Au8, or placebo treatment based on a pre-specified permuted-block randomization schedule, stratified by use of (i) riluzole, (ii) edaravone, (iii) both, or (iv) neither at the time of screening for the Master Protocol.

## 2.7 Treatment Administration

CNM-Au8 and placebo are supplied as pairs of bottles each containing 60 mL of liquid. Bottles of active study drug at the 30 mg/d dosage contain CNM-Au8 at a concentration of 250 µg/mL (15 mg per bottle). Bottles of active study drug at the 60 mg/d dosage contain CNM-Au8 at a

concentration of 500 µg/mL (30 mg per bottle). All formulations contain 32.8 mg sodium bicarbonate and 60 mL USP purified water.

A normal dosage is two (2) 60 mL bottles of study drug daily. If participants have difficulty tolerating study drug and with Site Investigator and Medical Monitor approval, participants may reduce the dosage to one (1) bottle daily. Participants who have down titrated may be rechallenged at full dosage with approval by the Site Investigator and Medical Monitor. Participants who cannot tolerate the rechallenge should remain at the dosage of one (1) bottle daily. Drug holidays (e.g., treatment-free periods) are not permitted.

The first dose of study drug should be administered while in the office/clinic on the day of the Baseline Visit after all visit assessments are complete. Additional details of treatment administration are described in the RGC RSA.

## 2.8 Allocation Concealment

Allocation concealment is the same as described in the M-SAP. All formations of study drug are identical in appearance (size, shape, volume, and color) and smell.

## 2.9 Master Protocol Schedule of Activities (SoA)

| Activity                                 | MP<br>Scrn <sup>1</sup> | RGC<br>Scrn <sup>2</sup> | Base-<br>line | Week<br>2   | Week<br>4 <sup>3</sup> | Week<br>8 <sup>3</sup> | Week<br>12  | Week<br>16       | Week<br>20   | Week<br>24 <sup>3</sup> | Final<br>Call <sup>4</sup> |
|------------------------------------------|-------------------------|--------------------------|---------------|-------------|------------------------|------------------------|-------------|------------------|--------------|-------------------------|----------------------------|
|                                          | Cln                     | Cln                      | Cln           | Phn         | Cln <sup>5</sup>       | Cln <sup>5</sup>       | Phn         | Cln <sup>5</sup> | Phn          | Cln                     | Phn                        |
|                                          | -42d<br>to -1d          | -41d<br>to 0d            | Day<br>0      | Day<br>14±3 | Day<br>28±7            | Day<br>56±7            | Day<br>84±3 | Day<br>112±7     | Day<br>140±3 | Day<br>168±7            | 28d ±3<br>ALD              |
| Written Informed Consent <sup>6</sup>    | X                       | X                        |               |             |                        |                        |             |                  |              |                         |                            |
| Inclusion/Exclusion Review <sup>7</sup>  | X                       | X                        |               |             |                        |                        |             |                  |              |                         |                            |
| ALS & Medical History                    | X                       |                          |               |             |                        |                        |             |                  |              |                         |                            |
| Demographics                             | X                       |                          |               |             |                        |                        |             |                  |              |                         |                            |
| Physical Examination                     | X                       |                          |               |             |                        |                        |             |                  |              |                         |                            |
| Neurological Exam                        | X                       |                          |               |             |                        |                        |             |                  |              |                         |                            |
| Vital Signs <sup>8</sup>                 | X                       |                          | X             |             | X                      | X                      |             | X                |              | X                       |                            |
| Slow Vital Capacity                      | X <sup>9</sup>          |                          | X             |             |                        | C                      |             | C                |              | X                       |                            |
| Home Spirometry                          | X <sup>9</sup>          |                          | X             |             |                        | X                      |             | X                |              | X                       |                            |
| Muscle Strength Assessment               |                         |                          | X             |             |                        | C                      |             | C                |              | X                       |                            |
| ALSFERS-R                                | X                       |                          | X             |             | X                      | X                      | X           | X                | X            | X                       |                            |
| ALSAQ-40                                 |                         |                          | X             |             |                        |                        |             |                  |              | X                       |                            |
| CNS-BFS                                  |                         |                          | X             |             |                        | X                      |             | X                |              | X                       |                            |
| 12-Lead ECG                              | X                       |                          |               |             |                        |                        |             |                  |              | X                       |                            |
| Clinical Safety Labs <sup>10</sup>       | X                       |                          | X             |             | X                      | X                      |             | X                |              | X                       |                            |
| CNM-Au8 PK Samples <sup>11</sup>         |                         |                          | X             |             | C                      | C                      |             |                  |              | X                       |                            |
| CNM-Au8 PD Samples <sup>11</sup>         |                         |                          | X             |             | C                      | C                      |             |                  |              | X                       |                            |
| Biomarker Blood Collection <sup>11</sup> |                         |                          | X             |             |                        | C                      |             | C                |              | X                       |                            |
| Biomarker Urine Collection <sup>11</sup> |                         |                          | X             |             |                        | C                      |             | C                |              | X                       |                            |
| DNA Collection <sup>12</sup> (optional)  |                         |                          | X             |             |                        |                        |             |                  |              |                         |                            |
| CSF Collection (optional)                |                         |                          | X             |             |                        |                        |             | C <sup>13</sup>  |              |                         |                            |
| Concomitant Medication Review            | X                       | X                        | X             | X           | X                      | X                      | X           | X                | X            | X                       |                            |
| Adverse Event Review <sup>14</sup>       | X                       | X                        | X             | X           | X                      | X                      | X           | X                | X            | X                       | X                          |
| Suicidality C-SSRS                       |                         |                          | X             |             | X                      | X                      |             | X                |              | X                       |                            |
| Install Smartphone Apps <sup>15</sup>    |                         |                          | X             |             |                        |                        |             |                  |              |                         |                            |
| Smartphone Voice Recording <sup>16</sup> |                         |                          | X             |             | X                      | X                      |             | X                |              | X                       |                            |
| Uninstall Smartphone App                 |                         |                          |               |             |                        |                        |             |                  |              | X                       |                            |

| Activity                       | MP<br>Scrn <sup>1</sup> | RGC<br>Scrn <sup>2</sup> | Base-<br>line   | Week<br>2       | Week<br>4 <sup>3</sup> | Week<br>8 <sup>3</sup> | Week<br>12      | Week<br>16       | Week<br>20      | Week<br>24 <sup>3</sup> | Final<br>Call <sup>4</sup> |
|--------------------------------|-------------------------|--------------------------|-----------------|-----------------|------------------------|------------------------|-----------------|------------------|-----------------|-------------------------|----------------------------|
|                                | Cln                     | Cln                      | Cln             | Phn             | Cln <sup>5</sup>       | Cln <sup>5</sup>       | Phn             | Cln <sup>5</sup> | Phn             | Cln                     | Phn                        |
|                                | -42d<br>to -1d          | -41d<br>to 0d            | Day<br>0        | Day<br>14±3     | Day<br>28±7            | Day<br>56±7            | Day<br>84±3     | Day<br>112±7     | Day<br>140±3    | Day<br>168±7            | 28d ±3<br>ALD              |
| Regimen Assignment             | X                       |                          |                 |                 |                        |                        |                 |                  |                 |                         |                            |
| Randomization within RGC       |                         |                          | X               |                 |                        |                        |                 |                  |                 |                         |                            |
| Administer/Dispense Study Drug |                         |                          | X <sup>17</sup> |                 |                        | X                      |                 | X                |                 |                         |                            |
| Drug Accountability/Compliance |                         |                          |                 | X <sup>18</sup> | X                      | X                      | X <sup>18</sup> | X                | X <sup>18</sup> | X                       |                            |
| Exit Questionnaire             |                         |                          |                 |                 |                        |                        |                 |                  |                 | X                       |                            |
| Vital Status                   |                         |                          |                 |                 |                        |                        |                 |                  |                 | X <sup>19</sup>         |                            |

Abbreviations: ALD = after last dose, ALS = amyotrophic lateral sclerosis, ALSAQ-40 = ALS Assessment Questionnaire, ALSFRS-R = ALS Functional Rating Scale Revised, BP = blood pressure, C = completed only if the visit is conducted in-clinic, CBC = complete blood count, Cln = Clinic visit, CNS-BFS = Center for Neurologic Study Bulbar Function Scale, CSF = cerebrospinal fluid, C-SSRS = Columbia-Suicide Severity Rating Scale, d = day, DNA = deoxyribonucleic acid, ECG = electrocardiogram, LFTs = liver function tests, MP = Master Protocol, PD = pharmacodynamic, Phn = Phone visit, RGC = the CNM-Au8 regimen, Scrn = Screening Visit.

<sup>1</sup> Master Protocol (MP) screening procedures must be completed between 42 days and 1 day prior to the Baseline Visit.

<sup>2</sup> RGC regimen-specific screening procedures must be completed no more than 41 days prior to the Baseline Visit. RGC regimen-specific screening procedures may be completed on the same day as the Baseline Visit.

<sup>3</sup> Participants should be instructed to hold the morning dose of study drug on the day of the Week 4, 8, and 24 study visits. Study drug should not be taken until after study visit procedures are complete.

<sup>4</sup> Participants who are participating into the Open-label Extension (OLE) at the time scheduled for the Follow-up Safety Call will not complete a Follow-up Safety Call. Participants who continue into the OLE and then terminate prior to the time scheduled for the Follow-up Safety Call should be asked to complete a Follow-Up Safety Call.

<sup>5</sup> The Week 4, 8, and 16 visits may be conducted via telemedicine (or via phone if telemedicine is not available) with remote services instead of in-clinic if this is needed to protect the safety of the participant due to a pandemic, or other reason. If a planned in-clinic visit is conducted via telemedicine (or via phone if telemedicine is not available) with remote services, only selected procedures will be performed: ALSFRS-R, home spirometry (Weeks 8 and 16 only), Center for Neurologic Study Bulbar Function Scale (CNS-BFS, Weeks 8 and 16 only), voice recording, adverse event review, Columbia Suicide Severity Rating Scale, concomitant medications, study drug accountability/compliance, and administer/dispense study drug. Additionally, selected vital signs (systolic and diastolic pressure, respiratory rate, heart rate, and temperature) and clinical safety labs will be performed by a home health agency.

<sup>6</sup> During the Master Protocol Screening Visit, participants will be consented using the Master Protocol informed consent form (ICF). After a participant is randomly assigned to RGC, he or she will be consented a second time using the RGC ICF.

<sup>7</sup> At the MP Screening Visit, participants will have MP inclusion and exclusion criteria assessed. At the RGC Screening Visit, participants will have regimen-specific inclusion and exclusion criteria assessed.

<sup>8</sup> At all visits, whether conducted in-clinic or via telemedicine (or via phone if telemedicine is not available), vital signs include systolic and diastolic blood pressure, heart rate, respiratory rate, and temperature. At all in-clinic visits, vital signs include weight. At the MP Screening Visit only, vital signs include height measured in cm.

<sup>9</sup> If required due to pandemic-related restrictions, forced vital capacity (FVC) performed by a Pulmonary Function Laboratory evaluator or with a study-approved home spirometer or sustained phonation using a study-approved method may be used for eligibility (Master Protocol Screening ONLY).

<sup>10</sup> Clinical safety labs include hematology (CBC with differential), complete chemistry panel, liver function tests, thyroid function, and urinalysis. Serum pregnancy testing will occur in women of child-bearing potential at the Master Protocol Screening Visit and as necessary during participation. Pregnancy testing is only repeated as applicable if there is a concern for pregnancy.

<sup>11</sup> Whole blood samples for CNM-Au8 PK, plasma samples for riluzole PK, and whole blood, plasma, and urine samples for PD biomarkers should be obtained prior to the first daily dose of study drug at each applicable clinic visit. The time of the most recent dose of study drug, the most recent meal prior to sampling, and the time of sample collection should be reported. Whole blood and urine will be collected only at the Baseline and Week 24/Early Termination visits.

<sup>12</sup> If the participant consents to DNA sample collection but a DNA sample is not obtained at the Baseline Visit or if that sample is not usable, a DNA sample can be collected after the Baseline Visit.

<sup>13</sup> If the participant consents to CSF sample collection but a CSF sample is not obtained at the Week 16 Visit, a CSF sample can be collected at the Week 24 Visit.

<sup>14</sup> Adverse events that occur after signing the RGC ICF will be recorded.

<sup>15</sup> Two smartphone apps should be installed on the participant's phone, one to collect home spirometry and one to collect voice recordings.

<sup>16</sup> In addition to study visits specified in the SoA for collection of voice recordings, participants should complete twice weekly voice recordings at home. During weeks when a participant completes a voice recording in-clinic, he or she should only complete one other voice recording at home.

<sup>17</sup> Administer first dose of study drug only after Baseline Visit procedures are completed. Participants should take the first dose of study drug while in the office/clinic on the day of the Baseline visit.

<sup>18</sup> Drug accountability will not be done at phone visits. A drug compliance check-in must be held during phone visits to ensure participant is taking drug per dose regimen and to note any report of missed doses.

<sup>19</sup> Vital status, a determination of date of death or death equivalent or date last known alive, will be made for each randomized participant at the end of their double-blind follow-up (generally the Week 24 Visit as indicated) and again, if previously alive, after the end of double-blind follow-



up of all participants randomized to RGC. Vital status may also be ascertained at later time points by using publicly available data sources.

### **3. General Considerations for Data Analysis**

#### **3.1 Statistical Software**

Statistical software use for analyses is the same as described in the M-SAP.

#### **3.2 Summary Statistics**

Data summaries are the same as described in the M-SAP.

#### **3.3 Precision**

Precision of reported results is the same as described in the M-SAP.

#### **3.4 Transformations**

Data transformations are the same as described in the M-SAP.

#### **3.5 Multiplicity Adjustments**

Handling of multiplicity adjustments is the same as described in the M-SAP with the addition that the primary contrast for any given analysis will be between all active arm participants (i.e., both 30 mg/d and 60 mg/d dosage groups) vs. placebo participants. Comparisons between the two dosage groups and between each dosage group and placebo will be exploratory.

#### **3.6 Missing Data**

Handling of missing data is the same as described in the M-SAP. Clinic-based assessments that are missing due to COVID-19 restrictions or disruptions are considered missing at random.

### **4. Study Endpoints**

#### **4.1 Efficacy Endpoints**

The primary endpoint is the same as described in the M-SAP. ALSFRS-R total score is considered the primary efficacy endpoint as described in the M-SAP. Combined assessment of function and survival (CAFS), slow vital capacity (SVC) change, and survival (time to death or death equivalent) are considered key secondary efficacy endpoints in RGC.

#### **4.2 Exploratory Endpoints**

The following categories of exploratory endpoints will be evaluated:

- Change in ALSFRS-R domain scores,
- Change in strength: hand-held dynamometry (HHD) global percentage, HHD upper extremity percentage, HHD lower extremity percentages, HHD0, and HHD0<sup>2</sup>,

- Change in quantitative voice characteristics as measured by Aural Analytics: maximum phonation time, pause rate, breathy vocal quality, pitch instability, regulation of voicing, articulatory precision, speaking rate, articulation rate, and monotonicity,
- Change in biofluid biomarkers of neurodegeneration,
- Change in patient-reported outcomes: ALSAQ-40 and CNS-BFS,
- Change in plasma and whole blood concentrations of CNM-Au8,
- Change in CNM-Au8 PD biomarkers,
- Change in respiratory function as assessed by home spirometry,
- Time to clinical events:
  - First hospitalization due to a serious adverse event (SAE),
  - First hospitalization due to an ALS-related SAE,
  - First use of assisted ventilation,
  - First placement of a feeding tube,
  - First time reaching King's stage 4a or 4b,
  - First instance of any of the following events: feeding tube placement, tracheostomy, initiation of permanent assisted ventilation (PAV), or death,
  - First instance of any of the following events: hospitalization for an SAE, feeding tube placement, tracheostomy, initiation of PAV, or death.
- Time to first occurrence of a  $\geq 6$ -point decline in ALSFRS-R total score
- A composite of the standardized slope estimates for rate of change in ALSFRS-R total score, SVC percent-predicted, and ALSAQ-40 total score

### 4.3 Safety Endpoints

In addition to the safety endpoints described in the M-SAP, the following RGC regimen-specific safety endpoint will be evaluated:

- Safety lab alerts: Proportion of participants with ALT  $>3\times$  upper limit of normal (ULN), AST  $>3\times$  ULN, creatinine  $>1.5\times$  level at the Baseline Visit, and platelet count  $<75,000/\text{mm}^3$ , and
- Pharmacokinetics of riluzole.

## 5. Measurement Definitions

### 5.1 ALSFRS-R

The definitions of ALSFRS-R scores are the same as described in the M-SAP. Pre-baseline slope in ALSFRS-R (delta-FRS) is defined as 48 minus the baseline ALSFRS-R total score then divided by the number of months from onset of symptomatic weakness to the Baseline Visit.

ALSFRS-R domain scores are exploratory measures of the primary efficacy endpoint ALSFRS-R.

Time to first occurrence of a  $\geq 6$ -point decline in ALSFRS-R total score is a composite endpoint with death or death equivalent, whichever occurs first, and will be censored at the last date at which an ALSFRS-R assessment was performed up to the end of the Week 24 Visit window.

## 5.2 SVC

The derivation of SVC percent-predicted of normal is the same as described in the M-SAP with age calculated as number of days from date of birth to the date of a given SVC assessment divided by 365.25 and with the following correspondence between self-identified race and race defined by Global Lung Initiative (GLI) classification:

| Self-identified Race                      | GLI-defined Race |
|-------------------------------------------|------------------|
| American Indian or Alaska Native          | Mixed/Other      |
| Asian                                     | South East Asian |
| Black or African American                 | African American |
| Native Hawaiian or Other Pacific Islander | Mixed/Other      |
| White                                     | Caucasian        |
| Unknown                                   | Caucasian        |
| Not reported                              | Caucasian        |
| More than one race indicated              | Mixed/Other      |

## 5.3 Home Spirometry

Home spirometry assesses FVC remotely using a smartphone app (ZEPHYRx, Albany, NY) and a handheld spirometer (Spirobank Smart, Medical International Research, Rome, Italy). Coordinators guide participants through 3 to 8 maneuvers with live-video coaching using the ZEPHYRx platform. Flow loops are classified for acceptability and repeatability using American Thoracic Society (ATS) criteria and are manually reviewed by the NEALS Outcomes Center (Barrow Neurological Institute, Phoenix, AZ). The maximum FVC accepted by the NEALS Outcomes Center is converted to percent of predicted normal using GLI norms based on sex, age at time of assessment, height at time of screening, and race. Age is calculated as number of days from date of birth to the date of a given home spirometry assessment divided by 365.25. Higher values indicate greater respiratory function.

## 5.4 HHD and Grip Strength

The derivation of HHD upper and lower extremity scores and HHD0 are the same as described in the M-SAP with the revision that HHD0 is a composite endpoint with death or death equivalent.

A second HHD time-to-event endpoint is defined as the time from the Baseline Visit to the second post-baseline occurrence of a muscle with a strength recording of 0 among those muscles that were non-zero at baseline, or time to death or death equivalent, whichever occurs first (HHD0<sup>2</sup>).

Time at risk for HHD0 and HHD0<sup>2</sup> will be censored at the last date at which an HHD assessment was performed up to the end of the Week 24 Visit window.

HHD lower extremity average percentage, HHD upper extremity average percentage, HHD global average percentage, HHD0, and HHD0<sup>2</sup> are exploratory measures.

## 5.5 Quantitative Voice Characteristics

Voice samples will be collected using the Aural Analytics app installed on either an Android or iOS-based smartphone. At each assessment, participants perform a set of speaking tasks: reading 5 prespecified sentences, reading 5 sentences chosen at random from a large sentence bank, repeating a consonant-vowel sequence, producing a sustained phonation, and counting on a single breath. Speech analysis will be performed by Aural Analytics to derive the following quantitative voice characteristics: maximum phonation time, pause rate, breathy vocal quality, pitch instability, regulation of voicing, articulatory precision, speaking rate, articulation rate, and monotonicity.

## 5.6 Biofluid Biomarkers of Neurodegeneration

Blood biomarkers of neurodegeneration, including biomarkers of neuromuscular dysfunction, will be assayed. These will include serum creatinine and serum neurofilament light chain (NfL). Details of the assay techniques will be added once the procedures for collecting and analyzing biofluid specimens are confirmed.

## 5.7 ALSAQ-40

The definition of ALSAQ-40 score is the same as described in the M-SAP.

## 5.8 CNS-BFS

The definition of CNS-BFS score is the same as described in the M-SAP.

## 5.9 Survival

The primary definition of survival time is the same as described in the M-SAP with the clarification that PAV is defined as more than 22 hours per day of noninvasive or invasive mechanical ventilation for more than seven consecutive days. A secondary survival endpoint of death alone, independent of any death equivalent, is also defined.

Time at risk for the composite endpoint of death or death equivalent will be censored at the Week 24 Visit as defined in the M-SAP. Time at risk for the endpoint of death alone will be censored at two time points in separate analyses: (1) at the Week 24 Visit, and (2) at a subsequent assessment of vital status scheduled at the end of follow-up of the last participant in each respective regimen.

## 5.10 Pharmacokinetics of CNM-Au8

Whole blood and plasma samples collected at the Baseline and Week 24 visits will be analyzed to quantify the concentration of CNM-Au8. Details of the CNM-Au8 assay technique and analysis plan will be specified in a separate PK analysis plan. This plan will be finalized prior to database lock.

## 5.11 King's ALS Clinical Staging System

The King's ALS Clinical Staging System (Roche et al. 2012) is a 4-level ordinal scale with the first three levels indicating the number (1, 2, or 3) of distinct central nervous system regions (bulbar, upper limb, and lower limb) with neuromuscular dysfunction and levels 4a and 4b indicating nutritional or respiratory failure secondary to ALS, respectively.

Participants will be classified to King's stage 1, 2, 3, 4a, or 4b based on scores from ALSFRS-R assessments according to a published derivation (Balendra et al. 2014). Bulbar involvement is defined as a score less than 4 on any of the ALSFRS-R questions in the bulbar domain (questions 1, 2, and 3). Upper limb involvement is defined as a score less than 4 on either of the ALSFRS-R questions related to hand function (questions 4 and 5A). Lower limb involvement is defined as a score less than 4 on the ALSFRS-R question about walking (question 8). Nutritional failure is defined as responding that the participant uses gastrostomy for greater than 50% of their nutrition. Respiratory failure is defined as a score of 0 on the ALSFRS-R question addressing dyspnea (question 10 or R-1) or a score less than 4 on the ALSFRS-R question about use of mechanical ventilation (question 12 or R-3). Participants without evidence by ALSFRS-R scores of involvement of any of the three central nervous system regions will be scored as King's stage 1 due to their confirmed diagnosis with ALS. Participants may meet criteria for both King's stage 4a and 4b.

### 5.12 Hospitalization and Other Clinical Events

Times to the following clinically relevant events are defined:

- Time to first hospitalization due to a serious adverse event (SAE),
- Time to first hospitalization due to an ALS-related SAE,
- Time to first use of assisted ventilation,
- Time to first placement of a feeding tube,
- Time to King's stage 4a or 4b,
- Time to first instance of feeding tube placement, tracheostomy, initiation of PAV, or death, and
- Time to first instance of hospitalization for an SAE, feeding tube placement, tracheostomy, initiation of PAV, or death.

Time at risk for each event will be measured from each participant's Baseline Visit. Time to first hospitalization excludes hospitalizations for elective procedures. ALS-related SAEs are those indicated as related to ALS disease progression by the site investigator. Participants will be excluded from respective analysis (as applicable) if: (i) they are already using assisted ventilation or have a feeding tube at the time of the Baseline Visit, (ii) they have answered Question 5B on the ALSFRS-R at the Baseline visit, or (iii) have scored 0 points on Question 10 or less than 4 points on Question 12 of the ALSFRS-R at the Baseline visit. Death or death equivalent will be considered an outcome for each of the events listed, forming a composite endpoint.

Time at risk for these events will be censored at the Week 24 Visit, if completed, the date of consent withdrawal, if withdrawn, or the last date at which the status of each endpoint is known prior to the end of the Week 24 Visit window for participants lost to follow-up. Time to King's stage 4a or 4b is interval censored between ALSFRS-R assessments.

### 5.13 Pharmacodynamic Biomarkers

Whole blood and urine PD samples collected at the Baseline and Week 24 visits and plasma PD samples collected at the Baseline, Week 4, Week 8, and Week 24 visits will be analyzed

according to a separate PD analysis plan. The PD analysis plan will be finalized prior to database lock.

#### 5.14 Pharmacokinetics of Riluzole

Plasma samples collected at the Baseline, Week 4, and Week 8 visits from the first forty (40) RGC participants taking riluzole who reach the Week 8 visit plus plasma samples collected at the Baseline Visit from an additional ten (10) RGC participants not taking riluzole will be analyzed to quantify riluzole concentration by high-performance liquid chromatography with ultraviolet detection at 263 nm [assay method to be confirmed].

#### 5.15 Clinical Safety Laboratory Tests

Clinical safety labs include hematology, blood chemistry panel, liver function tests, thyroid function, urinalysis, and pregnancy testing in women of childbearing potential as specified in Section 9.1.2 Clinical Safety Laboratory Tests of the Master Protocol:

- Hematology: hematocrit, hemoglobin, platelet count, red blood cell (RBC) count, mean corpuscular volume, mean corpuscular hemoglobin, mean corpuscular hemoglobin concentration, RBC distribution width (RDW), RBC morphology, white blood cell (WBC) count, and counts and percentages of basophils, eosinophils, lymphocytes, monocytes, and neutrophils;
- Blood chemistry panel: bicarbonate, chloride, potassium, sodium, calcium, magnesium, phosphate, blood urea nitrogen, creatinine, estimated glomerular filtration rate (eGFR) calculated using the Modification of Diet in Renal Disease (MDRD) four-variable equation, creatinine clearance calculated using the Cockcroft-Gault equation, and glucose;
- Liver function tests: alanine aminotransferase (ALT [SGPT]), aspartate aminotransferase (AST [SGOT]), alkaline phosphatase (ALP), albumin, total protein, total bilirubin (TBL);
- Thyroid function tests: thyroid-stimulating hormone (TSH);
- Urinalysis: clarity, color, specific gravity, pH, microalbumin, protein, glucose, ketones, bilirubin, urobilinogen, nitrite, leukocyte esterase, and blood; and
- Pregnancy: qualitative and quantitative serum human chorionic gonadotropin (hCG).

Clinical safety labs will also include derived measures of potential drug-induced liver injury (DILI), including those that potentially meet the Hy's law criteria, as distinct safety lab outcomes.

Three potential DILI criteria will be defined:

- ALT or AST >3x ULN with TBL >1.5x ULN
- AST or ALT >3x ULN with TBL >2x ULN
- AST or ALT >3x ULN with TBL >2x ULN and ALP <2x ULN (potential Hy's Law cases)

where ULN is upper level of normal and all levels are measured on the same day.



## 6. Statistical Methodology

### 6.1 Analysis Sets

The ITT analysis set is henceforth referred to as the Full Analysis Set (FAS) and defined as follows:

- Full Analysis Set (FAS): Participants who were randomized within RGC plus placebo participants from specified regimens, classified according to their randomized treatment assignment. Observations made after premature permanent discontinuation of study drug are included in this sample, should such participants remain on study. Observations completed after regimen data lock are excluded. Participants determined to not meet ALS diagnostic criteria are excluded.

The definition of the STF analysis set is revised as follows:

- Safety Full (STF) Set: Participants who initiated treatment within RGC plus placebo participants from specified regimens who are not known to be ineligible for RGC and who initiated treatment in their respective regimen, classified according to the treatment they actually received. Observations made after premature permanent discontinuation of study drug are included in this sample, should such participants remain on study. Observations completed after regimen data lock are excluded.

The definitions of the ECC, ERO, STN, and SRO analysis sets are the same as described in the M-SAP with reference to the ITT analysis set now referencing the FAS analysis set. The following analysis set is specific to RGC:

- Efficacy Per-protocol (EPP) Set: The subset of participants in the FAS analysis set who initiated study treatment and who were not involved in major protocol deviations related to eligibility, treatment, follow-up, or assessment, classified according to the treatment they actually received. Final specification of participants included vs. excluded from the EPP analysis set will be defined prior to data lock. Inclusion or exclusion from the EPP analysis set of any participant for whom treatment assignment was unblinded prior to data lock will be governed by the prespecified criteria above.

Applicable analysis sets (FAS, EPP, STF, and STN) will include shared placebo participants from regimens A, B, and D. Data from shared placebo participants will include visits and events that occurred on or before the date of the final placebo-controlled period follow-up of an RGA, RGB, or RGC participant. As only concurrently enrolling regimens are contributing to efficacy analyses, the FAS and ECC analysis sets are synonymous and only the FAS analysis set will be referenced.

### 6.2 Baseline Characterization

The baseline characteristics summarized for participants randomized within RGC are the same as specified in the M-SAP with the addition of ALSAQ-40 score, CNS-BFS score, King's stage, weight, body mass index (BMI), serum urate concentration, serum creatinine concentration, and serum NfL concentration.



### 6.3 Primary Efficacy Analysis and Supportive Analyses

The primary analysis for RGC is a Bayesian shared-parameter, repeated-measures model of ALSFRS-R that accounts for loss of follow-up due to mortality. Details of the model, including documentation of operating characteristics under a range of scenarios, are provided in the "ALS Master Protocol Recommended Statistical Analysis, Design and Simulation Report" (Appendix 1 to the Master Protocol). The Bayesian shared-parameter, repeated-measures model will be applied to the FAS analysis set as the primary analysis, to the ERO analysis set as a sensitivity analysis, and to the EPP analysis set as a supportive analysis.

The estimand of the primary analysis is the relative rate of disease progression (the "disease rate ratio" or DRR) of the two active treatment groups relative to placebo in the FAS population under an assumption that active treatment slows mean time to death or death equivalent by the same proportion as treatment slows the mean rate of functional progression as measured by change in ALSFRS-R total score over time. The estimand is defined by the following attributes:

- Treatment: CNM-Au8 administered orally at dosages of 30 mg/d and 60 mg/d vs. placebo.
- Population: FAS population as defined in Section 6.1.
- Variables: time to death or death equivalent and rate of change in ALSFRS-R total score from baseline to the Week 24 Visit.
- Intercurrent event 1: treatment discontinuation due to death: no ALSFRS-R data from participants who reach the death or death equivalent endpoint are included in the analysis; instead, death or death equivalent are handled by a mortality component within the model, composite variable strategy approach.
- Intercurrent event 2: treatment discontinuation not due to death: handled via treatment policy approach, all data will be used including data collected during the placebo-controlled period after treatment discontinuation regardless of concomitant medication, for those participants who have not been censored due to mortality. Missing data post-treatment will not be imputed, handled via missing at random assumption.
- Population-level summary: mean ratio of hazard or progression rate of active treatment relative to placebo.

### 6.4 Interim Analysis

RGC will be considered for early stopping for futility according to the interim analysis schedule and definition specified in the "ALS Master Protocol Recommended Statistical Analysis, Design and Simulation Report" (Appendix 1 to the Master Protocol). RGC will not be stopped early for success.

### 6.5 Secondary Efficacy Analyses

#### 6.5.1 Hierarchical Testing

Primary inference for secondary efficacy endpoints will be based on analysis of the FAS analysis set using a joint-rank comparison for the combined assessment of function and survival (CAFS (Section 6.5.2 below), a repeated-measures linear mixed model for SVC (see Section 6.5.3 below), and Kaplan-Meier product-limit estimates and log-rank test for the primary survival endpoint (see Section 6.5.4 below) with a comparison between all active arm participants (i.e.,

the average of both 30 mg/d and 60 mg/d dosage groups) vs. placebo participants. The sequence for testing secondary efficacy endpoints is the following:

1. CAFS,
2. SVC,
3. Survival.

If the primary analysis indicates a significant slowing in disease progression from the Bayesian shared-parameter, repeated-measures model of ALSFRS-R and mortality, then each secondary efficacy endpoint in succession would be declared significant in the specified sequence using a comparison-wise criterion of two-tailed  $p < 0.05$ . After the first failure to declare significance, no endpoints lower in the hierarchy can be significant. This sequential closed-testing procedure controls the overall type 1 error rate at 5%. Nominal comparison-wise p-values for secondary efficacy endpoints will also be reported.

### 6.5.2 CAFS

The test consists of calculating a rank-sum score for each individual relative to pair-wise comparisons with all other participants. Participants are first ranked according to time to death or death equivalent when that is observed for both members of a pair or when one is censored after the observed event time for the other. Pairs that cannot be ranked by time to death or death equivalent are subsequently ranked by absolute change from baseline in ALSFRS-R total score at the maximum follow-up time at which both participants have an observation.

CAFS analyses are the same as specified in the M-SAP with the clarification that the active treatment group pools participants randomized to 30 mg/d CNM-Au8 and 60 mg/d CNM-Au8.

The estimand is the stochastic probability that a randomly selected active participant will rank higher than a randomly selected placebo participant based on time to death or death equivalent or change from baseline in ALSFRS-R. The estimand is defined by the following attributes:

- Treatment: CNM-Au8 administered orally at dosages of 30 mg/d and 60 mg/d vs. placebo.
- Population: FAS population as defined in Section 6.1.
- Variables: ranksum for relative time to death or death equivalent up to the Week 24 Visit or absolute change from baseline in ALSFRS-R total score.
- Intercurrent event: treatment discontinuation: handled via treatment policy approach, with pair ranked comparisons up at the maximum follow-up time at which both participants have an observation.
- Population-level summary: difference in means of ranks of active treatment relative to placebo.

Inference from this analysis is supportive of inference from the Bayesian shared-parameter, repeated-measures model for the primary endpoint and is the primary analysis for this secondary endpoint.

The following SAS code specifies the analysis:

```
proc npar1way data=work.cafs wilcoxon;  
  class Trt;  
  var Rank;  
run;
```

where Trt indicates treatment group (active or placebo) and Rank is the sum of the ranks (-1, 0, 1) of each participant vs. every other participant in the FAS analysis set. A significant difference requires interpretation based on estimates from the survival analysis described in Section 6.5.5 and the repeated-measures analysis of ALSFRS-R total score described in Section 6.5.3.

A secondary CAFS analysis will use time to death independent of any death equivalent.

### 6.5.3 Repeated-measures Model

The specification of the shared-baseline, repeated-measures linear mixed model and the primary linear contrast for estimating differences in 24-week change in a given continuous efficacy endpoint (ALSFRS-R total and domain scores, HHD upper extremity, lower extremity, and global average percentages, SVC, FVC by home spirometry, serum creatinine, serum NfL, ALSAQ-40, and CNS-BFS) is similar to the specifications in the M-SAP with the modification that treatment group is a three-level variable (i.e., placebo, 30 mg/d CNM-Au8, and 60 mg/d CNM-Au8).

The estimand estimated by the primary linear contrast of the shared-baseline, repeated-measures linear mixed model is the mean difference in 24-week change from baseline of a given continuous efficacy endpoint in the two active treatment groups relative to the placebo group in the FAS population. The estimand is defined by the following attributes:

- Treatment: CNM-Au8 administered orally at dosages of 30 mg/d and 60 mg/d vs. placebo.
- Population: FAS population as defined in Section 6.1.
- Variables: absolute change in endpoint from baseline to the Week 24 Visit.
- Intercurrent event: treatment discontinuation: handled via treatment policy approach, all data will be used including data collected during the placebo-controlled period after treatment discontinuation. Missing data post-treatment, including data missing due to death, will not be imputed, handled via missing at random assumption.
- Population-level summary: difference in conditional means of active treatment relative to placebo.

Inference from this analysis is supportive of inference from the Bayesian shared-parameter, repeated-measures model for the primary endpoint and is the primary analysis for the secondary SVC endpoint.

The following SAS code specifies the linear contrast for an endpoint measured every 8 weeks and assuming that the sort order for treatment group is placebo, 30 mg/d CNM-Au8, and 60 mg/d CNM-Au8 and visits are sorted chronologically:

```
estimate "3|Act vs Plb|dWk 24"
      post*trtrnd*visit 0 0 0 -2 0 0 0 1 0 0 0 1 / cl divisor=2;
```

A significant difference in 24-week change in the direction of improved function would support inference that CNM-Au8 treatment reduced progression over 24 weeks for the efficacy endpoint being tested.

### 6.5.4 Survival

The primary estimate of the effect of treatment on survival as an efficacy endpoint will include both death and death equivalent as components and will pool participants randomized to 30 mg/d

CNM-Au8 and 60 mg/d CNM-Au8 in the FAS analysis set. Otherwise, summaries and analyses of time to death or death equivalent are the same as specified in the M-SAP.

The estimand estimated by the logrank test is the deviation in expected survival times in the FAS population. The estimand is defined by the following attributes:

Treatment: CNM-Au8 administered orally at dosages of 30 mg/d and 60 mg/d vs. placebo.

Population: FAS population as defined in Section 6.1.

Variables: time to earlier of death or death equivalent.

Intercurrent event: treatment discontinuation: handled via treatment policy approach, all data will be used including data collected during the placebo-controlled period after treatment discontinuation. Missing data post-treatment will not be imputed, handled via missing completely at random assumption.

Population-level summary: difference in survival curves of active treatment and placebo.

In the presence of a significant difference by the logrank test, inference on whether active treatment extended survival will be based on the hazard ratio estimated from an unadjusted Cox proportional hazards regression.

Survival to the end of the placebo-controlled period will also be analyzed in the ERO, EPP, STF, SFN, and SRO analysis sets. Survival analyses that include follow-up beyond the placebo-controlled period will be analyzed in the ERO analysis set.

Listings of all participants in the FAS, ERO, EPP, STF, and SRO analysis sets will document whether each participant had died, had reached a death equivalent endpoint, or was alive and free of a death equivalent and the study day of death, death equivalent, or last known alive and free of a death equivalent for each analysis timepoint.

### 6.5.5 Random-slopes Model

The specification of the shared-baseline, random-slopes linear mixed model and the primary linear contrast for estimating differences in mean rate of progression in a given continuous efficacy endpoint (ALSFRS-R total and domain scores, HHD upper extremity, lower extremity, and global average percentages, SVC, FVC by home spirometry, quantitative voice characteristics, serum creatinine, serum NfL, ALSAQ-40, and CNS-BFS) is similar to the specifications in the M-SAP with the modification that treatment group is a three-level variable (i.e., placebo, 30 mg/d CNM-Au8, and 60 mg/d CNM-Au8).

The estimand estimated by the primary linear contrast of the shared-baseline, random-slopes linear mixed model is the difference in mean rate of progression of a given continuous efficacy endpoint in the two active treatment groups relative to the placebo group in the FAS population. The estimand is defined by the following attributes:

Treatment: CNM-Au8 administered orally at dosages of 30 mg/d and 60 mg/d vs. placebo.

Population: FAS population as defined in Section 6.1.

Variables: mean rate of change in endpoint from baseline to the Week 24 Visit.

Intercurrent event: treatment discontinuation: handled via treatment policy approach, all data will be used including data collected during the placebo-controlled period after treatment

discontinuation. Missing data post-treatment will not be imputed, handled via missing at random assumption.

Population-level summary: difference in conditional mean slopes of active treatment relative to placebo.

Inference from these analyses is supportive of inference from the Bayesian shared-parameter, repeated-measures model for the primary endpoint and inference from the shared-baseline, repeated-measures linear mixed model for secondary endpoints.

The following SAS code specifies the linear contrast assuming that the sort order for treatment group is placebo, 30 mg/d CNM-Au8, and 60 mg/d CNM-Au8:

```
estimate "3|Act vs Plb|Slope (/mn)"  
        month 0 trtrnd*month -2 1 1 / cl divisor=2;
```

A significant difference in slopes in the direction of improved function would support inference that the treatment reduced rate of progression for the efficacy endpoint being tested.

### 6.5.6 Time to Clinical Events

Time to hospitalizations and clinical events will be analyzed in the FAS, ERO, and EPP analysis sets. The summaries and analyses will parallel the analyses described for survival described in Section 6.5.4. Analysis of time to King's stage 4a or 4b and time to a 6-point decline in ALSFRS-R total score will accommodate interval censoring between ALSFRS-R assessments. Analysis of time to King's stage 4a or 4b will be stratified by baseline King's stage.

### 6.5.7 Composite of ALSFRS-R, SVC, and ALSAQ-40 Slopes

Participant-specific rates of change in ALSFRS-R total score, SVC percent-predicted, and ALSAQ-40 total score will be estimated in separate, endpoint-specific unadjusted random-slopes linear mixed models in the FAS and ERO analysis sets. Participant-specific random slopes for each endpoint will be standardized to a mean of zero (0) and a variance of one (1) and then averaged for each participant. The composite average of standardized random slope estimates will be compared across treatment groups by one-way analysis of variance with treatment comparison between all active arm participants (i.e., the average of both 30 mg/d and 60 mg/d dosage groups) vs. placebo participants and between each active treatment group separate vs. placebo estimated using contrasts of least-square means.

### 6.5.8 HHD0 and HHD0<sup>2</sup>

Analyses of HHD0 are the same as specified in the M-SAP with the modification that treatment group is a three-level variable (i.e., placebo, 30 mg/d CNM-Au8, and 60 mg/d CNM-Au8), with addition of parallel analyses of HHD0<sup>2</sup>, and with the clarification that time to zero strength for both analyses is interval censored between HHD assessments.

Inference from these analyses is supportive of inference from the shared-baseline, repeated-measures linear mixed model for HHD upper and lower extremity scores.

### 6.5.9 Quantitative Voice Measures

Given the high frequency of voice recordings, a repeated-measures analysis with unstructured covariance is overly flexible but the assumption of linear change required by the random-slopes model may be overly rigid. To complement estimates from the shared-baseline, random-slopes

linear mixed model, quantitative voice characteristics will be analyzed in a shared-baseline linear mixed model in which the temporal profile for both fixed and random terms is modeled using cubic B-splines with knots at 8 and 16 weeks. The model will include fixed terms for B-splines (4 terms), treatment group (3 levels)  $\times$  B-spline interaction, centered time since symptom onset and centered time since symptom onset  $\times$  B-spline interaction, centered delta-FRS and centered delta-FRS  $\times$  B-spline interaction, centered baseline riluzole use and centered baseline riluzole  $\times$  B-spline interaction, and centered baseline edaravone use and centered baseline edaravone  $\times$  B-spline interaction. The model will include random regimen-specific intercepts and slopes with unstructured covariance, random participant-specific B-splines (5 terms) with unstructured covariance, and a first-order autoregressive structure for residuals. A simplified covariance structure assuming no regimen-level covariance, heterogeneous compound symmetric covariance among the random B-splines, conditional independence of residuals, or a combination of the three simplifying assumptions will be used if the full model fails to converge. The primary estimand will be the active (i.e., the average of both 30 mg/d and 60 mg/d dosage groups) vs. placebo treatment-dependent difference in 24-week change from baseline. The estimate and its 95% Wald confidence bounds will be obtained by a linear contrast of adjusted means.

#### **6.5.10 Placebo Multiple Imputation**

Placebo multiple imputation analyses are the same as specified in the M-SAP with the modification that treatment group is a three-level variable (i.e., placebo, 30 mg/d CNM-Au8, and 60 mg/d CNM-Au8). Placebo multiple imputation analyses will be applied to ALSFRS-R total score, HHD upper and lower extremity scores, and SVC.

Inference from these analyses is supportive of inference from the Bayesian shared-parameter, repeated-measures model for the primary endpoint and assess sensitivity to the missing data assumption of the shared-baseline, repeated-measures linear mixed model for secondary endpoints in the FAS analysis set.

#### **6.5.11 Additional Sensitivity Analyses of Primary and Key Secondary Outcomes**

Sensitivity analyses of primary and key secondary efficacy outcomes are the same as specified in the M-SAP with the modification that treatment group is a three-level variable (i.e., placebo, 30 mg/d CNM-Au8, and 60 mg/d CNM-Au8).

#### **6.5.12 Subgroup Analyses**

In addition to the subgroups specified in the M-SAP, the following additional subgroups will be analyzed in the random-slope model (see Section 6.5.2) for primary and secondary efficacy endpoints in the FAS analysis set:

- Baseline use of riluzole and edaravone (neither, riluzole only, edaravone only, both),
- Age (less than 65 years vs. 65 years or older),
- Sex (female vs. male),
- Race (white vs. any minority race with greater than 5% prevalence in the sample),
- Ethnicity (Hispanic or Latino vs. non-Hispanic or Latino),
- Weight (less than 43 kg, 43 to less than 56 kg, 56 to less than 77 kg, 77 to less than 150 kg, 150 kg or more),



- BMI (less than 18.5 kg/m<sup>2</sup>, 18.5 to less than 25 kg/m<sup>2</sup>, 25 to less than 30 kg/m<sup>2</sup>, 30 to less than 40 kg/m<sup>2</sup>, 40 kg/m<sup>2</sup> or more),
- Chronic kidney disease (CKD) stage (stage 1 or better [eGFR 90 mL/min/1.73m<sup>2</sup> or more], stage 2 [eGFR 60 to 89 mL/min/1.73m<sup>2</sup>], stage 3 [eGFR 30 to 59 mL/min/1.73m<sup>2</sup>], stage 4 [eGFR 15 to 29 mL/min/1.73m<sup>2</sup>], stage 5 [eGFR less than 15 mL/min/1.73m<sup>2</sup>]),
- Time since onset of weakness (less than 18 months vs. 18 months or longer),
- El Escorial and time since onset of weakness (El Escorial definite and less than 18 months vs. not both),
- Baseline symptom severity (all ALSFRS-R questions scored 2 or greater vs. any question scored 0 or 1),
- Early disease state (all ALSFRS-R questions scored 2 or greater, SVC 80%-predicted or greater, and time since symptom onset less than 24 months vs. not meeting all three criteria),
- Baseline serum urate concentration (less than 5.5 mg/dL vs. 5.5 mg/dL or greater),
- Baseline serum NfL concentration (by median split),
- Baseline delta-FRS score (less than 0.47 points/year, 0.47 to less than 1.11 points/year, 1.11 points/year or greater),
- Site (individual sites with at least 5 participants per treatment group and all participants from sites with fewer than 5 participants per treatment group pooled).

For each classification, unknown, not reported, and missing will be considered one group. All individuals not included in a specified subgroup will be combined into a mixed, "other" group. The "other" group will be included in analyses if its prevalence is greater than 5%; otherwise, the "other" group will be excluded.

In cases where a model for a given subgroup and endpoint fails to converge, the covariance terms for the regimen-specific random effects will be simplified from unstructured covariance of intercepts and slopes to separate, uncorrelated variance components for intercepts and slopes. If convergence still fails, regimen-specific intercepts and slopes will be modeled as fixed effects. If convergence still fails, the participant-specific random effects will be simplified from unstructured covariance of intercepts and slopes to separate, uncorrelated variance components for intercepts and slopes.

### 6.5.13 Comparison of Controls across Regimens

Comparisons of placebo participants across regimens are the same as specified in the M-SAP.

### 6.5.14 Pharmacokinetic Analyses of CNM-Au8

Pre-dose concentrations of CNM-Au8 in whole blood and plasma will be summarized by treatment group and visit in the ERO sample. Concentrations below the limit of quantitation (BLQ) will be replaced with one half of the lower limit of quantitation. Summaries will include number of observations, number and percentage with concentrations BLQ, arithmetic mean, median, standard deviation, minimum, maximum, geometric mean, geometric coefficient of variation (calculated as  $\sqrt{\exp(\text{variance of log-transformed concentrations}) - 1}$ ), and 95% confidence bounds for the geometric mean assuming log-normally distributed data.



A concentration response relationship based on observed CNM-Au8  $C_{\max}$  per active participant versus ALSFRS-R slope change will also be analyzed as an exploratory endpoint.

Whole blood and plasma concentration data of CNM-Au8 may be subjected to population pharmacokinetic analysis to derive population estimates of pharmacokinetic parameters and test the effect of various covariates such as age, weight, and sex. Details of the analysis will be described in a separate data analysis plan (DAP). This analysis may be performed by combining data from the current study with data from other studies of CNM-Au8, if deemed appropriate. The tests of association with ALSFRS-R and the population pharmacokinetic analysis will be performed by Clene Nanomedicine or its designee and reported in a separate modelling report.

#### **6.5.15 Pharmacodynamic Biomarker Analyses**

Pharmacodynamic biomarkers will be analyzed by Clene Nanomedicine according to a separate pharmacodynamic DAP and will be reported in a separate report.

#### **6.5.16 Pharmacokinetics of Riluzole**

Pharmacokinetic analysis of plasma riluzole concentrations will be analyzed according to a separate pharmacokinetic analysis plan authored and conducted by Dr. Charles Venuto, PharmD.

### **6.6 Safety Analyses**

#### **6.6.1 Treatment-emergent Adverse Events**

Summaries and analyses of treatment-emergent adverse events (TEAE) are the same as specified in the M-SAP with the following revisions.

TEAEs are defined as those adverse events with onset dates in the interval from double-blind treatment initiation to either the Final Safety Visit if completed, the date of death if the participant dies, 28 days after last dose of study drug if the participant early terminates or is lost to follow-up, or the date of first dose of study drug during participation in the OLE, if so exposed. Adverse events with onset on the day of double-blind treatment initiation and adverse events with incompletely specified onset date where the ambiguous date spans the day of double-blind treatment initiation will be assumed to be treatment emergent except for those known to precede first exposure to study drug.

In addition to summaries specified in the M-SAP, serious TEAEs will be summarized by MedDRA system organ class, high level term, and preferred term for all TEAEs. In addition, TEAEs will be summarized by Medical Dictionary for Regulatory Activities (MedDRA) system organ class and preferred term for fatal TEAEs, TEAEs that occurred during a participant's COVID-19 infection (defined as 5 days prior to symptom onset to end of COVID-19 symptoms or the earlier of 91 days after symptom onset or the end of double-blind follow-up, if ongoing), and TEAEs and serious TEAEs stratified by the following subgroups: baseline riluzole use, baseline edaravone use, age, sex, race, ethnicity, weight, BMI, and CKD stage. Subgroup classifications will be the same as described in Section 6.5.10 Subgroup Analyses except that the "other" group will be retained regardless of prevalence.

TEAEs indicating COVID-19 infection are the following: Asymptomatic COVID-19, COVID-19, COVID-19 pneumonia, COVID-19 treatment, Post-acute COVID-19 syndrome, SARS-CoV-2 antibody test positive, SARS-CoV-2 RNA increased, SARS-CoV-2 sepsis, SARS-CoV-2 test false negative, SARS-CoV-2 test positive, SARS-CoV-2 viraemia, Suspected COVID-19.

Treatment-dependent differences in the proportion of participants experiencing a given type of TEAE will not be tested. Treatment-dependent differences in TEAE incidence rates in units of number per 100 participant years will be estimated as differences rather than ratios and will include comparison-wise 95% confidence intervals with variance estimates obtained by the delta method.

### **6.6.2 Safety Labs**

Summaries and analyses of clinical safety labs are the same as specified in the M-SAP with the revision that abnormal levels will be classified to a toxicity grade based on quantitative grading using National Cancer Institute (NCI) Common Terminology Criteria for Adverse Events (CTCAE) version 5.0 and with the addition that maximum toxicity over all post-baseline visits will be included in shift tables along with visit-specific shifts.

The proportion of participants with ALT >3x ULN, AST >3x ULN, creatinine >1.5x level at the Baseline Visit, and platelet count <75,000/mm<sup>3</sup>, or with any of the four will be presented as shift tables vs. the status of each participant at baseline for each visit by treatment group in all safety samples. The absolute level and the absolute change from baseline for ALT, AST, creatinine, and platelet count will be summarized as means, standard deviations, medians, and ranges at each visit by treatment group in all safety samples.

### **6.6.3 ECG Results**

Summaries of ECG parameters and findings are the same as specified in the M-SAP.

### **6.6.4 Vital Signs and Weight**

Summaries and analyses of vital signs and weight are the same as specified in the M-SAP.

### **6.6.5 Suicidality**

Summaries of suicidality are the same as specified in the M-SAP.

## **6.7 Other Analyses**

### **6.7.1 Participant Disposition**

All participants consented to the Master Protocol between the time of the first and last consent of a participant assigned to a regimen included in the FAS analysis set will be summarized as a single set for the following events: consented to the Master Protocol, failed screening for the Master Protocol, other reasons not assigned to a regimen (including timing out of the screening window, death, withdrawal of consent, early termination, and administrative termination), and assigned to a regimen. Reasons for Master Protocol screen failure will be summarized.

All participants in the above sample assigned to a regimen will be summarized as two sets (final screening for RGC vs. final screening for a non-RGC regimen) for the following events: consented to a regimen, failed screening for a regimen, other reasons not randomized within a regimen (including timing out of the screening window, death, withdrawal of consent, early termination, and administrative termination), and randomized within a regimen. If a given individual is screened multiple times prior to randomization within a regimen, then the final screening experience of that individual will be summarized. Reasons for RGC screen failure will be summarized separately for all participants screened for RGC whether that was their final screening experience or not.

All participants in the FAS analysis set will be summarized as two sets (randomization to active study drug vs. randomization to placebo) for the following events: initiated regimen-specific study drug, prematurely terminated study participation due to death, withdrawal of consent, early termination, loss to follow-up, or administrative termination, completed 24-week follow up, and completed a safety follow-up visit vs. continued into the OLE. Reasons for withdrawal of consent or early termination after randomization will be summarized.

Any randomized participants excluded from the FAS and EPP analysis sets or included in the FAS analysis set but not contributing to the primary analysis will be identified in a listing together with the reason for their exclusion.

### **6.7.2 Study Drug Compliance and Tolerance**

Summaries of study drug compliance and tolerance are the same as specified in the M-SAP with the clarification that summaries will be reported for the ERO and SRO analysis sets and that date of permanent discontinuation of study drug is the date of last use of double-blind study drug among all participants in a given analysis set.

The number of days of exposure to study drug will be calculated in three ways:

- as the number of days from dose initiation to the final safety assessment during the placebo-controlled period, inclusive,
- as the number of days from dose initiation to drug withdrawal, inclusive, less any interval during which use of study drug was interrupted (individual missed doses will not be subtracted unless noted in the dosage management log), and
- as the number of days from dose initiation to the earlier of final contact during the placebo-controlled period or 28 days after last dose of study drug, inclusive.

The proportion of participants who interrupted study drug or reduced study drug dosage and the time to first study drug interruption or dosage reduction will be summarized. The number of days of exposure to a reduced dosage of study drug will be summarized.

### **6.7.3 Concomitant Medication Use**

Summaries of concomitant medication use are the same as specified in the M-SAP with the clarification that medications taken at baseline and those initiated after first dose of study drug will be separately summarized and will be classified by ATC Therapeutic class and WHODrug Preferred base name.

### **6.7.4 Medical History**

Medical histories will be summarized by MedDRA system organ class, high level term, and preferred term in the STF and SRO analysis sets.

### **6.7.5 Blindedness**

The proportions of participants and site investigators who report on the Exit Questionnaire a guess of active vs. placebo treatment assignment, each level of surety of that guess, and each of five pre-specified reasons for making a treatment assignment will be summarized by treatment group in the FAS and ERO analysis sets. Treatment-dependent differences in the proportion guessing active treatment assignment will be tested among all respondents and among those

stating they are at least somewhat sure of their guess by Fisher's exact test and the difference in proportion guessing active treatment assignment will be estimated with confidence bounds.

#### **6.7.6 Protocol Deviations**

The number of major and minor protocol deviations will be summarized by type of deviation and treatment group in all analysis sets. Listings of all protocol deviations will be produced.

#### **6.7.7 Impact of COVID-19 Pandemic**

The proportions of planned assessments missed due to COVID-19 restrictions or disruptions will be summarized by treatment group, visit, and type of assessment in the FAS and ERO analysis sets. Protocol deviations that resulted from COVID-19 restrictions or disruptions will be summarized by treatment group and type of deviation in the FAS and ERO analysis sets and listed.

### **7. Validation**

#### **7.1 Primary Efficacy Analysis**

Validation of the primary efficacy analysis is the same as specified in the M-SAP.

#### **7.2 Secondary, Exploratory, and Safety Analyses**

Validation of secondary, exploratory, and safety analyses are the same as specified in the M-SAP.

### **8. References**

The following references are cited in addition to those specified in the M-SAP:

- Balendra R, Jones A, Jivraj N, Knights C, Ellis CM, Burman R, Turner MR, Leigh PN, Shaw CE, Al-Chalabi A. Estimating clinical stage of amyotrophic lateral sclerosis from the ALS Functional Rating Scale. *Amyotroph Lateral Scler Frontotemporal Degener.* 2014 Jun;15(3-4):279-84.
- Roche JC, Rojas-Garcia R, Scott KM, Scotton W, Ellis CE, Burman R, Wijesekera L, Turner MR, Leigh PN, Shaw CE, Al-Chalabi A. A proposed staging system for amyotrophic lateral sclerosis. *Brain.* 2012 Mar;135(Pt 3):847-52.

**RGC REGIMEN-SPECIFIC STATISTICAL ANALYSIS PLAN (R-SAP)**

|                                |                                                                                                                                                                                                                          |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Master Protocol</b>         | Platform Trial for the Treatment of Amyotrophic Lateral Sclerosis (ALS): A perpetual multi-center, multi-regimen, clinical trial evaluating the safety and efficacy of investigational products for the treatment of ALS |
| <b>Regimen</b>                 | RGC: CNM-Au8                                                                                                                                                                                                             |
| <b>Regimen Partner</b>         | Clene Nanomedicine, Inc.                                                                                                                                                                                                 |
| <b>Regulatory Sponsor</b>      | Merit E. Cudkowicz, MD                                                                                                                                                                                                   |
| <b>Master Protocol Version</b> | 4.0, 31 Aug 2020                                                                                                                                                                                                         |
| <b>RSA Version</b>             | 5.0, 22 Nov 2021                                                                                                                                                                                                         |
| <b>Master SAP Version</b>      | 1.0, 24 Jun 2020                                                                                                                                                                                                         |
| <b>R-SAP Version</b>           | 3.0, 22 Jul 2022                                                                                                                                                                                                         |

**SAP APPROVAL SIGNATURES**

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Merit E. Cudkowicz, MD, MSc  
Principal Investigator and Regulatory Sponsor

Date

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Sabrina Paganoni, MD, PhD  
Co-Principal Investigator

Date

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Robert Glanzman, MD, FAAN  
Chief Medical Officer, Clene Nanomedicine, Inc.

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Michael Hotchkin  
Chief Development Officer, Clene Nanomedicine, Inc.

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Ben Saville, PhD  
Senior Statistical Scientist, Berry Consultants

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Eric A. Macklin, PhD  
Study Biostatistician

Date

## SAP REVISION HISTORY

| Version | Date        | Description of Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | 17 Mar 2022 | Initial version                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2.0     | 09 May 2022 | <p>Revision of Section 5.9 Survival to specify that both PAV-free survival and overall survival will be evaluated at both (i) the Week 24 Visit time point, and (ii) the last-participant-last-visit time point, and to specify that PAV-free survival to the Week 24 Visit time point is the primary analysis of survival in this analysis plan.</p> <p>Revision of Section 6.5.2 CAFS to specify the following:</p> <ol style="list-style-type: none"> <li>1. A joint-rank analysis of PAV-free survival and SVC will be used as a supportive analysis for the secondary efficacy endpoint of SVC,</li> <li>2. The secondary endpoint joint-rank analyses will use multiple imputation to extend follow-up for participants who early terminate, withdraw consent, or are lost to follow-up as additional supportive analyses,</li> <li>3. The secondary joint-rank analyses will use time to death alone independent of any death equivalent as additional supportive analyses, and</li> <li>4. Primary inference from the joint-rank analyses will compare survival by time to death or death equivalent and will compare change in function to the last jointly observed time point.</li> </ol> |
| 3.0     | 22 Jul 2022 | <p>Revision of Section 4.2 Exploratory Endpoints to (i) identify serum creatinine, serum and CSF neurofilament light chain (NfL), and urinary p75[ECD] as exploratory biomarkers of neurodegeneration and neuromuscular degeneration, and (ii) include ALSAQ-40 domain scores and symptom index (SI) as exploratory endpoints.</p> <p>Revision of Section 5.1 ALSFRS-R to specify details of the calculation of pre-baseline slope.</p> <p>Revision of Section 5.5 Quantitative Voice Characteristics to include predicted vital capacity as an additional metric at the Baseline Visit.</p> <p>Revision of Section 5.6 Biofluid Biomarkers of Neurodegeneration to specify assay techniques used to quantify serum creatinine, serum and CSF NfL, and urinary p75[ECD].</p>                                                                                                                                                                                                                                                                                                                                                                                                                         |



| Version            | Date                       | Description of Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.0<br>(continued) | 22 Jul 2022<br>(continued) | <p>Revision of Section 5.7 ALSAQ-40 to specify calculation of domain scores and to revise calculation of overall ALSAQ-40 SI.</p> <p>Revision of Section 5.8 CNS-BFS to specify that the total score is referenced.</p> <p>Revision of Section 5.9 Survival to specify that time at risk begins at each participant's Baseline Visit and to specify that the date of PAV initiation, where applicable, will be imputed as the fifteenth day of a month if not specified more precisely.</p> <p>Revision of Section 6.1 Analysis Sets to add the Efficacy Common Mode of Administration (ECM) analysis set, to remove the restriction on protocol deviations that could be considered for exclusion from the Efficacy Per-protocol (EPP) analysis set must be classified as major protocol deviations, to specify the time point at which data is excluded from the EPP analysis set in the case of time-dependent exclusions, and to specify that data from placebo participants from other regimens would not be excluded from the EPP analysis set due to non-adherence to protocol-specified dosing.</p> <p>Revision of Section 6.2 Baseline Characterization to include ALSAQ-40 domain scores and SI, urinary p75[ECD], and vital capacity predicted from quantitative voice characteristics.</p> <p>Revision of Section 6.5.2 CAFS to clarify that the primary CAFS analysis is specified in the ALS Master Protocol Recommended Statistical Analysis, Design and Simulation Report and is an adjusted analysis, to provide code for an adjusted analysis, and to add three additional sets of CAFS analyses, one set that adjusts rank scores in linear models for time from ALS symptom onset, delta-FRS, baseline use of riluzole, and baseline use of edaravone, one set that adjusts for the same set of covariates plus baseline serum NfL level, and one that makes no adjustment.</p> <p>Revision of Section 6.5.3 Repeated-measures Model to add a fixed term for treatment group (removing the shared-baseline assumption at the recommendation of the FDA) and to specify a separate supportive analysis that includes fixed terms for centered baseline serum NfL level, and centered baseline serum NfL level <math>\times</math> visit interaction for analysis of clinical endpoints.</p> <p>Revision of Section 6.5.4 Survival to include ECM as an additional analysis set, to include baseline age as an additional covariate in all adjusted models, and to specify an additional adjusted analysis that includes baseline serum NfL level.</p> |

| Version            | Date                       | Description of Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.0<br>(continued) | 22 Jul 2022<br>(continued) | <p>Revision of Section 6.5.5 Random-slopes Model to add a fixed term for treatment group (removing the shared-baseline assumption) and to specify a separate supportive analysis that includes fixed terms for centered baseline serum NfL level, and centered baseline serum NfL level <math>\times</math> study month interaction for analysis of clinical endpoints.</p> <p>Revision of Section 6.5.7 Composite of ALSFRS-R, SVC, and ALSAQ-40 Slopes to specify that one analysis will adjust for years since ALS symptom onset, delta-FRS, baseline riluzole use, and baseline edaravone use, and an additional analysis will adjust for the same covariates plus baseline serum NfL level.</p> <p>Revision of Section 6.5.8 HHD0 and HHD0<sup>2</sup> to remove reference to the shared-baseline assumption of the repeated-measures mixed model of Section 6.5.3 and to specify a separate analysis that adds baseline serum NfL level as an additional covariate.</p> <p>Revision of Section 6.5.9 Quantitative Voice Measures to remove reference to the shared-baseline assumption of the repeated-measures and random-slopes mixed models and to add a fixed term for treatment group (removing the shared-baseline assumption) and to specify a separate analysis that includes fixed terms for centered baseline serum NfL level, and centered baseline serum NfL level <math>\times</math> B-spline interaction.</p> <p>Revision of Section 6.5.10 Placebo Multiple Imputation to specify regression over sequential visits by the fully conditional specification method, to remove reference to the shared-baseline assumption of the repeated-measures mixed model of Section 6.5.3, and to specify a separate analysis that adds baseline serum NfL level as an additional covariate.</p> <p>Revision of Section 6.5.13 Comparison of Controls across Regimens to specify separate analyses that add baseline serum NfL level as an additional covariate.</p> |

## ABBREVIATIONS

|           |                                                                         |
|-----------|-------------------------------------------------------------------------|
| ALP       | Alkaline Phosphatase                                                    |
| ALS       | Amyotrophic Lateral Sclerosis                                           |
| ALSAQ-40  | Amyotrophic Lateral Sclerosis Assessment Questionnaire, 40-item version |
| ALSFRS-R  | Amyotrophic Lateral Sclerosis Functional Rating Scale, Revised          |
| ALT       | Alanine Transaminase                                                    |
| AST       | Aspartate Transaminase                                                  |
| ATC       | WHODrug Anatomical, Therapeutic, and Chemical class                     |
| ATS       | American Thoracic Society                                               |
| BLQ       | Below the Limit of Quantitation                                         |
| BMI       | Body Mass Index                                                         |
| C-SSRS    | Columbia Suicide Severity Rating Scale                                  |
| CAFS      | Combined Assessment of Function and Survival                            |
| CBC       | Complete Blood Count                                                    |
| CKD       | Chronic Kidney Disease                                                  |
| COVID-19  | Coronavirus Disease 2019                                                |
| CNS-BFS   | Center for Neurologic Study Bulbar Function Scale                       |
| CSF       | Cerebrospinal Fluid                                                     |
| CSR       | Clinical Study Report                                                   |
| CTCAE     | Common Terminology Criteria for Adverse Events                          |
| delta-FRS | Pre-baseline Slope in ALSFRS-R                                          |
| DAP       | Data Analysis Plan                                                      |
| DILI      | Drug-induced Liver Injury                                               |
| DNA       | Deoxyribonucleic Acid                                                   |
| DRR       | Disease Rate Ratio                                                      |
| ECC       | Efficacy Concurrent Control                                             |
| ECD       | Extra-cellular Domain                                                   |
| ECG       | Electrocardiography or Electrocardiogram                                |
| eGFR      | Estimated Glomerular Filtration Rate                                    |
| EPP       | Efficacy Per-protocol                                                   |
| ERO       | Efficacy Regimen-only                                                   |
| ELISA     | Enzyme-linked Immunosorbent Assay                                       |

## ABBREVIATIONS (continued)

|          |                                                                                    |
|----------|------------------------------------------------------------------------------------|
| FAS      | Full Analysis Set                                                                  |
| FVC      | Forced Vital Capacity                                                              |
| GLI      | Global Lung Initiative                                                             |
| hCG      | Human Chorionic Gonadotropin                                                       |
| HHD      | Hand-held Dynamometry                                                              |
| HLT      | MedDRA High Level Term                                                             |
| ICF      | Informed Consent Form                                                              |
| ITT      | Intention-to-treat Principle                                                       |
| M-SAP    | Master Statistical Analysis Plan                                                   |
| MDRD     | Modification of Diet in Renal Disease                                              |
| MedDRA   | Medical Dictionary for Regulatory Activities                                       |
| MP       | Master Protocol                                                                    |
| MPRDR    | ALS Master Protocol Recommended Statistical Analysis, Design and Simulation Report |
| NCI      | National Cancer Institute                                                          |
| NEALS    | Northeast ALS                                                                      |
| NfL      | Neurofilament Light Chain                                                          |
| NIV      | Noninvasive Ventilation                                                            |
| OLE      | Open-label Extension                                                               |
| p75[ECD] | Neurotrophin Receptor p75 Protein Extracellular Domain                             |
| PAV      | Permanent Assisted Ventilation                                                     |
| PD       | Pharmacodynamics                                                                   |
| PK       | Pharmacokinetics                                                                   |
| PT       | MedDRA Preferred Term                                                              |
| RBC      | Red Blood Cell                                                                     |
| RDW      | RBC Distribution Width                                                             |
| RGC      | Regimen C (CNM-Au8, 30 mg/d and 60 mg/d)                                           |
| RSA      | Regimen-specific Appendix                                                          |
| R-SAP    | Regimen-specific Statistical Analysis Plan                                         |
| SAE      | Serious Adverse Event                                                              |
| SAP      | Statistical Analysis Plan                                                          |

## ABBREVIATIONS (continued)

|         |                                                    |
|---------|----------------------------------------------------|
| SGOT    | Serum Glutamic Oxaloacetic Transaminase            |
| SGPT    | Serum Glutamic Pyruvic Transaminase                |
| SI      | Symptom Index                                      |
| SoA     | Schedule of Activities                             |
| SOC     | MedDRA System Organ Class                          |
| SRO     | Safety Regimen-only                                |
| STF     | Safety and Tolerability Full                       |
| STN     | Safety and Tolerability Narrow                     |
| SVC     | Slow Vital Capacity                                |
| TBL     | Total Bilirubin                                    |
| TEAE    | Treatment-emergent Adverse Event                   |
| TSH     | Thyroid Stimulating Hormone                        |
| ULN     | Upper Limit of Normal                              |
| WBC     | White Blood Cell                                   |
| WHODrug | World Health Organization Drug Dictionary Enhanced |

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## 1. Governing Documents

This Regimen-specific Statistical Analysis Plan (R-SAP) for the CNM-Au8 regimen (RGC) specifies any modification from the default outcome measures, analysis samples, and planned analyses for the placebo-controlled period of the HEALEY ALS Platform Trial as specified in the Master SAP (M-SAP). The M-SAP and this R-SAP supplement the Master Protocol, the "ALS Master Protocol Recommended Statistical Analysis, Design and Simulation Report" (Appendix 1 to the Master Protocol), and the RGC Regimen-specific Appendix (RGC RSA). Please refer to the Master Protocol and the RGC RSA for details on the rationale for the study design, eligibility criteria, conduct of the trial, clinical assessments and schedule of assessments, definitions and reporting of adverse events, data management conventions, and regulatory oversight and compliance procedures. The "ALS Master Protocol Recommended Statistical Analysis, Design and Simulation Report" (MPRDR) and any regimen-specific deviations described in the RGC RSA, and this R-SAP are authoritative in defining the primary and interim analyses. In case of discrepancies between the RGC RSA and this R-SAP concerning use of shared placebos, this R-SAP is authoritative. In case of discrepancies between either SAP and the Master Protocol and the RGC RSA concerning matters of analysis other than the primary and interim analyses and use of shared placebos, the M-SAP and this R-SAP are authoritative. In case of discrepancies between the M-SAP and this R-SAP, this R-SAP is authoritative. On all matters not related to analysis, the Master Protocol and the RGC RSA are authoritative. The following table describes relationships among the relevant documents in adjudicating possible discrepancies with higher numbers indicating greater authority.

| Issues potentially requiring adjudication                                                                | Master Protocol | RGC RSA | MPRDR | M-SAP | RGC R-SAP |
|----------------------------------------------------------------------------------------------------------|-----------------|---------|-------|-------|-----------|
| Use of shared placebos                                                                                   | 1               | 4       | 2     | 3     | 5         |
| Primary and interim analysis specifications not related to use of shared placebo                         | 1               | 5       | 4     | 2     | 3         |
| Statistical analysis specifications not related to use of shared placebo or primary and interim analyses | 1               | 3       | 2     | 4     | 5         |
| All matters not related to statistical analysis                                                          | 4               | 5       | 1     | 2     | 3         |

## 2. Study Design

### 2.1 Overview

The HEALEY ALS Platform Trial is a perpetual multi-center, multi-regimen clinical trial evaluating the safety and efficacy of investigational products for the treatment of ALS. RGC evaluates the safety and efficacy of CNM-Au8 administered orally at dosages of 30 mg/d and 60 mg/d vs. placebo. The RGC RSA describes the nature of the intervention and its mechanism of action, the mode and frequency of administration, additional eligibility criteria beyond those specified in the Master Protocol, additional enrollment procedures, and additions and modifications of safety and efficacy assessments relative to those outlined in the Master Protocol.

## 2.2 Study Objectives

Primary Efficacy Objective:

- To evaluate the efficacy of CNM-Au8 as compared to placebo on ALS disease progression.

Secondary Efficacy Objectives:

- To evaluate the effect of CNM-Au8 on selected secondary measures of disease progression, including survival.

Safety Objectives:

- To evaluate the safety of CNM-Au8 for ALS.

Exploratory Efficacy Objectives:

- To evaluate the effect of CNM-Au8 on selected biomarkers and endpoints.
- To explore CNM-Au8 pharmacokinetics (PK) and pharmacodynamic (PD) effects.

## 2.3 Study Population

In addition to eligibility criteria specified in the Master Protocol, participants in RGC must not have a history of allergy to gold, gold salts, or colloidal gold preparations.

Participants will be recruited from approximately 60 centers located throughout the US that are part of the Northeast ALS (NEALS) Consortium.

## 2.4 Participant Flow

Participants in RGC follow the consenting, Master screening, regimen assignment, regimen-specific screening, randomization to active or placebo treatment, and follow-up procedures and timing described in the M-SAP. Detailed descriptions of study procedures and timing are specified in the Master Protocol and the RGC RSA.

## 2.5 Regimen Allocation

Participants in RGC are those determined eligible for Master Protocol-level inclusion and exclusion criteria and randomly assigned to RGC, stratified by use of: (i) riluzole, (ii) edaravone, (iii) both, or (iv) neither at the time of screening for the Master Protocol. Details of regimen assignment are described in the Platform Trial Regimen Assignment Plan.

## 2.6 Treatment Allocation

Participants in RGC are randomly allocated in a 3:3:2 ratio to 30 mg/d CNM-Au8, 60 mg/d CNM-Au8, or placebo treatment based on a pre-specified permuted-block randomization schedule, stratified by use of (i) riluzole, (ii) edaravone, (iii) both, or (iv) neither at the time of screening for the Master Protocol.

## 2.7 Treatment Administration

CNM-Au8 and placebo are supplied as pairs of bottles each containing 60 mL of liquid. Bottles of active study drug at the 30 mg/d dosage contain CNM-Au8 at a concentration of 250 µg/mL (15 mg per bottle). Bottles of active study drug at the 60 mg/d dosage contain CNM-Au8 at a

concentration of 500 µg/mL (30 mg per bottle). All formulations contain 32.8 mg sodium bicarbonate and 60 mL USP purified water.

A normal dosage is two (2) 60 mL bottles of study drug daily. If participants have difficulty tolerating study drug and with Site Investigator and Medical Monitor approval, participants may reduce the dosage to one (1) bottle daily. Participants who have down titrated may be rechallenged at full dosage with approval by the Site Investigator and Medical Monitor. Participants who cannot tolerate the rechallenge should remain at the dosage of one (1) bottle daily. Drug holidays (e.g., treatment-free periods) are not permitted.

The first dose of study drug should be administered while in the office/clinic on the day of the Baseline Visit after all visit assessments are complete. Additional details of treatment administration are described in the RGC RSA.

## 2.8 Allocation Concealment

Allocation concealment is the same as described in the M-SAP. All formations of study drug are identical in appearance (size, shape, volume, and color) and smell.

## 2.9 RGC Schedule of Activities (SoA)

| Activity                                 | MP<br>Scr <sup>n1</sup> | RGC<br>Scr <sup>n2</sup> | Base-<br>line | Week<br>2   | Week<br>4 <sup>3</sup> | Week<br>8 <sup>3</sup> | Week<br>12  | Week<br>16       | Week<br>20   | Week<br>24 <sup>3</sup> | Final<br>Call <sup>4</sup> |
|------------------------------------------|-------------------------|--------------------------|---------------|-------------|------------------------|------------------------|-------------|------------------|--------------|-------------------------|----------------------------|
|                                          | Cln                     | Cln                      | Cln           | Phn         | Cln <sup>5</sup>       | Cln <sup>5</sup>       | Phn         | Cln <sup>5</sup> | Phn          | Cln                     | Phn                        |
|                                          | -42d<br>to -1d          | -41d<br>to 0d            | Day<br>0      | Day<br>14±3 | Day<br>28±7            | Day<br>56±7            | Day<br>84±3 | Day<br>112±7     | Day<br>140±3 | Day<br>168±7            | 28d ±3<br>ALD              |
| Written Informed Consent <sup>6</sup>    | X                       | X                        |               |             |                        |                        |             |                  |              |                         |                            |
| Inclusion/Exclusion Review <sup>7</sup>  | X                       | X                        |               |             |                        |                        |             |                  |              |                         |                            |
| ALS & Medical History                    | X                       |                          |               |             |                        |                        |             |                  |              |                         |                            |
| Demographics                             | X                       |                          |               |             |                        |                        |             |                  |              |                         |                            |
| Physical Examination                     | X                       |                          |               |             |                        |                        |             |                  |              |                         |                            |
| Neurological Exam                        | X                       |                          |               |             |                        |                        |             |                  |              |                         |                            |
| Vital Signs <sup>8</sup>                 | X                       |                          | X             |             | X                      | X                      |             | X                |              | X                       |                            |
| Slow Vital Capacity                      | X <sup>9</sup>          |                          | X             |             |                        | C                      |             | C                |              | X                       |                            |
| Home Spirometry                          | X <sup>9</sup>          |                          | X             |             |                        | X                      |             | X                |              | X                       |                            |
| Muscle Strength Assessment               |                         |                          | X             |             |                        | C                      |             | C                |              | X                       |                            |
| ALSFRS-R                                 | X                       |                          | X             |             | X                      | X                      | X           | X                | X            | X                       |                            |
| ALSAQ-40                                 |                         |                          | X             |             |                        |                        |             |                  |              | X                       |                            |
| CNS-BFS                                  |                         |                          | X             |             |                        | X                      |             | X                |              | X                       |                            |
| 12-Lead ECG                              | X                       |                          |               |             |                        |                        |             |                  |              | X                       |                            |
| Clinical Safety Labs <sup>10</sup>       | X                       |                          | X             |             | X                      | X                      |             | X                |              | X                       |                            |
| CNM-Au8 PK Samples <sup>11</sup>         |                         |                          | X             |             | C                      | C                      |             |                  |              | X                       |                            |
| CNM-Au8 PD Samples <sup>11</sup>         |                         |                          | X             |             | C                      | C                      |             |                  |              | X                       |                            |
| Biomarker Blood Collection <sup>11</sup> |                         |                          | X             |             |                        | C                      |             | C                |              | X                       |                            |
| Biomarker Urine Collection <sup>11</sup> |                         |                          | X             |             |                        | C                      |             | C                |              | X                       |                            |
| DNA Collection <sup>12</sup> (optional)  |                         |                          | X             |             |                        |                        |             |                  |              |                         |                            |
| CSF Collection (optional)                |                         |                          | X             |             |                        |                        |             | C <sup>13</sup>  |              |                         |                            |
| Concomitant Medication Review            | X                       | X                        | X             | X           | X                      | X                      | X           | X                | X            | X                       |                            |
| Adverse Event Review <sup>14</sup>       | X                       | X                        | X             | X           | X                      | X                      | X           | X                | X            | X                       | X                          |
| Suicidality C-SSRS                       |                         |                          | X             |             | X                      | X                      |             | X                |              | X                       |                            |
| Install Smartphone Apps <sup>15</sup>    |                         |                          | X             |             |                        |                        |             |                  |              |                         |                            |
| Smartphone Voice Recording <sup>16</sup> |                         |                          | X             |             | X                      | X                      |             | X                |              | X                       |                            |
| Uninstall Smartphone App                 |                         |                          |               |             |                        |                        |             |                  |              | X                       |                            |
| Regimen Assignment                       | X                       |                          |               |             |                        |                        |             |                  |              |                         |                            |

| Activity                       | MP<br>Scrn <sup>1</sup> | RGC<br>Scrn <sup>2</sup> | Base-<br>line   | Week<br>2       | Week<br>4 <sup>3</sup> | Week<br>8 <sup>3</sup> | Week<br>12      | Week<br>16       | Week<br>20      | Week<br>24 <sup>3</sup> | Final<br>Call <sup>4</sup> |
|--------------------------------|-------------------------|--------------------------|-----------------|-----------------|------------------------|------------------------|-----------------|------------------|-----------------|-------------------------|----------------------------|
|                                | Cln                     | Cln                      | Cln             | Phn             | Cln <sup>5</sup>       | Cln <sup>5</sup>       | Phn             | Cln <sup>5</sup> | Phn             | Cln                     | Phn                        |
|                                | -42d<br>to -1d          | -41d<br>to 0d            | Day<br>0        | Day<br>14±3     | Day<br>28±7            | Day<br>56±7            | Day<br>84±3     | Day<br>112±7     | Day<br>140±3    | Day<br>168±7            | 28d ±3<br>ALD              |
| Randomization within RGC       |                         |                          | X               |                 |                        |                        |                 |                  |                 |                         |                            |
| Administer/Dispense Study Drug |                         |                          | X <sup>17</sup> |                 |                        | X                      |                 | X                |                 |                         |                            |
| Drug Accountability/Compliance |                         |                          |                 | X <sup>18</sup> | X                      | X                      | X <sup>18</sup> | X                | X <sup>18</sup> | X                       |                            |
| Exit Questionnaire             |                         |                          |                 |                 |                        |                        |                 |                  |                 | X                       |                            |
| Vital Status                   |                         |                          |                 |                 |                        |                        |                 |                  |                 | X <sup>19</sup>         |                            |

Abbreviations: ALD = after last dose, ALS = amyotrophic lateral sclerosis, ALSAQ-40 = ALS Assessment Questionnaire, ALSFRS-R = ALS Functional Rating Scale Revised, BP = blood pressure, C = completed only if the visit is conducted in-clinic, CBC = complete blood count, Cln = Clinic visit, CNS-BFS = Center for Neurologic Study Bulbar Function Scale, CSF = cerebrospinal fluid, C-SSRS = Columbia-Suicide Severity Rating Scale, d = day, DNA = deoxyribonucleic acid, ECG = electrocardiogram, LFTs = liver function tests, MP = Master Protocol, PD = pharmacodynamic, Phn = Phone visit, RGC = the CNM-Au8 regimen, Scrn = Screening Visit.

<sup>1</sup> Master Protocol (MP) screening procedures must be completed between 42 days and 1 day prior to the Baseline Visit.

<sup>2</sup> RGC regimen-specific screening procedures must be completed no more than 41 days prior to the Baseline Visit. RGC regimen-specific screening procedures may be completed on the same day as the Baseline Visit.

<sup>3</sup> Participants should be instructed to hold the morning dose of study drug on the day of the Week 4, 8, and 24 study visits. Study drug should not be taken until after study visit procedures are complete.

<sup>4</sup> Participants who are participating into the Open-label Extension (OLE) at the time scheduled for the Follow-up Safety Call will not complete a Follow-up Safety Call. Participants who continue into the OLE and then terminate prior to the time scheduled for the Follow-up Safety Call should be asked to complete a Follow-Up Safety Call.

<sup>5</sup> The Week 4, 8, and 16 visits may be conducted via telemedicine (or via phone if telemedicine is not available) with remote services instead of in-clinic if this is needed to protect the safety of the participant due to a pandemic, or other reason. If a planned in-clinic visit is conducted via telemedicine (or via phone if telemedicine is not available) with remote services, only selected procedures will be performed: ALSFRS-R, home spirometry (Weeks 8 and 16 only), Center for Neurologic Study Bulbar Function Scale (CNS-BFS, Weeks 8 and 16 only), voice recording, adverse event review, Columbia Suicide Severity Rating Scale, concomitant medications, study drug accountability/compliance, and administer/dispense study drug. Additionally, selected vital signs (systolic and diastolic pressure, respiratory rate, heart rate, and temperature) and clinical safety labs will be performed by a home health agency.

<sup>6</sup> During the Master Protocol Screening Visit, participants will be consented using the Master Protocol informed consent form (ICF). After a participant is randomly assigned to RGC, he or she will be consented a second time using the RGC ICF.

<sup>7</sup> At the MP Screening Visit, participants will have MP inclusion and exclusion criteria assessed. At the RGC Screening Visit, participants will have regimen-specific inclusion and exclusion criteria assessed.

<sup>8</sup> At all visits, whether conducted in-clinic or via telemedicine (or via phone if telemedicine is not available), vital signs include systolic and diastolic blood pressure, heart rate, respiratory rate, and temperature. At all in-clinic visits, vital signs include weight. At the MP Screening Visit only, vital signs include height measured in cm.

<sup>9</sup> If required due to pandemic-related restrictions, forced vital capacity (FVC) performed by a Pulmonary Function Laboratory evaluator or with a study-approved home spirometer or sustained phonation using a study-approved method may be used for eligibility (Master Protocol Screening ONLY).

<sup>10</sup> Clinical safety labs include hematology (CBC with differential), complete chemistry panel, liver function tests, thyroid function, and urinalysis. Serum pregnancy testing will occur in women of child-bearing potential at the Master Protocol Screening Visit and as necessary during participation. Pregnancy testing is only repeated as applicable if there is a concern for pregnancy.

<sup>11</sup> Whole blood samples for CNM-Au8 PK, plasma samples for riluzole PK, and whole blood, plasma, and urine samples for PD biomarkers should be obtained prior to the first daily dose of study drug at each applicable clinic visit. The time of the most recent dose of study drug, the most recent meal prior to sampling, and the time of sample collection should be reported. Whole blood and urine will be collected only at the Baseline and Week 24/Early Termination visits.

<sup>12</sup> If the participant consents to DNA sample collection but a DNA sample is not obtained at the Baseline Visit or if that sample is not usable, a DNA sample can be collected after the Baseline Visit.

<sup>13</sup> If the participant consents to CSF sample collection but a CSF sample is not obtained at the Week 16 Visit, a CSF sample can be collected at the Week 24 Visit.

<sup>14</sup> Adverse events that occur after signing the RGC ICF will be recorded.

<sup>15</sup> Two smartphone apps should be installed on the participant's phone, one to collect home spirometry and one to collect voice recordings.

<sup>16</sup> In addition to study visits specified in the SoA for collection of voice recordings, participants should complete twice weekly voice recordings at home. During weeks when a participant completes a voice recording in-clinic, he or she should only complete one other voice recording at home.

<sup>17</sup> Administer first dose of study drug only after Baseline Visit procedures are completed. Participants should take the first dose of study drug while in the office/clinic on the day of the Baseline visit.

<sup>18</sup> Drug accountability will not be done at phone visits. A drug compliance check-in must be held during phone visits to ensure participant is taking drug per dose regimen and to note any report of missed doses.

<sup>19</sup> Vital status, a determination of date of death or death equivalent or date last known alive, will be made for each randomized participant at the end of their double-blind follow-up (generally the Week 24 Visit as indicated) and again, if previously alive, after the end of double-blind follow-

up of all participants randomized to RGC. Vital status may also be ascertained at later time points by using publicly available data sources.

### **3. General Considerations for Data Analysis**

#### **3.1 Statistical Software**

Statistical software use for analyses is the same as described in the M-SAP.

#### **3.2 Summary Statistics**

Data summaries are the same as described in the M-SAP.

#### **3.3 Precision**

Precision of reported results is the same as described in the M-SAP.

#### **3.4 Transformations**

Data transformations are the same as described in the M-SAP.

#### **3.5 Multiplicity Adjustments**

Handling of multiplicity adjustments is the same as described in the M-SAP with the addition that the primary contrast for any given analysis will be between all active arm participants (i.e., both 30 mg/d and 60 mg/d dosage groups) vs. placebo participants. Comparisons between the two dosage groups and between each dosage group and placebo will be exploratory.

#### **3.6 Missing Data**

Handling of missing data is the same as described in the M-SAP. Clinic-based assessments that are missing due to COVID-19 restrictions or disruptions are considered missing at random.

### **4. Study Endpoints**

#### **4.1 Efficacy Endpoints**

The primary endpoint is the same as described in the M-SAP. ALSFRS-R total score is considered the primary efficacy endpoint as described in the M-SAP. Combined assessment of function and survival (CAFS), slow vital capacity (SVC) change, and survival (time to death or death equivalent) are considered key secondary efficacy endpoints in RGC.

#### **4.2 Exploratory Endpoints**

The following categories of exploratory endpoints will be evaluated:

- Change in ALSFRS-R domain scores,
- Change in strength: hand-held dynamometry (HHD) global percentage, HHD upper extremity percentage, HHD lower extremity percentages, HHD0, and HHD0<sup>2</sup>,



- Change in quantitative voice characteristics as measured by Aural Analytics: maximum phonation time, pause rate, breathy vocal quality, pitch instability, regulation of voicing, articulatory precision, speaking rate, articulation rate, and monotonicity,
- Change in biofluid biomarkers of neurodegeneration and neuromuscular degeneration: serum creatinine, serum and cerebrospinal fluid (CSF) neurofilament light chain (NfL), and urinary neurotrophin receptor p75 protein extracellular domain (p75[ECD]) normalized to urinary creatinine,
- Change in patient-reported outcomes: ALSAQ-40 physical mobility, independence in activities of daily living, eating and drinking, communications, and emotional reactions domain scores and ALSAQ-40 symptom index, CNS-BFS total score,
- Change in plasma and whole blood concentrations of CNM-Au8,
- Change in CNM-Au8 PD biomarkers,
- Change in respiratory function as assessed by home spirometry,
- Time to clinical events:
  - First hospitalization due to a serious adverse event (SAE),
  - First hospitalization due to an ALS-related SAE,
  - First use of assisted ventilation,
  - First placement of a feeding tube,
  - First time reaching King's stage 4a or 4b,
  - First instance of any of the following events: feeding tube placement, tracheostomy, initiation of permanent assisted ventilation (PAV), or death,
  - First instance of any of the following events: hospitalization for an SAE, feeding tube placement, tracheostomy, initiation of PAV, or death.
- Time to first occurrence of a  $\geq 6$ -point decline in ALSFRS-R total score
- A composite of the standardized slope estimates for rate of change in ALSFRS-R total score, SVC percent-predicted, and ALSAQ-40 total score.

### 4.3 Safety Endpoints

In addition to the safety endpoints described in the M-SAP, the following RGC regimen-specific safety endpoint will be evaluated:

- Safety lab alerts: Proportion of participants with ALT  $>3\times$  upper limit of normal (ULN), AST  $>3\times$  ULN, creatinine  $>1.5\times$  level at the Baseline Visit, and platelet count  $<75,000/\text{mm}^3$ , and
- Pharmacokinetics of riluzole.



## 5. Measurement Definitions

### 5.1 ALSFRS-R

The definitions of ALSFRS-R scores are the same as described in the M-SAP. Pre-baseline slope in ALSFRS-R (delta-FRS) is defined as 48 minus the baseline ALSFRS-R total score then divided by the number of months from onset of symptomatic weakness to the Baseline Visit. The number of months will be calculated as the difference in days from onset of symptomatic weakness to the Baseline Visit multiplied by 12 / 365.25. The date of onset of symptomatic weakness will be imputed as the fifteenth day of a month if not specified more precisely.

ALSFRS-R domain scores are exploratory measures of the primary efficacy endpoint ALSFRS-R total score.

Time to first occurrence of a  $\geq 6$ -point decline in ALSFRS-R total score is a composite endpoint with death or death equivalent, whichever occurs first, and will be censored at the last date at which an ALSFRS-R assessment was performed up to the end of the Week 24 Visit window.

### 5.2 SVC

The derivation of SVC percent-predicted of normal is the same as described in the M-SAP with age calculated as number of days from date of birth to the date of a given SVC assessment divided by 365.25 and with the following correspondence between self-identified race and race defined by Global Lung Initiative (GLI) classification:

| Self-identified Race                      | GLI-defined Race |
|-------------------------------------------|------------------|
| American Indian or Alaska Native          | Mixed/Other      |
| Asian                                     | South East Asian |
| Black or African American                 | African American |
| Native Hawaiian or Other Pacific Islander | Mixed/Other      |
| White                                     | Caucasian        |
| Unknown                                   | Caucasian        |
| Not reported                              | Caucasian        |
| More than one race indicated              | Mixed/Other      |

### 5.3 Home Spirometry

Home spirometry assesses FVC remotely using a smartphone app (ZEPHYRx, Albany, NY) and a handheld spirometer (Spirobank Smart, Medical International Research, Rome, Italy). Coordinators guide participants through 3 to 8 maneuvers with live-video coaching using the ZEPHYRx platform. Flow loops are classified for acceptability and repeatability using American Thoracic Society (ATS) criteria and are manually reviewed by the NEALS Outcomes Center (Barrow Neurological Institute, Phoenix, AZ). The maximum FVC accepted by the NEALS Outcomes Center is converted to percent of predicted normal using GLI norms based on sex, age at time of assessment, height at time of screening, and race. Age is calculated as number of days from date of birth to the date of a given home spirometry assessment divided by 365.25. Higher values indicate greater respiratory function.

## 5.4 HHD and Grip Strength

The derivation of HHD upper and lower extremity scores and HHD0 are the same as described in the M-SAP with the revision that HHD0 is a composite endpoint with death or death equivalent.

A second HHD time-to-event endpoint is defined as the time from the Baseline Visit to the second post-baseline occurrence of a muscle with a strength recording of 0 among those muscles that were non-zero at baseline, or time to death or death equivalent, whichever occurs first (HHD0<sup>2</sup>).

Time at risk for HHD0 and HHD0<sup>2</sup> will be censored at the last date at which an HHD assessment was performed up to the end of the Week 24 Visit window.

HHD lower extremity average percentage, HHD upper extremity average percentage, HHD global average percentage, HHD0, and HHD0<sup>2</sup> are exploratory measures.

## 5.5 Quantitative Voice Characteristics

Voice samples will be collected using the Aural Analytics app installed on either an Android or iOS-based smartphone. At each assessment, participants perform a set of speaking tasks: reading 5 prespecified sentences, reading 5 sentences chosen at random from a large sentence bank, repeating a consonant-vowel sequence, producing a sustained phonation, and counting on a single breath. Speech analysis will be performed by Aural Analytics to derive the following quantitative voice characteristics: maximum phonation time, pause rate, breathy vocal quality, pitch instability, regulation of voicing, articulatory precision, speaking rate, articulation rate, and monotonicity. Aural Analytics will use data on quantitative voice characteristics and participant age, sex, race, height, and weight to derive a prediction of vital capacity at the Baseline Visit.

## 5.6 Biofluid Biomarkers of Neurodegeneration

Biofluid biomarkers of neurodegeneration, including biomarkers of neuromuscular dysfunction, will be assayed. These will include serum creatinine, serum and CSF neurofilament light chain (NfL), and urinary p75[ECD]. Serum creatinine will be assayed by the kinetic Jaffe method (test 001370, Labcorp, Burlington, NC). NfL will be assayed by single-molecule array (Simoa; Quanterix, Billerica, MA). Urinary p75[ECD] will be assayed by sandwich enzyme-linked immunoassay (Rogers lab, Flinders Univ, Adelaide, Australia) and normalized to urinary creatinine. Levels of serum and CSF NfL and urinary p75[ECD] that are reported to be below the limit of quantitation will be imputed at the limit of quantitation. Levels of serum and CSF NfL will be log-transformed in all analyses.

## 5.7 ALSAQ-40

The description of the ALSAQ-40 instrument and item-level scores are the same as described in the M-SAP. Each of the five domains will be scored as the mean of all domain-specific items multiplied by 25 (range 0 to 100). An overall symptom index (SI) will be scored as the mean of the five domain scores. A domain score will be missing if more than 20% of the items are missing; otherwise, item non-response will be mean-imputed from other completed items from the same assessment. The ALSAQ-40 SI will be missing if any domain scores are missing. Higher scores indicate worse quality of life.

## 5.8 CNS-BFS

The definition of CNS-BFS total score is the same as described in the M-SAP.

## 5.9 Survival

The primary definition of survival time is the same as described in the M-SAP with the clarification that PAV is defined as more than 22 hours per day of noninvasive or invasive mechanical ventilation for more than seven consecutive days. The date of PAV initiation, where applicable, will be imputed as the fifteenth day of a month if not specified more precisely. A secondary survival endpoint of death alone, independent of any death equivalent, is also defined.

Time at risk for the composite endpoint of death or death equivalent and time at risk for the endpoint of death alone will be measured from each participant's Baseline Visit. Time at risk will be censored at two time points: (1) at the Week 24 Visit as defined in the M-SAP, and (2) at a subsequent assessment of death or death equivalent scheduled approximately at the end of placebo-controlled follow-up of the last RGC participant, with the latter effectively a comparison of either no-treatment or 24-week delayed treatment (for those participants who enter the open-label extension). The primary analysis of survival will evaluate PAV-free survival to the Week 24 Visit time point.

## 5.10 Pharmacokinetics of CNM-Au8

Whole blood and plasma samples collected at the Baseline and Week 24 visits will be analyzed to quantify the concentration of CNM-Au8. Details of the CNM-Au8 assay technique and analysis plan will be specified in a separate PK analysis plan. This plan will be finalized prior to database lock.

## 5.11 King's ALS Clinical Staging System

The King's ALS Clinical Staging System (Roche et al. 2012) is a 4-level ordinal scale with the first three levels indicating the number (1, 2, or 3) of distinct central nervous system regions (bulbar, upper limb, and lower limb) with neuromuscular dysfunction and levels 4a and 4b indicating nutritional or respiratory failure secondary to ALS, respectively.

Participants will be classified to King's stage 1, 2, 3, 4a, or 4b based on scores from ALSFRS-R assessments according to a published derivation (Balendra et al. 2014). Bulbar involvement is defined as a score less than 4 on any of the ALSFRS-R questions in the bulbar domain (questions 1, 2, and 3). Upper limb involvement is defined as a score less than 4 on either of the ALSFRS-R questions related to hand function (questions 4 and 5A). Lower limb involvement is defined as a score less than 4 on the ALSFRS-R question about walking (question 8). Nutritional failure is defined as responding that the participant uses gastrostomy for greater than 50% of their nutrition. Respiratory failure is defined as a score of 0 on the ALSFRS-R question addressing dyspnea (question 10 or R-1) or a score less than 4 on the ALSFRS-R question about use of mechanical ventilation (question 12 or R-3). Participants without evidence by ALSFRS-R scores of involvement of any of the three central nervous system regions will be scored as King's stage 1 due to their confirmed diagnosis with ALS. Participants may meet criteria for both King's stage 4a and 4b.

### 5.12 Hospitalization and Other Clinical Events

Times to the following clinically relevant events are defined:

- Time to first hospitalization due to a serious adverse event (SAE),
- Time to first hospitalization due to an ALS-related SAE,
- Time to first use of assisted ventilation,
- Time to first placement of a feeding tube,
- Time to King's stage 4a or 4b,
- Time to first instance of feeding tube placement, tracheostomy, initiation of PAV, or death, and
- Time to first instance of hospitalization for an SAE, feeding tube placement, tracheostomy, initiation of PAV, or death.

Time at risk for each event will be measured from each participant's Baseline Visit. Time to first hospitalization excludes hospitalizations for elective procedures. ALS-related SAEs are those indicated as related to ALS disease progression by the site investigator. Participants will be excluded from respective analysis (as applicable) if: (i) they are already using assisted ventilation or have a feeding tube at the time of the Baseline Visit, (ii) they have answered Question 5B on the ALSFRS-R at the Baseline visit, or (iii) have scored 0 points on Question 10 or less than 4 points on Question 12 of the ALSFRS-R at the Baseline visit. Death or death equivalent will be considered an outcome for each of the events listed, forming a composite endpoint.

Time at risk for these events will be censored at the Week 24 Visit, if completed, the date of consent withdrawal, if withdrawn, or the last date at which the status of each endpoint is known prior to the end of the Week 24 Visit window for participants lost to follow-up. Time to King's stage 4a or 4b is interval censored between ALSFRS-R assessments.

### 5.13 Pharmacodynamic Biomarkers

Whole blood and urine PD samples collected at the Baseline and Week 24 visits and plasma PD samples collected at the Baseline, Week 4, Week 8, and Week 24 visits will be analyzed according to a separate PD analysis plan. The PD analysis plan will be finalized prior to database lock.

### 5.14 Pharmacokinetics of Riluzole

Plasma samples collected at the Baseline, Week 4, and Week 8 visits from the first forty (40) RGC participants taking riluzole who reach the Week 8 visit plus plasma samples collected at the Baseline Visit from an additional ten (10) RGC participants not taking riluzole will be analyzed to quantify riluzole concentration by high-performance liquid chromatography mass spectroscopy (LC/MS/MS) [Covance, Method ID: M10125, Study Number 8454-985].

### 5.15 Clinical Safety Laboratory Tests

Clinical safety labs include hematology, blood chemistry panel, liver function tests, thyroid function, urinalysis, and pregnancy testing in women of childbearing potential as specified in Section 9.1.2 Clinical Safety Laboratory Tests of the Master Protocol:

- Hematology: hematocrit, hemoglobin, platelet count, red blood cell (RBC) count, mean corpuscular volume, mean corpuscular hemoglobin, mean corpuscular hemoglobin concentration, RBC distribution width (RDW), RBC morphology, white blood cell (WBC) count, and counts and percentages of basophils, eosinophils, lymphocytes, monocytes, and neutrophils;
- Blood chemistry panel: bicarbonate, chloride, potassium, sodium, calcium, magnesium, phosphate, blood urea nitrogen, creatinine, estimated glomerular filtration rate (eGFR) calculated using the Modification of Diet in Renal Disease (MDRD) four-variable equation, creatinine clearance calculated using the Cockcroft-Gault equation, and glucose;
- Liver function tests: alanine aminotransferase (ALT [SGPT]), aspartate aminotransferase (AST [SGOT]), alkaline phosphatase (ALP), albumin, total protein, total bilirubin (TBL);
- Thyroid function tests: thyroid-stimulating hormone (TSH);
- Urinalysis: clarity, color, specific gravity, pH, microalbumin, protein, glucose, ketones, bilirubin, urobilinogen, nitrite, leukocyte esterase, and blood; and
- Pregnancy: qualitative and quantitative serum human chorionic gonadotropin (hCG).

Clinical safety labs will also include derived measures of potential drug-induced liver injury (DILI), including those that potentially meet the Hy's law criteria, as distinct safety lab outcomes.

Three potential DILI criteria will be defined:

- ALT or AST >3x ULN with TBL >1.5x ULN
- AST or ALT >3x ULN with TBL >2x ULN
- AST or ALT >3x ULN with TBL >2x ULN and ALP <2x ULN (potential Hy's Law cases)

where ULN is upper level of normal and all levels are measured on the same day.

## 6. Statistical Methodology

### 6.1 Analysis Sets

The ITT analysis set is henceforth referred to as the Full Analysis Set (FAS) and defined as follows:

- Full Analysis Set (FAS): Participants who were randomized within RGC plus placebo participants from specified regimens, classified according to their randomized treatment assignment. Observations made after premature permanent discontinuation of study drug are included in this sample, should such participants remain on study. Observations completed after regimen data lock are excluded. Participants determined to not meet ALS diagnostic criteria are excluded.

The definition of the STF analysis set is revised as follows:

- Safety Full (STF) Set: Participants who initiated treatment within RGC plus placebo participants from specified regimens who are not known to be ineligible for RGC and who initiated treatment in their respective regimen, classified according to the treatment they

actually received. Observations made after premature permanent discontinuation of study drug are included in this sample, should such participants remain on study. Observations completed after regimen data lock are excluded.

An analysis set restricting shared placebo participants to those regimens in which study drug is administered by the same route as RGC is defined as follows:

- Efficacy Common Mode of Administration (ECM) Set: The subset of participants in the FAS analysis set who are in regimens in which study drug is administered by the same route as RGC (oral administration).

The definitions of the ECC, ERO, STN, and SRO analysis sets are the same as described in the M-SAP with reference to the ITT analysis set now referencing the FAS analysis set. The following analysis set is specific to RGC:

- Efficacy Per-protocol (EPP) Set: The subset of participants in the FAS analysis set who initiated study treatment and who were not involved in protocol deviations that affected the scientific integrity of the trial as documented prior to data lock, classified according to the treatment they actually received. Inclusion or exclusion from the EPP analysis set of any participant for whom treatment assignment was unblinded prior to data lock will be governed by the prespecified criteria above. If a participant's data is truncated for inclusion in the EPP analysis set due to non-adherence to protocol-specified dosing, clinical events observed up to 28 days after the censoring event will be included in the EPP analysis set. For all other events leading to truncation of a participant's data, no events beyond that date will be included. Data from placebo participants shared from other regimens will not be truncated due to non-adherence to protocol-specified dosing.

Applicable analysis sets (FAS, ECM, EPP, STF, and STN) will include shared placebo participants from regimens A, B, and D. Data from shared placebo participants will include visits and events that occurred on or before the date of the final placebo-controlled period follow-up of an RGA, RGB, or RGC participant. As only concurrently enrolling regimens are contributing to efficacy analyses, the FAS and ECC analysis sets are synonymous and only the FAS analysis set will be referenced. As regimen A is administered by subcutaneous injection and regimens B and D are administered orally, only regimens B and D will contribute shared placebo participants for the ECM and STN analysis sets.

## 6.2 Baseline Characterization

The baseline characteristics summarized for participants randomized within RGC are the same as specified in the M-SAP with the addition of ALSAQ-40 domain scores and SI, CNS-BFS total score, King's stage, weight, body mass index (BMI), serum urate concentration, serum creatinine concentration, serum NfL concentration, urinary p75[ECD], and vital capacity predicted from quantitative voice characteristics.

## 6.3 Primary Efficacy Analysis and Supportive Analyses

The primary analysis for RGC is a Bayesian shared-parameter, repeated-measures model of ALSFRS-R that accounts for loss of follow-up due to mortality. Details of the model, including documentation of operating characteristics under a range of scenarios, are provided in the "ALS Master Protocol Recommended Statistical Analysis, Design and Simulation Report" (Appendix 1 to the Master Protocol). The Bayesian shared-parameter, repeated-measures model will be



applied to the FAS analysis set as the primary analysis, to the ECM and ERO analysis sets as sensitivity analyses, and to the EPP analysis set as a supportive analysis.

The estimand of the primary analysis is the relative rate of disease progression (the "disease rate ratio" or DRR) of the two active treatment groups relative to placebo in the FAS population under an assumption that active treatment slows mean time to death or death equivalent by the same proportion as treatment slows the mean rate of functional progression as measured by change in ALSFRS-R total score over time. The estimand is defined by the following attributes:

- Treatment: CNM-Au8 administered orally at dosages of 30 mg/d and 60 mg/d vs. placebo.
- Population: FAS population as defined in Section 6.1.
- Variables: time to death or death equivalent and rate of change in ALSFRS-R total score from baseline to the Week 24 Visit.
- Intercurrent event 1: treatment discontinuation due to death: no ALSFRS-R data from participants who reach the death or death equivalent endpoint are included in the analysis; instead, death or death equivalent are handled by a mortality component within the model, composite variable strategy approach.
- Intercurrent event 2: treatment discontinuation not due to death: handled via treatment policy approach, all data will be used including data collected during the placebo-controlled period after treatment discontinuation regardless of concomitant medication, for those participants who have not been censored due to mortality. Missing data post-treatment will not be imputed, handled via missing at random assumption.
- Population-level summary: mean ratio of hazard or progression rate of active treatment relative to placebo.

## 6.4 Interim Analysis

RGC will be considered for early stopping for futility according to the interim analysis schedule and definition specified in the "ALS Master Protocol Recommended Statistical Analysis, Design and Simulation Report" (Appendix 1 to the Master Protocol). RGC will not be stopped early for success.

## 6.5 Secondary Efficacy Analyses

### 6.5.1 Hierarchical Testing

Primary inference for secondary efficacy endpoints will be based on analysis of the FAS analysis set using a joint-rank comparison for the combined assessment of function and survival (CAFS (Section 6.5.2 below), a repeated-measures linear mixed model for SVC (see Section 6.5.3 below), and Kaplan-Meier product-limit estimates and log-rank test for the primary survival endpoint (see Section 6.5.4 below) with a comparison between all active arm participants (i.e., the average of both 30 mg/d and 60 mg/d dosage groups) vs. placebo participants. The sequence for testing secondary efficacy endpoints is the following:

1. CAFS,
2. SVC,
3. Survival.



If the primary analysis indicates a significant slowing in disease progression from the Bayesian shared-parameter, repeated-measures model of ALSFRS-R and mortality, then each secondary efficacy endpoint in succession would be declared significant in the specified sequence using a comparison-wise criterion of two-tailed  $p < 0.05$ . After the first failure to declare significance, no endpoints lower in the hierarchy can be significant. This sequential closed-testing procedure controls the overall type 1 error rate at 5%. Nominal comparison-wise p-values for secondary efficacy endpoints will also be reported.

### 6.5.2 CAFS

The test consists of calculating a rank-sum score for each individual relative to pair-wise comparisons with all other participants. Participants are first ranked according to time to death or death equivalent when that is observed for both members of a pair or when one is censored after the observed event time for the other. Pairs that cannot be ranked by time to death or death equivalent are subsequently ranked by absolute change from baseline in ALSFRS-R total score at the maximum follow-up time at which both participants have an observation.

The primary CAFS analysis is the same as specified in the MPRDR with the clarification that the active treatment group pools participants randomized to 30 mg/d CNM-Au8 and 60 mg/d CNM-Au8.

The estimand is the stochastic probability that a randomly selected active participant will rank higher than a randomly selected placebo participant based on time to death or death equivalent or change from baseline in ALSFRS-R. The estimand is defined by the following attributes:

- Treatment: CNM-Au8 administered orally at dosages of 30 mg/d and 60 mg/d vs. placebo.
- Population: FAS population as defined in Section 6.1.
- Variables: ranksum for relative time to death or death equivalent up to the Week 24 Visit or absolute change from baseline in ALSFRS-R total score.
- Intercurrent event: treatment discontinuation: handled via treatment policy approach, with pair ranked comparisons up at the maximum follow-up time at which both participants have an observation.
- Population-level summary: difference in means of ranks of active treatment relative to placebo.

Inference from this analysis is supportive of inference from the Bayesian shared-parameter, repeated-measures model for the primary endpoint and is the primary analysis for this secondary endpoint.

The following SAS code specifies the analysis:

```
proc mixed data=work.cafs;  
  class trtrnd(ref='0');  
  model Rank = sx2b1 dFRS rlz edv trtrnd / solution clparm;  
  estimate "Active vs. Placebo (one-sided)" Treat 1 -1 / cl upper;  
run;
```

where trtrnd indicates treatment group (active = 1 or placebo = 0), Rank is the sum of the ranks (-1, 0, 1) of each participant vs. every other participant in the FAS analysis set, sx2b1 is years

since ALS symptom onset, dFRS is pre-baseline slope, r1z is an indicator of riluzole use at baseline, and edv is an indicator of edaravone use at baseline. A significant difference requires interpretation based on estimates from the survival analysis described in Section 6.5.4 and the repeated-measures analysis of ALSFRS-R total score described in Section 6.5.3.

In addition to the joint-rank CAFS analysis of PAV-free survival and change in ALSFRS-R total score including adjustment for covariates, the following additional joint-rank analyses will also be performed:

1. SVC will be analyzed as a joint-rank test by substituting the change from baseline of SVC in place of ALSFRS-R total score as an additional supportive analysis,
2. An additional set of joint-rank analyses will use multiple imputation to extend follow-up of (i) ALSFRS-R total score and (ii) SVC for participants who early terminate, withdraw consent, or are lost to follow-up as supportive analyses,
3. An additional set of joint-rank analyses for (i) ALSFRS-R total score and (ii) SVC will use time to death independent of any death equivalent as supportive analyses,
4. An additional set of joint-rank analyses for (i) ALSFRS-R total score and (ii) SVC will adjust rank scores in a linear model with the following covariates: time from ALS symptom onset, delta-FRS, baseline use of riluzole, baseline use of edaravone, and baseline serum NfL level, and
5. An additional set of joint-rank analyses for (i) ALSFRS-R total score and (ii) SVC without adjustment and comparing pairs of participants at the last jointly observed time point as sensitivity analyses.

The multiple imputation model referenced as method 2 above and used to extend follow-up of functional scores for participants who early terminate, withdraw consent, or are lost to follow-up will use linear regression with covariates of time since symptom onset, delta-FRS, baseline riluzole use, baseline edaravone use, and each observed functional score prior to a missing assessment.

Inference from the joint-rank analyses of SVC are supportive of inference from the repeated-measures model. Primary inference from the joint-rank CAFS analyses will compare survival by time to death or death equivalent, will compare change in function to the last jointly observed time point, and will adjust for the specified covariates.

### 6.5.3 Repeated-measures Model

The specification of the repeated-measures linear mixed model and the primary linear contrast for estimating differences in 24-week change from baseline in a given continuous efficacy endpoint (ALSFRS-R total and domain scores, HHD upper extremity, lower extremity, and global average percentages, SVC, FVC by home spirometry, serum creatinine, serum NfL, urinary p75[ECD], ALSAQ-40 domain scores and SI, and CNS-BFS total score) are revised from those specified in the M-SAP to include a main effect of treatment and with the modification that treatment group is a three-level variable (i.e., placebo, 30 mg/d CNM-Au8, and 60 mg/d CNM-Au8).

The model will include fixed terms for discrete visit, treatment group (3 levels), treatment group  $\times$  visit interaction, centered time since symptom onset and centered time since symptom onset  $\times$  visit interaction, centered delta-FRS and centered delta-FRS  $\times$  visit interaction, centered baseline riluzole use and centered baseline riluzole  $\times$  visit interaction, and centered baseline edaravone

use and centered baseline edaravone  $\times$  visit interaction. The following equations describe the model with regimen random effects:

$$Y_{ij} = a_{k(i)} + \gamma_1 t_i + \gamma_{2,j} v_j + \gamma'_3 \mathbf{z}_i + \gamma_{4,j} t_i v_j + \gamma'_{5,j} \mathbf{z}_i v_j + \epsilon_{ij} \quad (\text{eqn. 1})$$

$$a_k \sim N(0, \sigma_r^2), \epsilon_{i.} \sim N(\mathbf{0}, \mathbf{R}), \text{Cov}(b_{k(i)}, \epsilon_{ij}) = 0$$

where  $Y_{ij}$  is a given efficacy endpoint measured in participant  $i$  at visit  $j$ ,  $a_{k(i)}$  is a random intercept for regimen  $k$  to which participant  $i$  was assigned,  $v_j$  is an indicator variable for visit  $j$ ,  $\mathbf{z}_i$  is the vector of covariates (centered time since onset, centered delta-FRS, centered baseline riluzole use, and centered baseline edaravone use) for participant  $i$ ,  $t_i$  is an indicator variable for treatment  $t$  to which participant  $i$  was assigned,  $\gamma_1$ ,  $\gamma_{2,j}$ ,  $\gamma_3$ ,  $\gamma_{4,j}$ , and  $\gamma_{5,j}$  are estimated parameters and vectors of parameters for the fixed effects, and  $\epsilon_{ij}$  is the residual for participant  $i$  at visit  $j$ . The regimen-specific random effects are normally distributed with mean 0 and variance  $\sigma_r^2$ . The vector of residuals for a given participant are normally distributed with mean  $\mathbf{0}$  and an unstructured covariance matrix  $\mathbf{R}$ . The regimen-specific random effect for a given participant and residuals for that participant are uncorrelated.

The following SAS code specifies the model:

```
proc mixed data=xxx method=reml;
  class regimen id trtrnd visit;
  model Value = trtrnd|visit
              sx2b1|visit dFRS|visit r1z|visit edv|visit / solution cl;
  random intercept / subject=regimen type=vc;
  repeated visit / subject=id type=un;
```

where  $id$  is a participant study identifier,  $trtrnd$  is the randomly assigned treatment group,  $visit$  is the visit identifier,  $Value$  is value of the efficacy endpoint being tested for a given participant at a given visit,  $sx2b1$  is years since ALS symptom onset centered at the sample median,  $dFRS$  is pre-baseline slope centered at the sample median,  $r1z$  is an indicator of riluzole use at baseline, and  $edv$  is an indicator of edaravone use at baseline. The primary estimate will be the treatment-dependent difference in change from baseline to the Week 24 Visit. The estimate and its 95% Wald confidence bounds will be obtained by a linear contrast of adjusted means.

The estimand estimated by the primary linear contrast of the shared-baseline, repeated-measures linear mixed model is the mean difference in 24-week change from baseline of a given continuous efficacy endpoint in the two active treatment groups relative to the placebo group in the FAS population. The estimand is defined by the following attributes:

- Treatment: CNM-Au8 administered orally at dosages of 30 mg/d and 60 mg/d vs. placebo.
- Population: FAS population as defined in Section 6.1.
- Variables: absolute change in endpoint from baseline to the Week 24 Visit.
- Intercurrent event: treatment discontinuation: handled via treatment policy approach, all data will be used including data collected during the placebo-controlled period after treatment discontinuation. Missing data post-treatment, including data missing due to death, will not be imputed, handled via missing at random assumption.

- Population-level summary: difference in conditional means of active treatment relative to placebo.

Inference from this analysis is supportive of inference from the Bayesian shared-parameter, repeated-measures model for the primary endpoint and is the primary analysis for the secondary SVC endpoint.

The following SAS code specifies the linear contrast for an endpoint measured every 8 weeks and assuming that the sort order for treatment group is placebo, 30 mg/d CNM-Au8, and 60 mg/d CNM-Au8 and visits are sorted chronologically:

```
estimate "3|Act vs Plb|dWk 24"
          post*trtrnd*visit 2 0 0 -2 -1 0 0 1 -1 0 0 1 / cl divisor=2;
```

A significant difference in 24-week change in the direction of improved function would support inference that CNM-Au8 treatment reduced progression over 24 weeks for the efficacy endpoint being tested.

A separate supportive analysis for clinical endpoints (ALSFRS-R total and domain scores, HHD upper extremity, lower extremity, and global average percentages, SVC, FVC by home spirometry, ALSAQ-40 domain scores and SI, and CNS-BFS total score) will include centered baseline serum NfL level and the interaction of centered baseline serum NfL level and visit as additional covariates.

#### 6.5.4 Survival

The primary estimate of the effect of treatment on survival as an efficacy endpoint will include both death and death equivalent as components, will pool participants randomized to 30 mg/d CNM-Au8 and 60 mg/d CNM-Au8 in the FAS analysis set, and will include baseline age as an additional covariate in adjusted models. Otherwise, summaries and analyses of time to death or death equivalent are the same as specified in the M-SAP.

The estimand estimated by the logrank test is the deviation in expected survival times in the FAS population. The estimand is defined by the following attributes:

Treatment: CNM-Au8 administered orally at dosages of 30 mg/d and 60 mg/d vs. placebo.

Population: FAS population as defined in Section 6.1.

Variables: time to earlier of death or death equivalent.

Intercurrent event: treatment discontinuation: handled via treatment policy approach, all data will be used including data collected during the placebo-controlled period after treatment discontinuation. Missing data post-treatment will not be imputed, handled via missing completely at random assumption.

Population-level summary: difference in survival curves of active treatment and placebo.

In the presence of a significant difference by the logrank test, inference on whether active treatment extended survival will be based on the hazard ratio estimated from an unadjusted Cox proportional hazards regression.

Survival to the end of the placebo-controlled period will also be analyzed in the ECM, ERO, EPP, STF, SFN, and SRO analysis sets. Survival analyses that include follow-up beyond the placebo-controlled period will be analyzed in the ERO analysis set.

Listings of all participants in the FAS, ECM, ERO, EPP, STF, and SRO analysis sets will document whether each participant had died, had reached a death equivalent endpoint, or was alive and free of a death equivalent and the study day of death, death equivalent, or last known alive and free of a death equivalent for each analysis timepoint.

### 6.5.5 Random-slopes Model

The specification of the random-slopes linear mixed model and the primary linear contrast for estimating differences in mean rate of progression in a given continuous efficacy endpoint (ALSFRS-R total and domain scores, HHD upper extremity, lower extremity, and global average percentages, SVC, FVC by home spirometry, quantitative voice characteristics, serum creatinine, serum NfL, urinary p75[ECD], ALSAQ-40 domain scores and SI, and CNS-BFS total score) are revised from those specified in the M-SAP to include a main effect of treatment and with the modification that treatment group is a three-level variable (i.e., placebo, 30 mg/d CNM-Au8, and 60 mg/d CNM-Au8).

The model will include fixed terms for month since the Baseline Visit, treatment group, treatment group  $\times$  month interaction, centered years since ALS symptom onset and centered years since ALS symptom onset  $\times$  month interaction, centered delta-FRS and centered delta-FRS  $\times$  month interaction, centered baseline riluzole use and centered baseline riluzole use  $\times$  month interaction, and centered baseline edaravone use and centered baseline edaravone use  $\times$  month interaction. The following equations describe the model with regimen random effects:

$$\begin{aligned}
 Y_{ij} = & \gamma_1 + a_{k(i)}^0 + b_i^0 + \gamma_2 t_i + \gamma_3' z_i \\
 & + (\gamma_4 + a_{k(i)}^1 + b_i^1 + \gamma_5 t_i + \gamma_6' z_i) m_{ij} + \epsilon_{ij} \\
 \{a_k^0, a_k^1\} \sim & N(\mathbf{0}, \Sigma_r), \{b_k^0, b_k^1\} \sim N(\mathbf{0}, \Sigma_p), \epsilon_{ij} \sim N(0, \sigma_\epsilon^2) \\
 \text{Cov}(\mathbf{a}_k, \mathbf{b}_k) = & \mathbf{0}, \text{Cov}(\mathbf{a}_k, \epsilon_{i.}) = \mathbf{0}, \text{and } \text{Cov}(\mathbf{b}_k, \epsilon_{i.}) = \mathbf{0}
 \end{aligned} \tag{eqn. 2}$$

where  $Y_{ij}$  is a given efficacy endpoint measured in participant  $i$  at visit  $j$ ,  $a_{k(i)}^0$  and  $a_{k(i)}^1$  are random intercept and slope for regimen  $k$  to which participant  $i$  was assigned,  $b_i^0$  and  $b_i^1$  are random intercept and slope for participant  $i$ ,  $z_i$  is the vector of covariates (centered time since onset, centered delta-FRS, centered baseline riluzole use, and centered baseline edaravone use) for participant  $i$ ,  $m_{ij}$  is the time from baseline to observation  $j$  for participant  $i$  in months calculated as days  $\times 12 / 365.25$ ,  $t_i$  is an indicator variable for treatment  $t$  to which participant  $i$  was assigned,  $\gamma_1, \gamma_2, \gamma_3, \gamma_4, \gamma_5$ , and  $\gamma_6$  are estimated parameters and vectors of parameters for the fixed effects, and  $\epsilon_{ij}$  is the residual for observation  $j$  for participant  $i$ . The regimen-specific random effects are normally distributed with mean  $\mathbf{0}$  and unstructured covariance matrix  $\Sigma_r$ . The participant-specific random effects are normally distributed with mean  $\mathbf{0}$  and unstructured covariance matrix  $\Sigma_p$ . The residuals for a given participant are normally distributed with mean 0 and variance  $\sigma_\epsilon^2$ . The regimen-specific random effects, participant-specific random effects, and residuals are uncorrelated.

The following SAS code specifies the model:

```

proc mixed data=xxx method=reml;
  class regimen id trtrnd;
  model Value = trtrnd|month
              sx2b1|month dFRS|month rlz|month edv|month / solution cl;
  random intercept month / subject=regimen type=un;

```

```
random intercept month / subject=id type=un;
```

where month is time in months from the Baseline Visit (assuming 12 months in an average of 365.25 days per year) and other fields are the same as identified above in Section 6.5.3. The primary estimand will be the treatment-dependent difference in slopes. The estimate and its 95% Wald confidence bounds will be obtained by a linear contrast of adjusted means.

The estimand estimated by the primary linear contrast of the random-slopes linear mixed model is the difference in mean rate of progression of a given continuous efficacy endpoint in the two active treatment groups relative to the placebo group in the FAS population. The estimand is defined by the following attributes:

Treatment: CNM-Au8 administered orally at dosages of 30 mg/d and 60 mg/d vs. placebo.

Population: FAS population as defined in Section 6.1.

Variables: mean rate of change in endpoint from baseline to the Week 24 Visit.

Intercurrent event: treatment discontinuation: handled via treatment policy approach, all data will be used including data collected during the placebo-controlled period after treatment discontinuation. Missing data post-treatment will not be imputed, handled via missing at random assumption.

Population-level summary: difference in conditional mean slopes of active treatment relative to placebo.

Inference from these analyses is supportive of inference from the Bayesian shared-parameter, repeated-measures model for the primary endpoint and inference from the repeated-measures linear mixed model for secondary endpoints.

The following SAS code specifies the linear contrast assuming that the sort order for treatment group is placebo, 30 mg/d CNM-Au8, and 60 mg/d CNM-Au8:

```
estimate "3|Act vs Plb|Slope (/mn)"
        month 0 trtrnd*month -2 1 1 / cl divisor=2;
```

A significant difference in slopes in the direction of improved function would support inference that the treatment reduced rate of progression for the efficacy endpoint being tested.

A separate supportive analysis for clinical endpoints (ALSFRS-R total and domain scores, HHD upper extremity, lower extremity, and global average percentages, SVC, FVC by home spirometry, quantitative voice characteristics, ALSAQ-40 domain scores and SI, and CNS-BFS total score) will include centered baseline serum NfL level and the interaction of centered baseline serum NfL level and study month as additional covariates.

### 6.5.6 Time to Clinical Events

Time to hospitalizations and clinical events will be analyzed in the FAS, ECM, ERO, and EPP analysis sets. The summaries and analyses will parallel the analyses described for survival described in Section 6.5.4. Analysis of time to King's stage 4a or 4b and time to a 6-point decline in ALSFRS-R total score will accommodate interval censoring between ALSFRS-R assessments. Analysis of time to King's stage 4a or 4b will be stratified by baseline King's stage.



### 6.5.7 Composite of ALSFRS-R, SVC, and ALSAQ-40 Slopes

Participant-specific rates of change in ALSFRS-R total score, SVC percent-predicted, and ALSAQ-40 total score will be estimated in separate, endpoint-specific unadjusted random-slopes linear mixed models in the FAS, ECM, and ERO analysis sets. Participant-specific random slopes for each endpoint will be standardized to a mean of zero (0) and a variance of one (1) and then averaged for each participant. The composite average of standardized random slope estimates will be compared across treatment groups by one-way analysis of variance with adjustment for the following median-centered covariates: years since ALS symptom onset, delta-FRS, baseline riluzole use, and baseline edaravone use. A separate analysis will include baseline serum NfL level as an additional covariate. Treatment comparison between all active arm participants (i.e., the average of both 30 mg/d and 60 mg/d dosage groups) vs. placebo participants and between each active treatment group separate vs. placebo will be estimated using contrasts of least-square means.

### 6.5.8 HHD0 and HHD0<sup>2</sup>

Analyses of HHD0 are the same as specified in the M-SAP with the modification that treatment group is a three-level variable (i.e., placebo, 30 mg/d CNM-Au8, and 60 mg/d CNM-Au8), with addition of parallel analyses of HHD0<sup>2</sup>, with a separate analysis that includes baseline serum NfL level as an additional covariate, and with the clarification that time to zero strength for both analyses is interval censored between HHD assessments.

Inference from these analyses is supportive of inference from the repeated-measures linear mixed model for HHD upper and lower extremity scores.

### 6.5.9 Quantitative Voice Measures

Given the high frequency of voice recordings, a repeated-measures analysis with unstructured covariance is overly flexible but the assumption of linear change required by the random-slopes model may be overly rigid. To complement estimates from the random-slopes linear mixed model, quantitative voice characteristics will be analyzed in a linear mixed model in which the temporal profile for both fixed and random terms is modeled using cubic B-splines with knots at 8 and 16 weeks. The model will include fixed terms for B-splines (4 terms), treatment group (3 levels), treatment group × B-spline interaction, centered time since symptom onset and centered time since symptom onset × B-spline interaction, centered delta-FRS and centered delta-FRS × B-spline interaction, centered baseline riluzole use and centered baseline riluzole × B-spline interaction, and centered baseline edaravone use and centered baseline edaravone × B-spline interaction. The model will include random regimen-specific intercepts and slopes with unstructured covariance, random participant-specific B-splines (5 terms) with unstructured covariance, and a first-order autoregressive structure for residuals. A simplified covariance structure assuming no regimen-level covariance, heterogeneous compound symmetric covariance among the random B-splines, conditional independence of residuals, or a combination of the three simplifying assumptions will be used if the full model fails to converge. The primary estimand will be the active (i.e., the average of both 30 mg/d and 60 mg/d dosage groups) vs. placebo treatment-dependent difference in 24-week change from baseline. The estimate and its 95% Wald confidence bounds will be obtained by a linear contrast of adjusted means.

A separate supportive analysis will include centered baseline serum NfL level and the interaction of centered baseline serum NfL level and B-splines as additional covariates.



### 6.5.10 Placebo Multiple Imputation

Placebo multiple imputation analyses are the same as specified in the M-SAP with the modification that treatment group is a three-level variable (i.e., placebo, 30 mg/d CNM-Au8, and 60 mg/d CNM-Au8). Placebo multiple imputation analyses will be applied to ALSFRS-R total score, HHD upper and lower extremity percentages, and SVC.

The following SAS code specifies the imputation for an endpoint measured every 8 weeks:

```
proc mi data=work.outcomes seed=xxx nimpute=50 out=work.out_mi01
    minimum=. . . . . x x x x maximum=. . . . . y y y y minmaxiter=1000;
    class rlz edv trtrnd;
    fcs reg(Wk08 = sx2b1 dFRS rlz edv Wk00);
    fcs reg(Wk16 = sx2b1 dFRS rlz edv Wk00 Wk08);
    fcs reg(Wk24 = sx2b1 dFRS rlz edv Wk00 Wk08 Wk16);
    mnar model( Wk08 Wk16 Wk24 / modelobs=(trtrnd="0"));
    var sx2b1 dFRS rlz edv Wk00 Wk08 Wk16 Wk24;
run;
```

where Wk00, Wk08, Wk16, and Wk24 are the values of a given efficacy endpoint at the Baseline, Week 8, Week 16, and Week 24 Visits, respectively, trtrnd has a value of zero (0) for participants randomized to placebo, and x and y take appropriate values to specify the range of a given outcome measure (i.e., 0 and 48 for ALSFRS-R total score; 0 and . for HHD upper and lower extremity percentages and SVC).

Inference from these analyses is supportive of inference from the Bayesian shared-parameter, repeated-measures model for the primary endpoint and assess sensitivity to the missing data assumption of the repeated-measures linear mixed model for secondary endpoints in the FAS analysis set. A separate analysis will include centered baseline serum NfL level as an additional covariate in both imputation stages.

### 6.5.11 Additional Sensitivity Analyses of Primary and Key Secondary Outcomes

Sensitivity analyses of primary and key secondary efficacy outcomes are the same as specified in the M-SAP with the modification that treatment group is a three-level variable (i.e., placebo, 30 mg/d CNM-Au8, and 60 mg/d CNM-Au8).

### 6.5.12 Subgroup Analyses

In addition to the subgroups specified in the M-SAP, the following additional subgroups will be analyzed in the random-slope model (see Section 6.5.5) for primary and secondary efficacy endpoints in the FAS analysis set:

- Baseline use of riluzole and edaravone (neither, riluzole only, edaravone only, both),
- Age (less than 65 years vs. 65 years or older),
- Sex (female vs. male),
- Race (white vs. any minority race with greater than 5% prevalence in the sample),
- Ethnicity (Hispanic or Latino vs. non-Hispanic or Latino),

- Weight (less than 43 kg, 43 to less than 56 kg, 56 to less than 77 kg, 77 to less than 150 kg, 150 kg or more),
- BMI (less than 18.5 kg/m<sup>2</sup>, 18.5 to less than 25 kg/m<sup>2</sup>, 25 to less than 30 kg/m<sup>2</sup>, 30 to less than 40 kg/m<sup>2</sup>, 40 kg/m<sup>2</sup> or more),
- Chronic kidney disease (CKD) stage (stage 1 or better [eGFR 90 mL/min/1.73m<sup>2</sup> or more], stage 2 [eGFR 60 to 89 mL/min/1.73m<sup>2</sup>], stage 3 [eGFR 30 to 59 mL/min/1.73m<sup>2</sup>], stage 4 [eGFR 15 to 29 mL/min/1.73m<sup>2</sup>], stage 5 [eGFR less than 15 mL/min/1.73m<sup>2</sup>]),
- Time since onset of weakness (less than 18 months vs. 18 months or longer),
- El Escorial and time since onset of weakness (El Escorial definite and less than 18 months vs. not both),
- Baseline symptom severity (all ALSFRS-R questions scored 2 or greater vs. any question scored 0 or 1),
- Early disease state (all ALSFRS-R questions scored 2 or greater, SVC 80%-predicted or greater, and time since symptom onset less than 24 months vs. not meeting all three criteria),
- Baseline serum urate concentration (less than 5.5 mg/dL vs. 5.5 mg/dL or greater),
- Baseline serum NfL concentration (by median split),
- Baseline delta-FRS score (less than 0.47 points/year, 0.47 to less than 1.11 points/year, 1.11 points/year or greater),
- Site of onset (any bulbar onset, no bulbar onset),
- Site (individual sites with at least 5 participants per treatment group and all participants from sites with fewer than 5 participants per treatment group pooled).

For each classification, unknown, not reported, and missing will be considered one group. All individuals not included in a specified subgroup will be combined into a mixed, "other" group. The "other" group will be included in analyses if its prevalence is greater than 5%; otherwise, the "other" group will be excluded.

In cases where a model for a given subgroup and endpoint fails to converge, the covariance terms for the regimen-specific random effects will be simplified from unstructured covariance of intercepts and slopes to separate, uncorrelated variance components for intercepts and slopes. If convergence still fails, regimen-specific intercepts and slopes will be modeled as fixed effects. If convergence still fails, the participant-specific random effects will be simplified from unstructured covariance of intercepts and slopes to separate, uncorrelated variance components for intercepts and slopes.

### 6.5.13 Comparison of Controls across Regimens

Comparisons of placebo participants across regimens are the same as specified in the M-SAP with separate analyses that include baseline serum NfL level as an additional covariate plus applicable interaction terms as relevant to a given model.

#### 6.5.14 Pharmacokinetic Analyses of CNM-Au8

Pre-dose concentrations of CNM-Au8 in whole blood and plasma will be summarized by treatment group and visit in the ERO sample. Concentrations below the limit of quantitation (BLQ) will be replaced with one half of the lower limit of quantitation. Summaries will include number of observations, number and percentage with concentrations BLQ, arithmetic mean, median, standard deviation, minimum, maximum, geometric mean, geometric coefficient of variation (calculated as  $\sqrt{\exp(\text{variance of log-transformed concentrations}) - 1}$ ), and 95% confidence bounds for the geometric mean assuming log-normally distributed data.

A concentration response relationship based on observed CNM-Au8  $C_{\max}$  per active participant versus ALSFRS-R slope change will also be analyzed as an exploratory endpoint.

Whole blood and plasma concentration data of CNM-Au8 may be subjected to population pharmacokinetic analysis to derive population estimates of pharmacokinetic parameters and test the effect of various covariates such as age, weight, and sex. Details of the analysis will be described in a separate data analysis plan (DAP). This analysis may be performed by combining data from the current study with data from other studies of CNM-Au8, if deemed appropriate. The tests of association with ALSFRS-R and the population pharmacokinetic analysis will be performed by Clene Nanomedicine or its designee and reported in a separate modelling report.

#### 6.5.15 Pharmacodynamic Biomarker Analyses

Pharmacodynamic biomarkers other than serum creatinine, serum and CSF NfL, and urinary p75[ECD] will be analyzed by Clene Nanomedicine according to a separate pharmacodynamic DAP (Regimen C Pharmacodynamic Analysis Plan) and will be reported in a separate report.

#### 6.5.16 Pharmacokinetics of Riluzole

Pharmacokinetic analysis of plasma riluzole concentrations will be analyzed according to a separate pharmacokinetic analysis plan authored and conducted by Dr. Charles Venuto, PharmD.

### 6.6 Safety Analyses

#### 6.6.1 Treatment-emergent Adverse Events

Summaries and analyses of treatment-emergent adverse events (TEAE) are the same as specified in the M-SAP with the following revisions.

TEAEs are defined as those adverse events with onset dates in the interval from double-blind treatment initiation to either the Final Safety Visit if completed, the date of death if the participant dies, 28 days after last dose of study drug if the participant early terminates or is lost to follow-up, or the date of first dose of study drug during participation in the OLE, if so exposed. Adverse events with onset on the day of double-blind treatment initiation and adverse events with incompletely specified onset date where the ambiguous date spans the day of double-blind treatment initiation will be assumed to be treatment emergent except for those known to precede first exposure to study drug.

In addition to summaries specified in the M-SAP, serious TEAEs will be summarized by MedDRA system organ class, high level term, and preferred term for all TEAEs. In addition, TEAEs will be summarized by Medical Dictionary for Regulatory Activities (MedDRA) system organ class and preferred term for fatal TEAEs, TEAEs that occurred during a participant's COVID-19 infection (defined as 5 days prior to symptom onset to end of COVID-19 symptoms

or the earlier of 91 days after symptom onset or the end of double-blind follow-up, if ongoing), and TEAEs and serious TEAEs stratified by the following subgroups: baseline riluzole use, baseline edaravone use, age, sex, race, ethnicity, weight, BMI, and CKD stage. Subgroup classifications will be the same as described in Section 6.5.12 Subgroup Analyses except that the "other" group will be retained regardless of prevalence.

TEAEs indicating COVID-19 infection are the following: Asymptomatic COVID-19, COVID-19, COVID-19 pneumonia, COVID-19 treatment, Post-acute COVID-19 syndrome, SARS-CoV-2 antibody test positive, SARS-CoV-2 RNA increased, SARS-CoV-2 sepsis, SARS-CoV-2 test false negative, SARS-CoV-2 test positive, SARS-CoV-2 viraemia, Suspected COVID-19.

Treatment-dependent differences in the proportion of participants experiencing a given type of TEAE will not be tested. Treatment-dependent differences in TEAE incidence rates in units of number per 100 participant years will be estimated as differences rather than ratios and will include comparison-wise 95% confidence intervals with variance estimates obtained by the delta method.

### **6.6.2 Safety Labs**

Summaries and analyses of clinical safety labs are the same as specified in the M-SAP with the revision that abnormal levels will be classified to a toxicity grade based on quantitative grading using National Cancer Institute (NCI) Common Terminology Criteria for Adverse Events (CTCAE) version 5.0 and with the addition that maximum toxicity over all post-baseline visits will be included in shift tables along with visit-specific shifts.

The proportion of participants with ALT >3x ULN, AST >3x ULN, creatinine >1.5x level at the Baseline Visit, and platelet count <75,000/mm<sup>3</sup>, or with any of the four will be presented as shift tables vs. the status of each participant at baseline for each visit by treatment group in all safety samples. The absolute level and the absolute change from baseline for ALT, AST, creatinine, and platelet count will be summarized as means, standard deviations, medians, and ranges at each visit by treatment group in all safety samples.

### **6.6.3 ECG Results**

Summaries of ECG parameters and findings are the same as specified in the M-SAP.

### **6.6.4 Vital Signs and Weight**

Summaries and analyses of vital signs and weight are the same as specified in the M-SAP.

### **6.6.5 Suicidality**

Summaries of suicidality are the same as specified in the M-SAP.

## **6.7 Other Analyses**

### **6.7.1 Participant Disposition**

All participants consented to the Master Protocol between the time of the first and last consent of a participant assigned to a regimen included in the FAS analysis set will be summarized as a single set for the following events: consented to the Master Protocol, failed screening for the Master Protocol, other reasons not assigned to a regimen (including timing out of the screening window, death, withdrawal of consent, early termination, and administrative termination), and assigned to a regimen. Reasons for Master Protocol screen failure will be summarized.

All participants in the above sample assigned to a regimen will be summarized as two sets (final screening for RGC vs. final screening for a non-RGC regimen) for the following events: consented to a regimen, failed screening for a regimen, other reasons not randomized within a regimen (including timing out of the screening window, death, withdrawal of consent, early termination, and administrative termination), and randomized within a regimen. If a given individual is screened multiple times prior to randomization within a regimen, then the final screening experience of that individual will be summarized. Reasons for RGC screen failure will be summarized separately for all participants screened for RGC whether that was their final screening experience or not.

All participants in the FAS analysis set will be summarized as two sets (randomization to active study drug vs. randomization to placebo) for the following events: initiated regimen-specific study drug, prematurely terminated study participation due to death, withdrawal of consent, early termination, loss to follow-up, or administrative termination, completed 24-week follow up, and completed a safety follow-up visit vs. continued into the OLE. Reasons for withdrawal of consent or early termination after randomization will be summarized.

Any randomized participants excluded from the FAS and EPP analysis sets or included in the FAS analysis set but not contributing to the primary analysis will be identified in a listing together with the reason for their exclusion.

### **6.7.2 Study Drug Compliance and Tolerance**

Summaries of study drug compliance and tolerance are the same as specified in the M-SAP with the clarification that summaries will be reported for the ERO and SRO analysis sets and that date of permanent discontinuation of study drug is the date of last use of double-blind study drug among all participants in a given analysis set.

The number of days of exposure to study drug will be calculated in three ways:

- as the number of days from dose initiation to the final safety assessment during the placebo-controlled period, inclusive,
- as the number of days from dose initiation to drug withdrawal, inclusive, less any interval during which use of study drug was interrupted (individual missed doses will not be subtracted unless noted in the dosage management log), and
- as the number of days from dose initiation to the earlier of final contact during the placebo-controlled period or 28 days after last dose of study drug, inclusive.

The proportion of participants who interrupted study drug or reduced study drug dosage and the time to first study drug interruption or dosage reduction will be summarized. The number of days of exposure to a reduced dosage of study drug will be summarized.

### **6.7.3 Concomitant Medication Use**

Summaries of concomitant medication use are the same as specified in the M-SAP with the clarification that medications taken at baseline and those initiated after first dose of study drug will be separately summarized and will be classified by ATC Therapeutic class and WHODrug Preferred base name.

#### **6.7.4 Medical History**

Medical histories will be summarized by MedDRA system organ class, high level term, and preferred term in the STF and SRO analysis sets.

#### **6.7.5 Blindedness**

The proportions of participants and site investigators who report on the Exit Questionnaire a guess of active vs. placebo treatment assignment, each level of surety of that guess, and each of five pre-specified reasons for making a treatment assignment will be summarized by treatment group in the FAS and ERO analysis sets. Treatment-dependent differences in the proportion guessing active treatment assignment will be tested among all respondents and among those stating they are at least somewhat sure of their guess by Fisher's exact test and the difference in proportion guessing active treatment assignment will be estimated with confidence bounds.

#### **6.7.6 Protocol Deviations**

The number of major and minor protocol deviations will be summarized by type of deviation and treatment group in all analysis sets. Listings of all protocol deviations will be produced.

#### **6.7.7 Impact of COVID-19 Pandemic**

The proportions of planned assessments missed due to COVID-19 restrictions or disruptions will be summarized by treatment group, visit, and type of assessment in the FAS and ERO analysis sets. Protocol deviations that resulted from COVID-19 restrictions or disruptions will be summarized by treatment group and type of deviation in the FAS and ERO analysis sets and listed.

### **7. Validation**

#### **7.1 Primary Efficacy Analysis**

Validation of the primary efficacy analysis is the same as specified in the M-SAP.

#### **7.2 Secondary, Exploratory, and Safety Analyses**

Validation of secondary, exploratory, and safety analyses are the same as specified in the M-SAP.

### **8. References**

The following references are cited in addition to those specified in the M-SAP:

- Balendra R, Jones A, Jivraj N, Knights C, Ellis CM, Burman R, Turner MR, Leigh PN, Shaw CE, Al-Chalabi A. Estimating clinical stage of amyotrophic lateral sclerosis from the ALS Functional Rating Scale. *Amyotroph Lateral Scler Frontotemporal Degener*. 2014 Jun;15(3-4):279-84.
- Roche JC, Rojas-Garcia R, Scott KM, Scotton W, Ellis CE, Burman R, Wijesekera L, Turner MR, Leigh PN, Shaw CE, Al-Chalabi A. A proposed staging system for amyotrophic lateral sclerosis. *Brain*. 2012 Mar;135(Pt 3):847-52.